

Governing Environmental Markets: Evidence From Irrigation In Water Markets*

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Abstract

This paper provides evidence of how property rights institutions, particularly local irrigators' organizations, impact water markets' efficiency. Our analysis is based on a unique dataset that integrates administrative and hydrological records, geographic information, and satellite imagery. We develop a novel misallocation test, which suggests that these organizations reduce misallocation caused by the natural capacity of upstream users to over-extract. Using different identification strategies, we show that these efficiency gains are a result of both water reallocation and individual adaptation. In the short run, downstream farmers increase their water consumption and agricultural yield. Consistent with this large-scale reallocation, we find increases in river stream-flow. In the long run, these farmers plant otherwise infeasible water-intensive crops using capital-intensive irrigation technologies, suggesting that enforcement and capital investment are complements. Meanwhile, although upstream farmers reduce their water consumption, their productive outcomes remain mostly unchanged. Our results suggest that environmental markets require institutions that do more than assign property rights: they must also solve the enforcement challenges inherent in the underlying resource. *JEL codes: D23, D24, H41, O13, Q12, Q15, Q25.*

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1 Introduction

Climate change is closely intertwined with the Tragedy of the Commons. It arises from a commons problem—the unrestricted emission of greenhouse gases—and generates new challenges around other common pool resources, such as the increasing frequency of droughts worldwide (IPCC, 2021). The establishment of environmental markets over those resources aims to address these issues by restricting open access and forcing agents to internalize the negative externalities of their actions (Greenhill et al., 2026). However, our understanding of the institutions necessary to sustain the operation of these markets remains limited. This is particularly important since managing common pool resources presents inherent challenges—such as high exclusion costs and monitoring difficulties (Ostrom, 1990, 2009)—that complicate market implementation. In this paper, we use intra-country variation in property rights institutions to show empirically that water markets may fail to allocate water efficiently despite legally well-defined property rights, and further study how specialized enforcement institutions can improve their operation.

We study water allocation in 12 large-scale river basins in Chile, which share a unique institutional setting that allows us to isolate the role of local enforcement institutions. In Chile, water is allocated through water rights that, unlike many natural resource markets in developing countries (e.g., land, see Chari et al., 2021; Manyшева, 2022; Bolhuis et al., 2026), are full property rights: they are perpetual, tradable, inheritable, independent of land tenure rights, and constitutionally protected against expropriation. Yet these strong protections have led regulations and courts to restrain government action, making it difficult for state agencies to enforce property rights, particularly to protect downstream users from upstream over-extraction. In this context, local formal irrigation organizations called Water Boards (*Juntas de Vigilancia*) have been established with both the goal and power to enforce water rights and resolve conflicts between users.

Our analysis shows that water markets may achieve within-basin allocative efficiency only when supported by these specialized enforcement institutions, despite property rights being legally well defined across the territory. In a competitive market equilibrium, the marginal value of the resource should be equal across locations, with deviations from this

benchmark indicating a Pareto inefficient allocation. To empirically assess whether this condition holds and where, we develop a novel misallocation test that exploits idiosyncratic variation in rainfall timing and volume to compare the shadow value of water at different locations within a basin. This “sufficient statistic approach” measure identifies economic misallocation even after accounting for adaptation, entry and exit decisions, and private arrangements made by the agents. Our results show that the average shadow value of water remains constant within basins governed by Water Boards. In contrast, in areas without such boards, the shadow value of water is significantly higher in downstream than in upstream locations, indicating over-extraction by upstream users relative to a socially optimal allocation.

We then examine the mechanisms behind these differences in allocative efficiency, which fall into two families: the reallocation of water toward downstream users, and the private adaptations and investments it enables, such as changes in crop choice and irrigation technology. We first provide evidence that board establishment causes large-scale reallocation in the short run—with immediate consequences in production and efficiency. We also show that farmers governed by boards adopt productive decisions and investments in the long run that are complements of improved water reliability—meaning that cost-effective private adaptations are not enough to compensate for the lack of governance.

To establish that Water Boards causally affect water allocation, we exploit the staggered adoption of Water Boards across basins in an event-study difference-in-differences design, focusing on the dry season (December-February) when water scarcity is most acute and irrigation demand peaks. Using remote sensing-based estimates of water consumption and agricultural yields at the farm-plot level, we measure the distributive impacts of Water Board establishment across locations within treated areas. We find that following the establishment of a Water Board, downstream farms increase their water consumption by 1 standard deviation of their individual historical distribution, growing up to 2 standard deviation after 7 years; upstream farms, instead, see reductions up to 0.8 standard deviations after 7 years. We also show how this addresses a source of misallocation directly: downstream farms increase their yields by 0.6 standard deviations after 7 years, while upstream farms do not see significant reductions in this measure.

These farm-level reallocation patterns are driven by changes in river streamflows, the main physical channel through which Water Boards reallocate water for irrigation. Exploiting the staggered adoption of Water Boards in a difference-in-differences design at the river segment level, we find that Water Board establishment increases river streamflows during the dry season, when incentives to over-extract are strongest, by 13% on average over the post-establishment period and up to 25% at its peak. The effect is concentrated in areas with high agricultural activity, and is stronger at upstream monitoring stations, consistent with Water Boards preventing upstream over-extraction by physically controlling water flows through canal gates,¹ and enabling water to reach downstream users. We do not observe effects in the rest of the year, when the absence of enforcement is not binding.

To shed light on long-run effects, we employ an instrumental variable strategy based on the cost of reaching the specific courts with jurisdiction to initiate Water Board establishment—costs that are irrelevant for other legal matters affecting downstream users. We recenter this distance against the distances implied by alternative jurisdictional rules, so that the instrument captures only the arbitrary component of court assignment (Borusyak and Hull, 2023). Our IV estimates suggest that enforcement effects extend beyond the mechanical reallocation of water. The greater stability and predictability of water access enabled by Water Boards allows downstream farms to make complementary investments: they increase the planting of fruit trees by 20 pp , extend their growing season by 0.7 months, and allow numerous farms to switch to summer crops. The increase in fruit tree planting is driven by the planting of water-intensive varieties, and accompanied by the usage of capital-intensive irrigation techniques. These induced private responses help explain the pattern observed in our event-study estimates, where increases in water consumption among downstream users exceed the reduction among upstream farmers, a pattern consistent with enforcement generating net gains through both direct reallocation and the productive investments it enables.

A simple model featuring unobservable productivity differences and unverifiable water consumption formalizes this intuition.² We focus on two equilibria. Without governance,

¹See Section 2 for a detailed description of the enforcement mechanism and Figure I for an illustration.

²See Appendix B.

responsibility for any downstream shortfall is dispersed among all upstream users, leading to misallocation and a lack of litigation. With governance, responsibility for downstream water access is centralized in an agent that any user denied their allotment can sue, which makes the failure attributable to a single defendant and generates incentives to invest in costly enforcement.

Overall, our results indicate that legally well-defined property rights alone are insufficient to ensure efficient resource allocation when enforcement is weak. At first glance, Chile’s water rights system appears to satisfy the Coase Theorem’s prerequisites: property rights are clearly defined, tradable, and protected in a country with strong Rule of Law.³ Yet our findings reveal that these conditions are not sufficient for markets to allocate water efficiently (Coase, 1960; Medema, 2020; Deryugina et al., 2021).

The key missing element is effective enforcement—a gap that emerges directly from Chile’s regulatory design, which deliberately limits government intervention in favor of decentralized transactions (Bauer, 2004; Tamayo and Carmona, 2019b). This reflects a fundamental tension in the design of environmental markets: while reducing state oversight may lower some transaction costs, it can simultaneously undermine the enforcement mechanisms necessary for markets to function. Our results show that institutional arrangements empowering local user communities to monitor and enforce rights can bridge this gap (Ostrom, 1990), improving allocative efficiency even when formal property rights are already well established.

Our paper contributes to a range of literatures, including management of water and common pool resources in general, frictions in environmental and developing markets, and the economic impacts of agricultural infrastructure. We first contribute to the literature on common pool resource management.⁴ Ostrom (1990) identifies local communities as an alternative to state management or privatization for governing common pool resources, with subsequent work testing these ideas in small-scale settings through case studies and lab

³In 2024, Chile ranked 25th worldwide in the V-DEM Rule of Law Index with a score of 0.945, comparable to the United Kingdom, France, and Singapore (Herre et al., 2013).

⁴Water exhibits the classic features of a common pool resource—difficulty enforcing exclusion and rivalry in consumption—creating incentives for over-extraction in the absence of effective governance (Hardin, 1968). Conventional approaches allocate decision rights to either the state or private agents through privatization, but implicitly assign monitoring and enforcement tasks to the state.

experiments (e.g. Ostrom and Gardner, 1993; Ostrom, 2009; Meinzen-Dick, 2007; Ostrom, 2010; Cardenas and Carpenter, 2008; Cárdenas and Ostrom, 2004; Henrich et al., 2006).⁵

We extend this literature by showing how local enforcement organizations can address free-riding across multiple communities in large-scale river basins by wielding state-like authority to resolve disputes and sanction violations. More broadly, we contribute to work on water institutions and property rights enforcement (Libecap, 2011; Bauer, 2004; Jacoby et al., 2020; Chong and López-de Silanes, 2006; Edwards, 2015; Edwards et al., 2021; Edwards, 2016; Mesías and Torralba, 2023), providing the first quasi-experimental evidence on the causal impact of governance institutions in settings with well-defined, tradable property rights over common pool resources, and for water in general.

We also contribute to the literature on the efficiency and operation of markets for natural resources and environmental goods. The closest work to ours studies the efficiency of water allocation by quantifying gains from trade and comparing market-based allocations to alternative mechanisms: Rafey (2023) estimates substantial gains from trading in Australia under strong government enforcement, while other work on water compares markets to alternative allocation mechanisms, finding efficiency losses from rationing (Ryan and Sudarshan, 2022), mixed results relative to quota systems (Donna and Espín-Sánchez, 2023), and net gains relative to open access (Ayres et al., 2021). Research on misallocation in other natural resources shows how property rights reforms in land markets enable efficient reallocation and productivity gains (De Janvry et al., 2015; Chari et al., 2021; Gollin and Udry, 2021; Manyasheva, 2022; Bolhuis et al., 2026). Concurrently, recent work on the operation of environmental markets has focused on pollution permits, highlighting that enforcement is of particular concern (Calel et al., 2025b; Gonzalez-Lira and Mobarak, 2022) and that institutional design has first-order effects on market outcomes (Colmer et al., 2024; Greenstone et al., 2025; Calel et al., 2025a).

We build on this literature by making two distinct contributions. Methodologi-

⁵Among the conditions Ostrom (1990) identifies for successful community governance are monitoring capability, availability of sanctions among community members, and closed access to outsiders. However, in our context—distribution of water within basins encompassing multiple communities—these conditions do not directly apply: agents cannot observe actions outside their community, cannot exclude others from purchasing water rights, and asymmetric externalities (upstream actions harm downstream users, but not vice versa) prevent informal cooperation through repeated interactions.

cally, we develop a misallocation test for surface water that recovers the marginal product of water from output responses to exogenous availability shocks. This sufficient statistics approach does not require direct consumption measurement—critical since water use is not metered in many contexts. Substantively, we provide the first direct causal evidence on how enforcement of property rights affects water allocation. Most related work identifies misallocation from legal limits on exercising property rights (e.g. Bolhuis et al., 2026); we instead demonstrate how limits to government action—designed to prevent market interference—can undermine enforcement capacity and generate misallocation (Bauer, 2004). We are able to isolate this mechanism because, unlike pollution spillovers that disperse across space, water externalities in river basins are directional and geographically contained, allowing us to isolate and measure enforcement impacts in ways that are not feasible in other environmental markets.

We further contribute to research on the economic impacts of water infrastructure. Studies of large-scale irrigation infrastructure find increases in agricultural productivity alongside substantial distributional consequences (Duflo and Pande, 2007; Blakeslee et al., 2021; Asher et al., 2022; Burlig et al., 2026; Cisse et al., 2026). We document large differences in efficiency and benefits from infrastructure and markets within a country, and show how these productivity and distributional impacts depend critically on the interaction between local institutions and geography, documenting substantial heterogeneity by location (upstream vs. downstream) and farm characteristics.

Methodologically, we join the first wave of research using remote sensing to measure agricultural water consumption at the farm level (Boser et al., 2024), enabling causal analysis at high spatial and temporal resolution previously infeasible.

The rest of the paper is organized as follows: Section 2 provides an overview of Chile’s water property rights system and the role of Water Boards. After describing our data in Section 3, we present our misallocation test and its results in Section 4. Section 5 analyzes the key mechanisms driving our findings—water reallocation and farmers’ private responses. Section 6 concludes.

2 Background

We study the introduction of Water Boards, a local governance institution that manage rivers in periods of water scarcity and solve legal conflicts among users. Thanks to their local nature, Water Boards know and interact directly with water users, in contrast to most centralized bureaucracies in charge of water management. In this section, we provide background information on the study area, the system of property rights over water, and the characteristics of Water Boards.

Geography. The area under study covers latitudes -30 to -38 and the full longitudinal range of Chile in this area (approximately longitudes -68 to -72.5) as shown in the central panel of Figure II. This area covers 87% of Chile’s population and 85% of its agricultural GDP. The geography is marked by both the Andes –which defines the eastern border of the country– and Coastal Mountain Range that extend in a North-South axis. Most agricultural activity takes place in the Central Valley that separates both ranges, and most rivers run from the Andes (East) to the Coast (West)(Fernández and Gironás, 2021). The rugged topography makes the construction of infrastructure connecting basins extremely costly. In our analysis, we focus on 12 large-scale rivers that run across the full longitudinal range of Chile in this area –i.e. with river heads in the Andes, at the border with Argentina, to the river mouth in the Pacific. We present these basins in Figure III.

The climate in this area is characterized as Mediterranean, with rainfall increasing in a North-South gradient; and a dry season extending from November to March. Rivers in this area are mostly fed by both rainfall and snowmelt (Varas and Varas, 2021; CNR, 2018a). This implies that rivers reach their maximum stream levels in the boreal winter and spring, then decline to reach minimum levels in summer and early fall (between February and April).⁶ Importantly, longer days make summer a key period for agricultural production, implying that irrigation is most important in the driest months.⁷

⁶In the period we are studying (1981-2010), groundwater was not a significant substitute for surface water during these low-flow periods; well-based irrigation remained marginal until the 2010 Megadrought prompted wider adoption. See footnote 50 for further discussion.

⁷Appendix Table A.I presents the irrigation calendar for Chile’s major crops, which are concentrated during the summer months. Appendix Figure A.I illustrates how monthly water distribution closely mirror this seasonal pattern in irrigation demand.

Chilean System of Private Property Rights. Since 1981, Chile has been the only country in the world where perpetual private property rights over water (water rights in what follows) have explicit constitutional protection against expropriation, which has resulted in limited administrative action by governments (Bauer, 2004; World Bank, 2011a, 2021; Tamayo and Carmona, 2019a). These rights are fully transferable, separated from land, and legally considered real estate, such that a water rights purchase is legally equivalent to a purchase of land (CNR, 2018a).⁸⁹ These rights are defined in terms of a stream of water (measured in liters per second) to be extracted from a specific location and source and following a monthly schedule; each of these attributes are defined during the creation of each water right. Appendix Figure A.II presents an example of a water right.

Water rights can be claimed for free through public requests to the Directorate of Water (DGA, a national public institution similar to the US Bureau of Reclamation), the regulatory agency in charge of assessing water resources and enforcing laws governing water issues. These rights can be generated until the DGA declares the river exhausted. After a source is declared exhausted, any user needing water rights in the area must purchase them from other users. Water rights can be freely traded among both individuals and firms, without any government interference (Biblioteca del Congreso Nacional, 1981).

The legal body that regulates water matters is the Water Code of 1981. Enforcement in principle relies on the DGA, which is supposed to address water stealing and over-extraction. However, Chile’s higher courts have overruled and systematically limited the scope of DGA’s enforcement capacities (Bauer, 2004). A second enforcement layer is that the infrastructure in place shall be built consistently with the water rights owned. The diameter of the pipe—checked by DGA agents at the time of reclamation—connecting the farm to the canal or well limits the total extraction capacity (CNR, 2018a). This coarse measure limits over-extraction in normal times by limiting the maximum water intake, but

⁸Until 2022, water rights carried no fixed term. A reform in 2022 (Law N° 21.435) converted all newly constituted rights into temporary concessions of up to 30 years, renewable based on objective criteria such as source availability. Rights constituted before the reform retain their indefinite duration.

⁹The titles also include the property over the infrastructure that allows the distribution of water, but there is legal recourse that allows to mandate one user to share the infrastructure with other users that own water rights (CNR, 2018a).

it does not during droughts: while the law establishes that users should limit their water extraction proportionally to the reduction in total streamflow (Biblioteca del Congreso Nacional, 1981), the infrastructure cannot adapt accordingly. The DGA guidelines establish that permanent water rights should have an exceedance probability of 85%, meaning that they can be created up to the point where the total volume claimed in a water source is available 85% of the time. This implies that water is insufficient to satisfy all rights—and so, that infrastructure should not bind—for almost two months per year on average.

Water Boards. Droughts reduce the total stream flow, and Chilean law establishes that these reductions should be prorated proportionally among all users, such that a reduction of 50% of the total river streamflow should imply a 50% reduction in maximum extractions by each user. Until recently, public agencies have not been able to intervene effectively in the allocation of water under scarcity, due to restrictions on administrative government action and lack of resources, leaving a void in the enforcement of drought-induced reductions (Bauer, 2004).

In response to droughts, early in the 20th century agricultural users created Water Boards as a representative body of water users (Peña, 2021). With the passing of the Water Code of 1981, Water Boards moved to an enforcement role: their statutory object is to administer and distribute the water to which their members hold rights,¹⁰ and they gained the legal authority to 1) determine and enforce water allocations across legal users under extraordinary circumstances, such as drought, 2) adjudicate disputes among users within their jurisdiction, 3) keep track of Water Rights claims, and 4) provide common goods such as legal assistance and common infrastructure, while defining its own funding sources.¹¹ Their design combines features of private associations and local regulators: they maintain registries of water rights, mediate disputes, and enforce allocations during scarcity by physically controlling water flows through canal gates (Bauer, 2004; Donoso, 2016; Engler et al., 2021; Blanco and Donoso, 2024). This arrangement reflects a governance model where the

¹⁰Article 266 of the Water Code (Biblioteca del Congreso Nacional, 1981)

¹¹Appendix Figures A.III, A.IV, and A.V illustrate core functions typically performed by Water Boards, including (1) enforcement of water rights through proration during scarcity, (2) organization of general assemblies, and (3) maintenance of shared irrigation infrastructure such as canal cleaning and repairs. The images were retrieved on July 29, 2025, from LinkedIn posts and official websites of Water Boards in Chile.

practical administration of water and the enforcement of rights are largely exercised by user organizations within the legal framework established by the State (Vergara Blanco, 2014).

This enforcement mechanism becomes critical during periods of water stress—typically summer months—when Chilean law mandates that water shortages be distributed proportionally among users according to their rights. Figure I illustrates this: in normal times (Panel A), fixed irrigation infrastructure provides enforcement. During droughts, without Water Boards (Panel B), upstream farmers can over-extract, leaving downstream users without their water allotment. With Water Boards (Panel C), instead, they enforce proportional allocation based on water rights, stopping upstream over-extraction and allowing water to reach downstream farmers.

Water Boards enforce water allocations during droughts by implementing a system of irrigation shifts, in which the Boards calculate the number of days or hours of unrestricted irrigation that correspond to each farmer, given each farmer’s water allotment and the total water available for distribution. The Boards enforce this delineated irrigation time by opening and locking canal gates, such as the one depicted in Figure IV. This technology is not unique to the Chilean context (e.g., the case of Murcia, see Ostrom, 1990, p. 77), nor costly: in 2005, membership fees represented less than 4% of the total average cost of production for farms growing the most common crops in the area of study (Jara et al., 2009).

The creation of Water Boards is triggered by either an agreement among water rights owners within the area under consideration, or by a lawsuit from at least one water user. Until 2005, the private-agreement route required unanimity among all users, so a single objection forced the process into court.¹² This process is under the jurisdiction of an ordinary judge housed in the most upstream province capital city within the basin in question.¹³ During this process, each community agrees on the final jurisdiction and statutes, which are subject to restrictions by the Water Code. The location and establishment date of these boards are presented in Figure V.

During their 2 or 3-year tenure, Water Boards report only to their constituents, who

¹²Articles 269 and 270 of the Water Code. The 2005 reform (Law 20.017) lowered the private-agreement threshold to an absolute majority; all boards in our sample were established under the earlier rule.

¹³Articles 269th and 270th of the Water Code of 1981 (Biblioteca del Congreso Nacional, 1981).

elect them with votes weighted by their Water Rights streamflow property. Water Boards are further subject to regulation by the DGA, but courts have curtailed the DGA’s ability to intervene (Bauer, 2004). Therefore, Water Boards are effectively the highest administrative authority in water-related issues in the basins under their jurisdiction, except for emergency situations.

Administrative and Legal Jurisdiction. Water Boards’ jurisdictions covers surface water bodies within their boundaries.¹⁴ Figure VI presents flowcharts of how Water Boards relate to Chilean administrative (Panel a) and legal institutions (Panel b) on water matters. Administrative measures, such as cutting allotments in the context of drought, are decided by each Water Board, for water rights within their jurisdiction.¹⁵ If any user wants to dispute a given decision, they can appeal to the DGA; however, in practice, DGA’s capacity is limited, and its decisions have been overruled by courts in several lawsuits (Bauer, 2004; World Bank, 2011b).

In the case of legal actions, any people and firms owning water rights should ask for a ruling from the Water Board that has jurisdiction over the water source in question.¹⁶ Part of the duties of a Water Board is to appoint a “Judge of Waters,” who is usually part of the board or an employee of the Board. This judge has full authority to solve legal disputes and to enforce their rulings with the authority of the Water Board. In the absence of a Water Board, instead, the only option available to users is to initiate legal action through ordinary courts (civil or penal courts, depending on the nature of the conflict). Water Boards substitute ordinary courts on water matters, with additional field expertise.

Appeals to Water Board rulings –or lawsuits against the Boards themselves– must be made to the Appeals Courts, and eventually can be escalated to the Supreme Court. Bauer

¹⁴In 2005, their jurisdiction expanded, to include groundwater (CNR, 2018a; Fernández and Gironás, 2021).

¹⁵For water rights registered in canals, Water Boards make decisions regarding allotments for the full canal, and the corresponding Canal Association will solve the matter within the canal. Users willing to dispute their Canal Association decisions may direct their complaints to the Water Board.

¹⁶If the users under conflict own water rights linked to a canal, their first step is to address their Canal Association, which manages water issues within a given canal. If the agents are unsatisfied with their ruling, they can appeal to their Water Board, or ordinary courts, if there is no Water Board with jurisdiction in the area. Water Boards also have jurisdiction over all conflicts that may arise among canals themselves, as long as they are within the Boards’ jurisdiction.

(2004) discusses how higher courts lacked water-specific knowledge and how their rulings have ignored substantive water issues, instead focusing exclusively on the legal issues at hand and emphasizing the “letter of the law.”

3 Data

We gathered a richness of information that reflects the *de jure* and *de facto* allocation of water across space and time, together with detailed agricultural information to measure outcomes and climatic controls. Our analysis has three stages: first, our misallocation test, implemented using farm level census data, combined with weather data at the county level; then a farm level analysis based on remote sensing and administrative data, and finally our basin level analysis.

3.1 Farm level, Misallocation Test

2007 Agricultural Census. The misallocation test uses data from the 2007 Agricultural Census, collected by the National Statistic Bureau (INE, the official statistical office of Chile). This Census includes operation-level information on land use and extension, crop choice, capital and employment decisions, managerial characteristics, and legal organization. Importantly, includes information on production for more than 20 annual crops, and self-reported information on the use of irrigation and the sources and legal status of irrigation water, together with affiliation to agricultural organizations (including specifically Canal Associations).

Climate Data. The Center for Climate and Resilience Research (CR²) created daily climatic estimates for the entire Chilean territory at a $0.05^\circ \times 0.05^\circ$ resolution, by calibrating satellite measures with local input from climatic monitoring stations (Alvarez-Garreton et al., 2018). These estimates include precipitation, potential evapotranspiration and minimum and maximum temperatures. We aggregate these climatic estimates at the plot, county or the drainage basin level, according to the analysis on which the data is being used.

3.2 Farm Level, DID and IV Analysis

Water Organizations. The information on the jurisdictions and establishment date of Water Boards was provided by the DGA. This institution also provided the maps of the jurisdictions of each board, and also information on the location and jurisdiction of Canal Associations.

Land Plot Limits and Canal Locations. SII (the Chilean Tax Authority) maintains for tax purposes a Land Cadaster, with detailed information on each plot of land in the country. CIREN geocoded the Land Cadaster for 2013. CIREN also has information on soil quality, measured as their suitability for irrigation. Our sample corresponds to land plots located less than 1km away from a canal. We obtained the canal locations and data from the DGA and CIEDESS, a local research center focused on natural resources.

Satellite Information on Evapotranspiration and Greenness. We use EVI and Evapotranspiration estimates by the USGS based on LANDSAT 5 images. USGS estimates and maps Evapotranspiration using the SSEBop model (Senay, 2018; Senay et al, 2023). This method recovers Evapotranspiration from an Energy Balance condition that equates the measured sun radiation on the surface to the calculated surface reflectance, estimated soil heat absorption, and Evapotranspiration (which is recovered as a residual)(Allen et al., 2015). We use images captured since the year 1986 using as input Landsat-5 images, with a resolution of $30m \times 30m$, a resolution fine enough to allow us to perform plot-level analysis. We also use EVI estimates based on Landsat 5 images from the same source, also with a resolution of $30m \times 30m$.

Fruit Cadaster. CIREN, jointly with ODEPA—a technical institution dependent on the Ministry of Agriculture — performs each year a Fruit Cadaster on at least one Chilean region, with the objective of updating information on the production of fruit trees. On each round, they collect information on cultivated surface by species and irrigation technology

for each plot of land in the region. We use all geocoded rounds with fieldwork between 2005 and 2013.

3.3 Basin Level Analysis

Basins, Streamflows and Climate. The DGA maintains a network of 803 monitoring stations in rivers and canals across the country since 1913. Our main sample is composed by 306 of these stations that have been created before 1980 in the Study Area. CR² has identified the drainage areas of each monitoring station and their characteristics. These characteristics include the cultivated surface within the drainage area, and also the amount of water rights created in the area each year.

Figure VII summarizes the correspondence between data sources, stage of analysis, and units of analysis employed throughout the paper.

4 Misallocation Test

In this section, we provide evidence that water misallocation occurs only in areas without Water Boards. We propose a misallocation test based on the idea that if irrigation water can be reallocated within a basin through a frictionless market, the marginal product of water (MPW) should be equalized within the basin.

The argument proceeds as follows. First, consider the problem of a farmer choosing the amount of water rights to acquire at the beginning of the season; this corresponds to the maximum amount of irrigation the farmer could use during the irrigation season. Rainfall substitutes for irrigation water at a fixed rate, but follows a known random distribution (Rafey, 2023). The First Order Condition of this problem is that the farmer acquires water rights such that the expected marginal product of water is equal to the expected shadow value of water in the irrigation season. Second, the effect of an unexpected rainfall shock during the irrigation season is equal to the marginal product of water (times the marginal rate of technical substitution between rainfall and irrigation water), as a consequence of the Envelope Theorem, combined with the presence of fixed inputs (Hsiang, 2016; Deryugina

and Hsiang, 2017). Finally, a benevolent Social Planner maximizing the total value of the production by society will equate the expected shadow values of water across users¹⁷.

To test empirically the null hypothesis of equal average marginal product of water across locations, we identify unexpected rainfall shocks at different positions within the basin. By measuring how these shocks affect profits, we can calculate the semielasticity of profits to rainfall, which equals the marginal value of water. Our results show that in areas without Water Boards, the marginal value of water grows from upstream to downstream locations, showing a pattern of misallocation consistent with a natural advantage of upstream users to over-extract. In contrast, areas governed by Water Boards show no significant differences in marginal returns across locations.

4.1 Model of Agricultural Production and Irrigation under Water Rights

Environment. Consider a Social Planner’s problem of allocating water rights to N farmers, who can freely choose inputs for agricultural production of annual crops. Agricultural production follows a cycle over the year, with 3 seasons: a Planting season $s = 0$, a Growing season ($s = 1$) and Harvest time ($s = 2$). Water supply has different impacts depending on this stage; in what follows, we assume that irrigation is only useful in $s = 1$.

At stage $s = 0$ each farmer $i \in N$ chooses crop c , capital K_i and land S_i , which are fixed over the full production cycle. At stage $s = 1$, the farmer chooses the flexible inputs, namely labor L_i and effective irrigation w_i . Effective irrigation is capped by the amount of water rights allocated to the farmer $\bar{w}_i$. Rainfall r is a perfect substitute for irrigation water, up to a technical rate of substitution constant θ . Rainfall is a random variable with a distribution known by all agents. We assume that input and output prices are known in advance, and all markets are competitive.

There is a Social Planner who allocates Water Rights to each user; each user will

¹⁷We focus on the allocation of water rights instead of effective water because our setting has limited short-term (i.e. intra-seasonal) trading due to limited storage capacity in most basins (Bauer, 2004; Hadjigeorgalis and Lillywhite, 2004). Hence, most trading decisions happen when there is still uncertainty about how much water will be effectively available. Nevertheless, this analysis is equivalent to a first-order approximation to a similar analysis for the allocation of water itself.

extract after rainfall uncertainty is realized. The timeline of decisions is therefore:

Time= 0 : Social planner allocates water rights. Farmers choose crops, capital and land.

Time= 1 : Farmers hire labor and apply irrigation water subject to their Water Rights caps

Time= 2 : Profits are realized

Finally, each production function F_c is increasing, continuous, strictly concave, and monotone¹⁸.

Farmers' Problem. We solve by backward induction: the problem of user i at stage $s = 1$ is to choose the optimal irrigation and labor quantities to maximize profits:

$$\max_{L_i, w_i} \quad p_c F_i^c(S_i, K_i, L_i, w_i + \theta r_i) - \lambda_i^s (w_i - \bar{w}_i) - c_L L_i$$

The First Order Conditions of this problem are

$$FOC(w_i) : p_c F_{i w}^{c'} = \lambda_i^w$$

$$FOC(L_i) : p_c F_{i L}^{c'} = c_L$$

Under the assumptions above, each farmer will just use the total amount of water rights allocated to them. The shadow value of water will be equal to the marginal product of irrigation water.

¹⁸While the first two properties are assumed to keep the analysis simple (i.e. to guarantee that the demands for all factors are functions and not correspondences) and it is possible to replace them without loss of generality, the last assumption may be more controversial, as it rules out scenarios where excessive rainfall adversely affects production. While such a scenario is certainly possible, in the area under study—with mostly dry Mediterranean weather with a well-marked rainfall season in the winter—is rare, and it did not take place in the period under analysis (2006-2007 Austral agricultural year).

Consider the case where irrigation increases production until a total water input threshold, after which water damages production: any rainfall that falls below this threshold will just supplement the water input provided by the farmer up to this threshold, after which it will just crowd out the farmer's input (i.e. the farmer will reduce its water input up to keep the total water input below the threshold). In this scenario, the shadow value of water is zero, and so, both the problem and the shadow value preserve their meaning.

In stage $s = 0$ the problem of the farmer is to choose the optimal Capital, Land, and crop: The fixed inputs are chosen based on

$$\max_{K_i, S_i} \mathbb{E}_r \{ p_c F_i^c (S_i, K_i, L_i (K_i, S_i, \bar{w}_i, r_i), w_i (K_i, S_i, \bar{w}_i, r_i) + \theta r_i) - c_S S_i - c_K K_i - c_L L_i (K_i, S_i, \bar{w}_i, r_i) \mid I_0 \}$$

The First Order Conditions of this problem are

$$\begin{aligned} FOC (S_i) &: \mathbb{E}_r \{ p_c F_i^{c'} S \mid I_0 \} = c_S \\ FOC (K_i) &: \mathbb{E}_r \{ p_c F_i^{c'} K \mid I_0 \} = c_K \end{aligned}$$

where the Envelope Theorem rules out any indirect effects on any flexible inputs. Given the choices for each input, we can define the expected profits for farmer i conditional on choosing crop c :

$$\begin{aligned} \pi_i^c(\bar{w}_i) \equiv & \mathbb{E}_r \{ p_c F_i^c (S_i(\bar{w}_i), K_i(\bar{w}_i), L_i(\bar{w}_i), w_i(\bar{w}_i) + \theta r_i) \\ & - c_S S_i(\bar{w}_i) - c_K K_i(\bar{w}_i) - c_L L_i(\bar{w}_i) - \lambda_i^w (w_i(\bar{w}_i) - \bar{w}_i) \mid I_0 \} \end{aligned} \quad (1)$$

The farmers, therefore, will choose the crop with maximum expected profits. The farmer's expected profits are

$$\bar{\pi}_i(\bar{w}_i) \equiv \max_k \{ \pi_i^k(\bar{w}_i) \} \quad (2)$$

Social Planners' Problem. Let's define the Social Welfare Function as the sum of the expected production of all farmers within the basin:

$$\Omega(\bar{\mathbf{w}}) \equiv \sum_i \bar{\pi}_i(\bar{w}_i)$$

The problem of the social planner is to allocate water rights across users to maximize the expected total production value, subject to the total availability of water:

$$\max_{\{\bar{w}_i\}_{i=1}^N} \mathcal{L}(\mathbf{w}) = \sum_i \bar{\pi}_i(\bar{w}_i) - \lambda^W \left(\sum_i \bar{w}_i - \bar{W} \right) \quad (3)$$

Note that the social planner's objective function is just the sum of value functions of all farmers; therefore, as a consequence of the Maximum Theorem, the social planner's objective function is continuous on each water right $\bar{w}_i$. The first order condition with respect to $\bar{w}_i$ implies¹⁹

$$\frac{\partial \bar{\pi}_i}{\partial \bar{w}_i} = \lambda^W \quad (4)$$

Therefore, the optimal allocation satisfies $\frac{\partial \bar{\pi}_i}{\partial \bar{w}_i} = \frac{\partial \bar{\pi}_j}{\partial \bar{w}_j}$, $\forall i, j \in N$.²⁰ Note that:

$$\frac{\partial \bar{\pi}_i}{\partial \bar{w}_i} = \frac{\partial \mathbb{E}_r \{p_K F_i^K | I_0\}}{\partial \bar{w}_i} = \mathbb{E} \{\lambda_i^w | I_0\}$$

where K represents the crop chosen by the farmer, and the second equality is a consequence of the Envelope Theorem.²¹ In the socially optimal allocation, therefore, the expected shadow value of water is equal across farmers; any deviation from that implies the opportunity to increase expected welfare by reallocating water rights from users with a low shadow value of water to users with a high shadow value.

The Effect of a Rainfall Shock and the Marginal Product of Water. Consider the effect on social welfare of an unanticipated rainfall shock over farm i at $s = 1$. The Envelope Theorem implies that any induced adjustments in flexible inputs have a net zero effect on realized profit. As the water rights allocation is fixed and the marginal cost of surface irrigation is zero, the marginal effect of the shock is given by the partial derivative of realized production:

$$\frac{\partial p_K F_i^K}{\partial r_i} = \theta \lambda_i^w$$

¹⁹This equality holds at all generic allocations where the farmer's optimal crop choice is strictly preferred and the value function is locally smooth. Because exact crop-switching thresholds have measure zero, the function is differentiable almost everywhere.

²⁰This is not true in the presence of fixed costs or other causes of corner solutions. In such cases, the previous statement is true for every pair of farms i, j that are optimally receiving positive water allocations (e.g., their fixed cost is sunk).

²¹Specifically, the Envelope Theorem implies that any induced adjustments in flexible inputs have a net zero effect on profit. Additionally, in this case, the marginal cost of irrigation water is zero, leading to the first equality. Therefore, the application of the theorem is direct in this case; a more general discussion can be found in Hsiang (2016); Deryugina and Hsiang (2017).

and so taking expectations yields:

$$\mathbb{E} \left[\frac{\partial p_K F_i^K}{\partial r_i} \mid I_0 \right] = \theta \mathbb{E} [\lambda_i^w \mid I_0] = \theta \frac{\partial \Omega}{\partial \bar{w}_i}$$

So the expected total effect of an unexpected rainfall shock on welfare is equal to the expected shadow value of water of the affected farmer, times the marginal rate of technical substitution between irrigation water and rainfall. Note that Rafey (2023) estimate θ to be equal to 1.048 for annual irrigated crops, which is approximately equal to 1.²²

Two key conclusions emerge from the previous discussion. First, the optimal allocation of water rights equalizes the expected marginal product of irrigation water across users. Second, the expected effect of an unexpected rainfall shock is equal to the expected marginal product of irrigation water.

4.2 Test Implementation

We can test the null hypothesis of equal average marginal product of water across locations by identifying unexpected rainfall shocks by position within the basin, for treated and control areas, and then to measure their impact over profits; the semielasticity of profits to these rainfall shocks will equate the marginal value of water (Deryugina and Hsiang, 2017). Our Agricultural Census data do not measure the effective water input for each parcel. However, rainfall is a perfect substitute for irrigation water, up to an absorption rate estimated to be approximately 1 for annual crops (Rafey, 2023). We therefore exploit the timing of rainfall to get within county variation in water input received during the irrigation season—which we call “useful rainfall”—at the parcel level across the production cycle.

We illustrate the construction of this variable in Figure VIII. We define useful rainfall as the total rainfall falling in a farmer’s county during the months when their crop requires irrigation in the current agricultural year. In this example, Farms A and B are both located in County 1, with rainfall over the agricultural year represented by the blue bars. Farm A

²²The estimates of θ for other crop choices are 1.081 for perennial crops, for annual non-irrigated crops is 0.591 and for dairy is 0.148.

cultivates a crop that follows the yellow irrigation schedule, while Farm B cultivates one that follows the green schedule. Consequently, all rainfall received in December, January, and February is useful for Farm A but not for Farm B. Another source of variation comes from geography. Consider a farm with the same irrigation schedule as Farm A, but located in a nearby county where rainfall is represented by the red bars. This farm experiences a slightly different water input than Farm A, due to the distinct monthly rainfall pattern in its county. This variation, conditional on fine spatial fixed effects and agricultural choices, allows us to test for differences in the average shadow value of water among farms that rely on canal-based irrigation and those with water rights, even when located within the same basin.

We implement our misallocation test using the 2007 Chilean Agricultural Census, which contains a rich set of input and technology choices (including irrigation technology, planted surface, temporary and total workers, machinery use, and property of water rights), which we combined with soil quality estimates and daily climate data at the county level, including precipitation and temperature by calendar day. The sample for the estimation includes farms with irrigation from canals, owning or renting water rights, and with a cultivated area below 50 hectares, located in counties whose centroids belong to one of the 12 basins under study.²³

We estimate

$$\begin{aligned}
\log \left(\frac{Y}{\text{Hectares}} \right)_{irc}^{2007} &= \beta_1 \text{Board}_c + \beta_2 \text{Useful Rain}_{rc} + \beta_3 \text{Distance to Sea}_c \\
&+ \beta_4 \text{Board}_c \times \text{Useful Rain}_{rc} + \beta_5 \text{Board}_c \times \text{Distance to Sea}_c \\
&+ \beta_6 \text{Useful Rain}_{rc} \times \text{Distance to Sea}_c \\
&+ \beta_7 \text{Board}_c \times \text{Useful Rain}_{rc} \times \text{Distance to Sea}_c \\
&+ \beta_X \mathbb{X}_i^{2007} + \mu_c + \mu_r + \varepsilon_{irc}
\end{aligned} \tag{5}$$

where $\log(Y/\text{Hectares})$ is the logarithm of the value of output per hectare obtained by farm i on planting crop r in county c , Useful rainfall $_{rc}$ is the rainfall received during the irriga-

²³We eliminate farms above 50 hectares to eliminate outliers; including farms above this threshold does not change the results qualitatively, but the standard errors are higher.

tion season of crop r in county c , in megaliters per hectare per months ($ML/ha/month$), $Distance\ to\ Coast_c$ is the distance to the river mouth from the centroid of each county c through the river, in kilometers.²⁴ X_i^{2007} is the set of controls, which includes the logarithm of the total labor hired during the 2007 agricultural year, a vector of capital and technology choices and the irrigated surface. $Board_c$ equals 1 if the farms belong to a county whose centroid falls inside the jurisdiction of a Water Board. Finally, μ_c is a county fixed effect, and μ_r is a crop fixed effect.

On estimating equation 5, we are exploiting within-county, within-crop variation in the timing of rainfall, which is arguably exogenous on most determinants of agricultural production.²⁵ County fixed effects will capture common shocks to all farms and average expectations, and individual farm controls—including crop fixed effects—will capture long term and short term determinants of output. One threat to our identification strategy is the potential presence of imbalances: farmers may try to match the pattern of rainfall to optimize their water input (Kala, 2019); if systematic differences in productivity and accuracy of rainfall expectations correlate with our measure of useful rainfall, then it is possible to have biased estimates.

In Table I we present a balance table for useful rainfall, after including our main controls: basin fixed effects, 0.5×0.5 -degree-cell fixed effects, and crop fixed effects. There are no systematic differences for our sample. For our main sample (irrigated fields) we only observe a decrease in rainfall as we move from the coast to upstream locations, but there is no significant differential rainfall pattern between farms located in counties with and without Water Boards. In the case of our placebo sample, there are some differences in useful rainfall that disappear once we include finer spatial controls (i.e. 0.5×0.5 -degree-cell fixed effects). In any case, any such difference would be absorbed by the county fixed effects included in our preferred specification.

Equation 5 allows us to estimate directly the functions needed for our Misallocation Test. First, we estimate the Average Shadow Value of Water as a function for the distance

²⁴This distance corresponds to the length of the least cost path connecting each county centroid to the river mouth, through the river network representing the hydrology of each basin. For details, see Annex C.

²⁵In other words, useful rainfall is as good as random, conditional on locational, crop, input, and technology choices.

to the coast, in the absence of Water Boards:

$$\frac{\partial \mathbb{E} \{ \pi_i | I_0, \text{Distance to Sea, No Board} \}}{\partial w_i} = \beta_2 + \beta_6 \times \text{Distance to Sea} \quad (6)$$

Second, we estimate the Average Shadow Value of Water as a function for the distance to the coast, under Water Boards:

$$\frac{\partial \mathbb{E} \{ \pi_i | I_0, \text{Distance to Sea, Board} \}}{\partial w_i} = (\beta_2 + \beta_4) + (\beta_6 + \beta_7) \times \text{Distance to Sea} \quad (7)$$

We test for misallocation by comparing the shadow value of water as a function of each farm's position in the river. We investigate how the shadow value of water varies with distance from the river mouth, specifically examining whether the marginal value of water differs between the top of the river (its head) and the location where it drains to the sea.

Results. In Table II we present the results of estimating equation 5, considering an array of location-fixed effects. Our preferred specification is in column 4, which we also present graphically in Figure IX. The estimated coefficient associated with rainfall (β_2) implies that for farms in our sample—water right owners, affiliated to canal associations and with irrigation—an extra unit of water ($ML/ha/month$) would increase yield by 42% if they are at the river mouth. When taking into account the coefficient associated with the interaction of useful rainfall and distance to the mouth, we see that for farms located approximately 200km upstream, the effect becomes a non-significant reduction of 12%. The estimated average MPW is presented as the red function in Figure IX and clearly displays a higher average shadow value of water in locations downstream versus upstream.

This result is consistent with misallocation: farms located downstream are water-constrained, while farms located upstream are not; a marginal displacement of water through the river from upstream locations to downstream locations would increase the total value of production, but the lack of enforcement prevents such reallocation. At the bottom of Table II we present the p-values of the test of equality of MPW across locations; for all specifica-

tions, we reject the null hypothesis of no misallocation at the 10% confidence threshold.^{26,27}

In contrast, for counties within the jurisdiction of Water Boards, the average shadow value of water is similar, regardless of their distance to the river mouth. For counties located next to the coast, there is a non-significant reduction in value per hectare of around 5% per extra unit of water ($ML/ha/month$), which is approximately the same for farms located 200km upstream. The average shadow value of water as a function of the distance to the coast—presented as the blue function in Figure IX—is flat compared to the function for places with boards, and never significantly different from zero. These results do not allow us to reject the null hypothesis of no misallocation in counties under the jurisdiction of a Water Board, as reflected in the last row of Table II. The results are similar controlling for basin, 0.5-degree cells, and county fixed effects.

Robustness Checks: Placebo Exercise with Rainfed Farms. To address concerns regarding potential confounders that may drive the former cross-sectional results, we present a placebo exercise in which we estimate equation 5, including the same set of controls, but for rainfed farms in the same counties as our main sample. These agricultural operations display different technology choices but are exposed to the same geographies and climates. These parcels rely exclusively on natural precipitation, so there is no additional water input on top of rainfall; Water Boards have no mechanism to affect their input. Therefore, we expect to find no effect of Water Boards on these farms.²⁸

Table III presents the results of this exercise. Our preferred specification is in column 4, where we again exploit within-county, within-crop variation in useful rainfall across farms. The results suggest that the yield per hectare increases by around 8% per extra unit of water ($ML/ha/month$) at the coast. The increase in yield for farms located the farthest from the coast (at 219km) is 11%, which is not statistically different from the effect on the coast. For counties with Water Boards, the corresponding estimates are 12% and 6%; they

²⁶This test was performed considering the standard errors clustered by county; the results do not change by considering the SE clustered by county and irrigation season crop type.

²⁷Appendix Table A.II reports estimates of equation 5 by Poisson pseudo-maximum likelihood, with the value of output per hectare entering in levels rather than in logs—a specification closer to the formulation of the model. The results are the same as in the main specification.

²⁸Also, our estimates will correspond to the marginal product of water for these farms. Manysheva (2022) and Rafey (2023) use this strategy to estimate the production functions of rainfed farms

are not statistically different either. Importantly, all interaction terms involving the Water Board dummy are not significant and economically small. The last rows of Table III show that after including our finer spatial fixed effects –either 0.5-degree cells or county fixed effects–, we cannot reject that the MPW is equal for all locations, either with or without Water Boards, as expected.

Figure X presents the estimated functions of Average MPW for areas with and without Water Boards. The most important conclusion from the figure is that none of these functions exhibit a significant slope, and despite the existence of some (statistically non-significant) differences in levels, both functions are contained in the confidence intervals of the other.

This placebo exercise suggests that the estimated effects of rainfall on irrigated parcels (our main exercise) recover a structural relationship between water and yield across different locations, and so our Misallocation Test identifies the underlying misallocation existing in the absence of water rights enforcement. More importantly, our test fails to find such misallocation in places with Water Boards exerting property rights enforcement, suggesting that we cannot reject the efficiency of water markets supported by Water Boards.

Additional Robustness Checks. We present additional robustness checks addressing different potential concerns in Appendix Table A.III. In column 1 we include our preferred estimate for reference. The first concern is that the relationship between useful rainfall and production might be confounded by spatial sorting and crop choice; therefore, it is ideal to preserve the within crop comparisons as local as possible. In column 2, we present the results of estimating equation 5 but including Crop $\times$ 0.5° $\times$ 0.5° bin fixed-effects—effectively, keeping comparisons within the farms growing the same crops within the same spatial bin. The results remain unchanged—if anything, there is a reduction in standard errors.

A second concern is that crops with different water requirements may exhibit different elasticities with respect to irrigation; if so, the misallocation test results could reflect crop choices rather than misallocation. In column 3 we re-estimate equation 5, now allowing for interactions between crop fixed effects and useful rainfall. This effectively permits elasti-

ties to useful rainfall to vary by crop. The results are mostly unchanged.²⁹

Another potential concern is that the relationship between revenue and useful rainfall may be confounded by other inputs, technology choices, or determinants of productivity. In columns 4–6 of Table A.III, we re-estimate our main equation, interacting useful rainfall, and useful rainfall $\times$ distance to the sea, with three potential confounders: in column 4, with a dummy equal to one if a farm reports the use of groundwater; in column 5, with a dummy equal to one if a farm uses fertigation³⁰, a technology complementary to irrigation; and in column 6, with a measure of potential productivity of the land.³¹ Reassuringly, the results for all three exercises are unchanged relative to the main estimates.

We assess sensitivity related to the linear functional form in equation 5 by replacing the continuous distance variable with quintile indicators (Appendix Figure A.VI). The marginal product of water is substantially higher for farms without Water Boards located closest to the coast (quintile 1), and converges across quintiles under Water Board jurisdiction. This is consistent with the interpretation that the semi-elasticity to useful rainfall estimated above reflects the shadow value of water, and that downstream farmers are more exposed to water scarcity than farmers located upstream them.

Our test assumes that farmers cannot anticipate the level of useful rainfall of the coming irrigation season; if some could, the precision of their forecasts might correlate with productivity and crop choice, potentially confounding our estimates. In our last robustness check, we provide evidence in favor of this assumption—that useful rainfall is unpredictable for farmers—by exploring the relationship between useful rainfall and the timing of hiring. Temporary labor—the only flexible input we measure at the intensive margin—is uncorrelated with useful rainfall during the Fall and Winter, before the irrigation season begins, and responds to it only once the season is underway in Spring and Summer, when that uncertainty is resolved (Table A.IV, conditioning on farm size, county, and crop fixed effects).

²⁹The main coefficient of useful rainfall is missing because it is collinear with the crop-specific elasticities.

³⁰The application of fertilizer through the irrigation system

³¹The measure is the average potential revenue value per hectare, according to FAO/GAEZ, for the 20 most common crops in Chile available in this source.

5 Water Reallocation and Complementary Private Investments

In the previous section, we documented intra-basin water misallocation between farms except where Water Boards operate. Here, we show that Water Boards correct this misallocation and raise agricultural productivity through two families of mechanisms: the reallocation of water toward downstream users, and the private adaptations it enables. We establish water reallocation in the short run; over the long run, our central finding is that downstream farmers undertake complementary private investments—changes in crop choice and irrigation technology.

Our analysis focuses on the dry season (December-February), when water scarcity is most acute and enforcement of property rights becomes critical. During these months, precipitation is minimal while irrigation demand for high-value crops remains high, creating a seasonal mismatch that generates the strongest incentives for upstream over-extraction. This makes Water Board enforcement most consequential for agricultural outcomes.

Our main analysis relies on a novel database, containing information from more than 75,000 land parcels located less than 1km away from a canal in the whole area of study (i.e. regions *IV* to *IX*, in Central Chile). We follow these farms between 1986 and 2012. We estimate plot-level water consumption using more than 15,000 images of a new LANDSAT 5-based product that provides estimates of Evapotranspiration at a 30m resolution every 16 days since 1986.³² In addition, we also estimate agricultural yield and crop choice using Enhanced Vegetation Index (EVI) estimates from LANDSAT 5 from the same period. To illustrate the detailed nature of this data, Figure [XI](#) present our estimates of water consumption and agricultural yield (proxied by actual evapotranspiration and EVI, respectively) for all farms in the Aconcagua Basin. In Panel (a) we observe a decline in water consumption when comparing upstream (right) to downstream (left) locations. Panel (b) presents a similar decline in yield. However, there is substantial intra-location variation, especially in upstream locations.

³²Although LANDSAT 5 was launched in 1984, it actually started operating in 1985. Starting in 1986 there is stable supply of frequent images of our area of study (Senay, 2018; Senay et al., 2023).

The previous gradient of water usage and position in the basin is not a unique feature of the Aconcagua Basin: Figure XII presents a scatter plot of the average evapotranspiration for third-level basins within basins that eventually adopted boards, over the 5 years before (in red) and between 5 and 10 years after the establishment event (in blue). While the previous gradient is noticeable in both periods, the slope flattens after board establishment, as is visible from the fitted lines. More formally, the local polynomial regression of the difference (after minus before) in average evapotranspiration versus distance to the coast (in gray) shows reductions in water usage upstream and increases downstream. We will formally estimate a difference-in-differences model in the next subsection.

In Table IV we present summary statistics over our full sample. Farms within the jurisdiction of Water Boards have slightly lower EVI and evapotranspiration levels than farms outside them, but they also face substantially dryer conditions.

5.1 Reallocation of Water and Agricultural Yield

In our first analysis, we exploit the staggered adoption of Water Boards across basins to estimate the causal impact of property rights enforcement and governance on the allocation of water. Table V presents the year of establishment of Water Boards, and the number of farm plots under their jurisdiction. Given the data available and the institutional design in place, we focus our analysis on the boards established after 1981 and before 2010. This window captures Water Board establishments under Chile’s 1981 Water Code framework and before the severe hydrological changes experienced from 2010 onwards, ensuring our estimates reflect a consistent institutional and environmental context. We consider 5 years before and 7 years after the establishment of the board for our analysis.

Identification Strategy. The establishment of Water Boards is triggered by either an agreement among water users or a lawsuit by users who perceive their interests are being affected by the lack of governance in the basin. In both cases, there is latent competition over the resource, driven by both water supply and demand factors.

We explore the potential role of water supply factors in Figure XIII, where we show the standardized total precipitation over our sample of farms for each year around a Board

Establishment event. This figure shows that, indeed, the years surrounding a Board Establishment event are drier: during the year of a Board Establishment, the farms under their jurisdiction experience almost 10% of a standard deviation lower precipitation. Therefore, although the timing of Board Establishment is linked to environmental conditions, these conditions seem to be particularly adverse for agricultural activity, so that our estimates of the benefits from Water Boards are, if anything, a lower bound of their true values.

Exploring demand side-drivers of Water Boards adoption is more complex, as they are mostly unobservable. The process, however, is long and irreversible, implying that the most likely driver of Water Boards adoption is the long-run congestion of the basin, as a growing number of users compete for a finite resource.³³ Motivated by this, we will focus our analysis on a sample of ever-treated units, and therefore, for any Board Establishment event, we will consider as a control group only not-yet-treated plots.

Econometric Specification of the DID Analysis. Our main equation is:

$$Y_{it} = \sum_{k=-5}^7 \delta_k \text{Board Establishment}_{it} \times 1[t - t^* = k] + \alpha_i + \mu_{t,g(i)} + \varepsilon_{it} \quad (8)$$

where Y_{it} is either Average Evapotranspiration in the Summer or the Peak Enhanced Vegetation Index (EVI) over the season at plot i in year t . These are satellite-derived variables averaged at the parcel level that proxy for water consumption and agricultural yield, respectively³⁴. In our preferred implementation, we use standardized versions of these variables³⁵ to address the substantial heterogeneity across units in space and time. Board_{it} equals 1 if plot i is under any Water Board jurisdiction. t^* denotes the establishment year of the Water Board with jurisdiction over i . α_i is a plot-level fixed effect, while $\mu_{t,g(i)}$ denotes flexible

³³There is no systematic documentation of the length of the full constitution process, but one testimony about Water Board established through a lawsuit in 2015 describes a process of at least two years: the first year for lawsuit resolution and the second year for executing the resolution and establishing the board (CNR, 2018b). Similar legal processes in the 1980s and 90s were most likely longer, due to higher communication and transportation costs.

³⁴Evapotranspiration is measured in average millimeters per pixel during summer season (December-February). Peak EVI corresponds to the maximum value of Enhanced Vegetation Index over the growing season. Both variables are derived from LANDSAT 5 satellite imagery.

³⁵Obtained by subtracting each plot's mean from its observed value and dividing by its standard deviation.

time trends by geographical grid cell, where $g(i)$ represents the $2^\circ \times 2^\circ$ latitude-longitude cell. We estimate equation 8 separately for each tercile of distance to the river mouth for plot i , defining downstream, mid-section, and upstream farm locations.

Our main hypothesis is that Water Boards reallocate water within the basin, which in our case translates into possible heterogeneous treatment effects across upstream and downstream locations. We therefore use the staggered difference-in-differences imputation estimator proposed by Borusyak et al. (2024). This approach handles within-cohort heterogeneous effects in a context of staggered adoption, and provides a standard errors estimator that allows for correct coverage under clustering and small sample size.³⁶

Difference-in-Differences Results. Figure XIV presents our main set of results for the effect of Water Board establishment on water consumption. It shows heterogeneous effect estimates by location within the basin. Downstream farms experience an immediate 1-standard-deviation increase in water consumption, growing to 2 standard deviations after 7 years. Upstream plots show a reduction in water consumption that reaches 0.8 standard deviations after 7 years.³⁷ There is no evidence of significant pre-trends for either group. Appendix Figure A.VIII shows similar patterns using non-standardized evapotranspiration: downstream farms increase water consumption by 0.7mm initially, reaching 1.5mm after 7 years, while upstream farms experience reductions of 0.3mm by year 7.³⁸

In Figure XV we present our estimates of the effect of Water Boards establishment on agricultural yield. It presents heterogeneous effects estimates by location within basin. Downstream farms increase their yield by 0.4 standard deviations in the first year, and the gains increase to more than 0.6 standard deviations after 7 years. Upstream plots show lower average yields in the years after the event, with reductions smaller in magnitude (averaging 0.25 standard deviations), not significant for all years, and seeming to dissipate

³⁶Similar to our setting, Cisse et al. (2026) apply the same estimator to evaluate irrigation projects in the Senegal River Valley, where treatment timing is staggered and the panel is unbalanced due to missing satellite imagery, a challenge also present in our context.

³⁷We present results including midsection farms in Appendix Figure A.VII. There is no discernible effect over them.

³⁸While boards seem to form during droughts (Figure XIII), mean reversion—in the form of a natural recovery in precipitation—would increase overall water availability and therefore cannot explain the divergence in water consumption, particularly the reductions observed upstream.

at year 7.³⁹ We observe some weak negative pre-trends among downstream farms, while no pre-trends for upstream farmers. Appendix Figure A.X presents estimates using non-standardized EVI, showing downstream farms increase peak EVI by 0.04 immediately and 0.06 after 7 years, while upstream farms experience smaller reductions.

Appendix Figure A.XI shows that these results are materially unchanged when a full set of climatic controls is included⁴⁰: downstream farms sustain sizable gains in yield and water consumption after 7 years, while upstream farms experience persistent reductions in water consumption and yield losses that dissipate by year 7.

Overall, the previous results suggest that the introduction of Water Boards has heterogeneous effects: while there are significant gains among downstream farmers in water usage—that translate into increases in yield—upstream farmers reduce their water consumption without a substantial reduction in yields.

5.2 River Streamflows

The previous subsection documented increases in water consumption among downstream farms and reductions upstream following Water Board establishment. We now examine evidence of the physical footprint of the underlying reallocation: changes in river streamflows. Using a difference-in-differences design that leverages the staggered adoption of Water Boards across basins, we analyze their impact on streamflows during the dry season (December-February)—when water scarcity is most acute and incentives to over-extract are strongest.

Our empirical strategy follows a staggered difference-in-differences approach as in the previous subsection, but now applied at the river segment-month level rather than farm-year level. Our unit of analysis is the river segment, operationalized through monitoring stations that measure streamflows. We use monthly streamflow data from the DGA’s network of monitoring stations. From the DGA’s network of 803 stations nationwide, our sample consists of 306 stations established before 1980 in the study area, allowing us to analyze

³⁹We present results including midsection farms in Appendix Figure A.IX. There is no discernible effect over them.

⁴⁰A vector of annual climatic characteristics: total precipitation over the year and over the Summer, and the number of days with maximum temperatures in the following temperature ranges: $[25, 29]$, $[29, 31]$, $[31, 35]$, and $[35, \infty^+)$, in Celsius degrees.

the impacts of Water Board establishments that occurred after 1981. Table VI presents the year of establishment of Water Boards for the monitoring stations within our study area.

Difference-in-Differences design for River Streamflows. Our main equation is:

$$\begin{aligned} \text{Stream}_{gmt} = & \sum_{i=-3}^3 \delta_i \text{Board Establishment}_{gst} \times 1[t - t^* = i] \\ & + \beta X_{gmt} + \gamma \text{WR}_{gmt} + \mu_t + \eta_{gm} + \varepsilon_{gmt} \end{aligned} \quad (9)$$

where Stream_{gmt} is streamflow at monitoring station g in month m and year t , Board_{gt} equals 1 if segment g is under Water Board jurisdiction in year t , and t^* denotes the Water Board establishment year. X_{gmt} includes monthly climatic controls (rainfall, temperature, and potential evapotranspiration), and WR_{gmt} is total water rights claimed in the drainage area. μ_t and η_{gm} are year and station-month fixed effects.⁴¹

Identification. Rivers that eventually adopt Water Boards differ systematically from those that do not. Figure XVI shows that Water Boards emerge in rivers with lower annual streamflow but relatively high summer streamflow—characteristic of nivo-glacial regimes where snowmelt provides water during the irrigation season. Given these systematic differences, we focus on an event study sample of monitoring stations in basins that eventually adopt Water Boards, relying on the timing of adoption for identification. We report full sample results as a robustness check.

Our identification assumes timing of adoption is as good as random conditional on fixed effects and covariates. The long and irreversible establishment process—requiring lawsuits or agreements—implies that adoption is driven by long-term water availability rather than short-term shocks. We rule out two additional concerns. First, we discard that mean reversion in precipitation explains positive impacts of boards on streamflows: if Water Boards were established during unusually dry periods, we would expect precipitation to increase afterward, but Figure XVII shows precipitation is actually lower after Board establishment. Second, regarding potential endogeneity of water rights creation, our estimates are robust

⁴¹Equation 9 is derived from a water balance equation, where inflows equalize outflows in a basin.

to including or excluding controls for new water rights claimed in each basin.⁴²

We use the staggered difference-in-differences estimator of Borusyak et al. (2024), which handles heterogeneous effects in staggered adoption settings and provides correct standard errors under clustering.

Results. Water Boards significantly increase river streamflows during the dry season, consistent with water reallocation from upstream to downstream users. Figure XVIII presents our main estimates. For our event study sample (Panel a), streamflows increase by approximately 25% in years 2-3 post-establishment, averaging 13% across all post-establishment years (0-3). Full sample estimates (Panel c) yield similar results: 27% in years 2-3, and 16% on average. Results are robust to including water rights as a control (Panel b and d). We find no evidence of pre-trends in either sample. These summer-specific effects contrast sharply with estimates for the entire year (Figure A.XII), where no coefficients are statistically significant.

These results reflect large-scale reallocation of water. Assuming half of summer streamflow is used for irrigation, the 25% increase in years 2-3 implies a 50% increase in water availability for downstream farms.⁴³

To further test whether Water Boards prevent upstream over-extraction, we examine heterogeneous effects by location. If enforcement works through this mechanism, effects should be stronger at: 1) upstream monitoring stations (positioned between upstream diverters and downstream users), and 2) basins with higher agricultural intensity. Our estimates in Figure XIX are consistent with these predictions. Upstream stations (Panel a) show increases of approximately 28% in years 2-3, compared to 25% at downstream stations (Panel b). Stations in high-cropland areas (Panel c) show increases of over 40% in years 2-3, compared to 20% in low-cropland areas (Panel d). The strongest effects occur for upstream stations in high-cropland basins (Figure XX, Panel d), while downstream stations in low-cropland areas show no significant effects (Figure XX, Panel a).

⁴²Moreover, we find no evidence that Water Boards affect new water rights creation.

⁴³This back-of-the-envelope calculation aligns with our short-run farm-level DID estimates (Subsection 5.1), which show similar magnitudes. Our long-run IV estimates (Subsection 5.3) show similar effects.

5.3 Long-Run Private Investment in Fruit Trees

In this subsection, we explore whether reliable water access stimulates long-run investments, or if farmers privately adapt to compensate for the lack of governance. We focus on fruit-tree planting, a decision with high upfront costs and multi-year payoffs that responds to persistent rather than transitory conditions (e.g. Burlig et al., 2026). An orchard is a largely irreversible, water-dependent asset where a single season of scarcity can damage the trees or the harvest, leaving the investment highly exposed to water-supply uncertainty.⁴⁴ However, the potential returns in international markets are correspondingly large. By providing monitoring and enforcement, Water Boards raise the reliability of downstream supply, making these investments feasible where they otherwise would not be. Secure water access has been shown to shift land use toward water-intensive crops in other settings (Hornbeck and Keskin, 2014). Conversely, if private adaptation is a close substitute for collective enforcement, board presence should have little effect on planting decisions.

To test this, we draw on the Fruit Cadaster (*Catastro Frutícola*), a census of fruit-growing operations recording the species planted and irrigation technology used.⁴⁵ We assign to each plot its earliest available regional census between 2005 and 2013. Because fruit trees are a durable stock, this cross-section reflects the accumulated, long-run outcome of past planting decisions rather than a transitional response, capturing the equilibrium well after the free trade agreements that reshaped Chilean agriculture.

Board presence is an equilibrium outcome, arising where the benefits of coordinated enforcement outweigh the establishment costs. Because these net benefits correlate with local agricultural conditions, cross-sectional OLS estimates are likely confounded. We therefore adopt an instrumental-variables strategy exploiting geographic variation in the legal costs of establishing a board. Given the nature of this instrument, we read the long-run estimates as evidence on the direction and order of magnitude of the response, rather than as precise point estimates.

⁴⁴An almond orchard, for instance, begins bearing only in its third year and reaches full production around the seventh, and establishing it costs roughly \$15,000 per acre before the first harvest (University of California Cooperative Extension, 2024).

⁴⁵This measure is available only for operations with fruit trees, precluding an analysis over the universe of farms.

Instrument and Identification. We exploit a unique institutional feature of the Water Code: board establishment may be initiated either by agreement among users or through a lawsuit.⁴⁶ Crucially, this lawsuit must be filed in the provincial capital with jurisdiction over the upstream section of the basin (Biblioteca del Congreso Nacional, 1981). If the water source lies entirely within one province, the filing must occur in that province’s capital; otherwise, it must be filed in the most upstream provincial capital. This rule generates systematic variation in the cost to downstream users of initiating a Water Board.

Provinces are third-order administrative units in Chile, a unitary rather than a federal country. Roughly, regions occupy the position that states do in the United States as the top subnational tier, and municipalities that of counties, with provinces in between. A river basin typically spans several provinces, so it usually contains more than one provincial capital. Which one is relevant for a given plot is determined solely by the upstream jurisdictional rule described earlier. To our knowledge, this is the only legal process in Chilean law that allocates competence to a lower-level court different from the one of the plaintiff or defendant based on this geographic criterion.⁴⁷

We leverage this variation by using the driving distance from each plot to the relevant upstream provincial capital (the easternmost capital, as water flows from the Andes to the Pacific). Nothing about hydrology makes that capital the natural forum: competence could as easily have been assigned to the most downstream capital, to the province where the plaintiff’s farm is located, or under any of several other rules. Which capital a plot faces is therefore a realized draw from a set of alternatives, and what identifies our estimates is how far that draw falls from what the plot’s location alone would predict. Following Borusyak and Hull (2023), we recenter the realized distance on the average distance under

⁴⁶Until 2005, agreement required unanimity among all users, so a single objection forced the process into court; the 2005 reform lowered this to an absolute majority. All boards in our sample were established under the earlier rule.

⁴⁷By general rule, the *Código Orgánico de Tribunales* establishes that lawsuits are presented in the lower court (*Juzgado de Letras*) with competent jurisdiction over the address of the plaintiff, and exceptionally in the address of the defendant. The former is true for non-contentious issues, where jurisdiction is voluntary. For real estate disputes, a lawsuit may be housed in the court with jurisdiction over the address of the property. Appeals to a lower level court are housed in their corresponding Court of Appeals, typically located in the regional capital city.

those counterfactual jurisdictional rules.⁴⁸ Our instrument is:

$$\tilde{ucd}_i = ucd_i - \frac{1}{\#K} \sum_{k \in K} cd_i^k, \quad (10)$$

where ucd_i is the distance to the most upstream capital city in the basin, cd_i^k is the distance from farm i to the capital that would have housed the competent court under counterfactual rule k , and K is the set of counterfactual rules considered.⁴⁹

Figure XXI illustrates the construction of the instrument for one basin (Itata). Panels (a)–(c) map each plot’s driving distance to the realized upstream capital (San Carlos) and to the capitals selected under counterfactual rules; the recentered instrument is the realized distance net of their average. Realized and average-counterfactual distances are correlated at 0.53 (panel d): a plot that is remote within the basin is likely far from every candidate capital. This common component is what recentering removes, and what remains is substantial (panel e): plots at the same realized distance receive very different instrument values. Recentering also shifts the distribution and compresses it modestly, from a standard deviation of 26.3 to 22.4 km (panel f).

Water Board establishment is a collective-action decision: downstream users weigh the benefits of coordinated enforcement against the costs of filing the lawsuit. The outside option—relying on state enforcement or rainfall—depends on alternative water availability.⁵⁰

Distance to the upstream capital shifts the cost side of this decision. This cost-based vari-

⁴⁸Concretely, the list of counterfactual rules considered here is: 1) the most downstream one, 2) the closest to the geographic center of the basin, 3) the closest to the average location of the irrigated plots in the basin (robust to using their convex-hull centroid), 4) the northernmost, and 5) the southernmost. In Appendix Table A.V we present as a robustness check the same results for an instrument based on the same set of rules plus another one: the capital of the province where each farm is located.

⁴⁹Distances are measured in kilometers, along the optimal route through the road network.

⁵⁰Groundwater was not yet a widespread alternative when most of these boards were established. In Central Chile, well-based groundwater usage remained very limited until 2010 (Taucare et al., 2024). In our 2007 Agricultural Census sample, only 5% of the plots reported some groundwater use. There are no subsidies for groundwater extraction, so groundwater-based irrigation has a positive marginal cost—as opposed to gravitational surface irrigation, with zero marginal cost—on top of the upfront investments required, in a context with prevalent liquidity constraints. Moreover, lack of access to energy probably imposed larger barriers to its usage: in 1990, only 54% of the rural population in Chile had access to electricity (Iorio and Sanin, 2019). Our water-consumption, yield, and alternative crop-choice measures are drawn from 2005–2009 and so predate this expansion. While our fruit census sample extends to 2013 to include all regions in the area of study, our results for downstream farms are robust to restricting the sample to regions surveyed in the 2005–2010 period, while the first stage for upstream farms weakens, leaving estimates imprecise (Appendix Table A.VI).

ation identifies a local average treatment effect for compliers (Angrist and Pischke, 2009): farms in areas where a Water Board is established when the upstream capital is relatively close, but would not be established if it were farther away.

The exclusion restriction requires that this recentered distance affects outcomes only through board establishment. Recentering is designed precisely to satisfy this: Appendix Table A.VII confirms that the recentered instrument (controlling for $1^\circ \times 1^\circ$ cell fixed effects) is uncorrelated with climatic variables, market access, and land productivity.

Econometric Specification of the IV Analysis. We estimate separate IV models by terciles of distance to the river mouth of each basin. Because the instrument has little power in the mid-section (a weak first stage and uninformative Anderson–Rubin intervals), throughout this section we report estimates only for the downstream and upstream terciles.⁵¹ This allows both the treatment effect and the complier population to differ across terciles, and mitigates potential SUTVA violations⁵².

Our main specification for the long-run estimates is

$$\begin{aligned} Y_{it} &= \alpha \text{Board}_i + \gamma X_{it} + \mu_g(i) + \psi_t + \varepsilon_i \\ \text{Board}_i &= \beta \tilde{ucd}_i + \delta X_{it} + \eta_g(i) + \zeta_t + u_{it} \end{aligned} \tag{11}$$

where Y_{it} denotes the outcome for farm i in year t . Our main outcome is a dummy equal to one if the farm has fruit trees planted; as alternative measures we use the growing-season length (the number of months from the start of the season to the month of peak Enhanced Vegetation Index, EVI) and a dummy for whether the EVI peak occurs in the Summer.⁵³ Although the outcome Y_{it} and controls X_{it} vary across years, the treatment variable Board_i is time-invariant: it indicates whether farm i is under the jurisdiction of a Water Board by

⁵¹The instrument’s first stage is non-monotonic across the basin (Appendix Table A.VIII). The recentered distance captures the cost of reaching the competent court, which downstream users incur to *establish* a board and upstream users incur to *oppose* one; the two forces therefore push board status in opposite directions at the extremes. Consistent with this, the coefficient is negative downstream and positive upstream, and the two forces roughly cancel in the mid-section, where the first stage is very weak and imprecisely signed.

⁵²Estimating separate models mitigates SUTVA violations related to water externalities that depend on aggregate extraction by all upstream users rather than on extraction by nearby users at similar locations within the basin.

⁵³We also examine Evapotranspiration and the peak of the EVI index—proxies for water consumption and agricultural yield, respectively, and outcomes of the previous section—which we report in Table VIII.

2005, and remains constant thereafter. Board_i is instrumented by the recentered distance $\tilde{ucd}_i$ (equation 10). X_{it} is a vector of farm-level controls including market access measures (driving distance to Santiago and the main ports); annual, combined winter and spring, and summer precipitation; and temperature, measured as the number of days with maximum temperatures in the following temperature ranges: $[25, 29]$, $[29, 31]$, $[31, 35]$, and $[35, \infty^+)$, in Celsius degrees (Hsiang, 2016).⁵⁴ $\mu_g(i)$ and $\eta_g(i)$ are $1^\circ \times 1^\circ$ latitude-longitude cell fixed effects, and ψ_t and ζ_t are year fixed effects. We address spatial correlation using clustered standard errors by county.⁵⁵ Following Andrews et al. (2019), we assess first-stage strength using the Effective F-statistic (Olea and Pflueger, 2013)⁵⁶ and address potential weak instrument issues by reporting Anderson-Rubin (AR) confidence intervals for all IV estimates (Keane and Neal, 2024).⁵⁷

Fruit Tree Planting. In Table VII, we present estimates of Equation 11 for a dummy equal to one if a plot of land has fruit trees. Downstream parcels under Water Board jurisdiction are substantially more likely to have planted trees: OLS (Column 1) yields a positive but imprecise 2.8 pp, while the IV estimate implies a 20.3 pp increase for complier farms, against a mean planting rate of 8 percent. The Anderson-Rubin test rejects a zero effect. Upstream, board presence is associated with a 3.8 pp lower probability of planting (Column 2), and the IV estimate carries the same negative sign, with a weaker first stage (effective F-statistic of 6.9 versus 11.6 downstream); the Anderson-Rubin interval contains zero. The magnitude of the downstream response suggests that private adaptation and collective enforcement are complements rather than substitutes: were farms able to secure reliable

⁵⁴All of our main estimates are robust to alternative control sets—adding soil-quality quartiles and ground-water rights, or the potential land-productivity value as a control (Appendix Tables A.XI, A.XII, A.XIII, A.XIV, and A.XV).

⁵⁵Counties (*comunas*) are the most decentralized tier of local government in Chile: they concentrate local administration, span local labor markets, and deliver public services, so they are the natural level at which shocks common to nearby farms operate. The theoretically correct level of clustering is the level at which treatment is assigned, which in our setting is not known ex ante: as described above, jurisdiction is defined in terms of the “water source” rather than an administrative unit. We address this below, where we complement these standard errors with randomization inference over three nested hydrological levels (Table A.X).

⁵⁶This is equal in exactly identified models to the robust F-statistic (Kleibergen and Paap, 2006).

⁵⁷As discussed by Keane and Neal (2024), standard 2SLS t-tests suffer from severe power asymmetries, yielding artificially inflated standard errors when IV estimates diverge substantially from OLS. The AR test, instead, maintains correct size regardless of instrument strength, so its confidence intervals have correct coverage whether or not the first stage is strong.

water supply on their own, board presence would leave planting decisions largely unchanged.

Alternative Measures. We corroborate the fruit census data using independent remote-sensing measures of crop choice over 2005–2009. Table A.IX presents IV estimates showing that Water Boards significantly lengthen the growing season. Downstream farms extend their season by 0.72 months (Column 3) and are 23 pp more likely to experience peak greenness during the summer (Column 7). Neither outcome shows upstream reductions comparable to the downstream increases: the upstream change in growing season is an order of magnitude smaller, and for summer peak the upstream estimate is positive. These independent measures corroborate the extensive-margin response documented in the fruit census.

Consistency between the DID and IV Estimates. Table VIII presents the IV estimates of equation 11 for both water consumption (evapotranspiration, columns 1–4) and agricultural productivity (peak EVI, columns 5–8). Within each outcome, the first two columns show OLS estimates and the last two IV estimates, by section of the river (downstream and upstream, respectively); Panel B reports the corresponding first stage. Water Boards increase downstream water consumption by 1.68mm (57%) and reduce upstream consumption by 1.30mm (38%). Overall, the IV estimates reveal economically significant reallocation of water from upstream to downstream farms, with net gains concentrated downstream⁵⁸.

Columns 5–8 of Table VIII present results for agricultural yield (peak EVI per pixel during the season). The results are consistent with the water reallocation pattern but suggest decreasing returns to water consumption: Water Boards increase downstream yield by 53% with no detectable effects upstream.

Our Difference-in-Differences and IV estimates yield consistent results across both water consumption and agricultural productivity. Downstream farms increase water consumption by 1.5mm after seven years (DID) and 1.7mm in the long run (IV), while upstream

⁵⁸Notably, we cannot reject equality between OLS and IV estimates for upstream locations, consistent with the idea that Water Board establishment is driven by downstream users and is therefore plausibly exogenous for upstream users.

farms face smaller reductions of $-0.3mm$ (DID) to $-1.3mm$ (IV). For agricultural productivity, downstream farmers increase yields by 60% of a standard deviation after seven years (DID) and 53% of the control group mean in the long run (IV), while upstream farmers experience no significant effects.

While these estimates seem large, they align with findings from other contexts, such as Cisse et al. (2026)’s analysis of irrigation infrastructure impacts in Senegal. They document a sixfold increase in dry season cultivation rates. Our estimates indicate that institutions can generate substantial variation in the impacts of infrastructure investments.

The asymmetric pattern —where downstream gains substantially exceed upstream losses—can be explained by different, non-mutually exclusive mechanisms; we discuss two here. First, downstream gains may occur along the extensive margin—shifting farms from no irrigation to irrigation, which imply large gains according to Cisse et al. (2026)—whereas upstream losses may occur along the intensive margin, limiting previously unrestricted access. Second, net increases in water consumption and yield may reflect complementarity between reliable water provision and individual or collective investments, which we examine below.

Randomization Inference. Because the geographic unit at which Water Board status is effectively assigned is not obvious, we subject our three main IV estimates—water consumption, agricultural productivity, and fruit-tree planting—to randomization inference at three nested levels: the basin, the sub-basin, and the sub-sub-basin (or 1st-, 2nd-, and 3rd-level basins). At each level we build placebo treatments by holding the number of treated units fixed at its actual value and randomly permuting which units are treated; each placebo-treated unit is assigned, by random permutation, one of the observed within-unit treatment shares, and its plots are then treated at random at that share. For each of 1,000 draws we re-estimate the specification and collect the placebo t -statistic, and we compare the observed t -statistic against the resulting distribution. Table A.X reports the results, which are mostly consistent with the main results in Tables VII and VIII: the downstream increase in fruit-tree planting lies outside the placebo distribution at all three levels, and the effects on water consumption and yield do so at two of the three. Upstream, only the reduction

in water consumption is distinguishable from placebo, consistent with the main results, but only for one of the levels (randomization over 1st-level basin).

Water Intensity. We further explore how boards presence shape productive decisions by estimating equation 11 for another dummy variable that takes value one when farm i has planted a fruit tree of a water-intensive species⁵⁹, and zero otherwise (either no water-intensive species, or no tree at all). We also define water-saving species as those that are not water-intensive. In Table IX, we show that most of the increase in fruit-tree planting is in water-intensive species: of the 20.3 pp increase in fruit tree planting downstream, 17.7 pp correspond to water-intensive species and 2.6 pp to water-saving ones. Upstream, the corresponding estimate for water-intensive species is negative and its Anderson-Rubin interval excludes zero, while the estimate for water-saving species is positive and imprecise. The data are consistent with a compositional shift away from water-intensive planting upstream, though the water-saving estimate is too imprecise to establish one.

Overall, we consider the previous results evidence that the presence of water boards expands the feasible production set: planting water-intensive trees may not be profitable or feasible under uncertain water supply; the presence of boards, by addressing over-extraction, improve the reliability of water supply, allowing downstream farms to perform this risky investments.

Complementarity between capital and enforcement: Irrigation Technology and farm size. Finally, we explore the joint decisions of tree planting and irrigation technology. Table X shows estimates of equation 11 for dummy variables equal to 1 if farm i planted a fruit tree and irrigates it using three different families of irrigation technology: micro-irrigation (such as dripping), macro-irrigation (such as large-scale sprinklers) and traditional (furrow irrigation). Our results show that most of the increase in tree planting

⁵⁹We identified water-intensive fruit tree species using the water requirements in Ferreyra et al. (2010), complemented with Lobos et al. (2017) and crop coefficient tables in Allen et al. (1998, Table 12). Water-intensive species are those above the median in the ranking of fruit trees planted in our study sample, ranked by their water requirement through the whole season. Under this definition, the list of water-intensive species is: Avocado, Kiwi, Walnut, American blueberry, Cherry, Lemon, Orange, Mandarin, Raspberry, Clingstone/canning peach, Nectarine, and Fresh-market peach. The results do not change if we include Table Grape in this list.

is accompanied by the use of micro-irrigation, a water-saving, capital-intensive technology: our results imply that out of a 20pp increase in tree planting, almost 15pp are accompanied by the use of micro-irrigation, while only 6pp are irrigated using traditional irrigation—a technology that requires more water to provide the same effective water input to the tree.⁶⁰ Although we cannot distinguish directly the role of capital investments and water reliability in the expansion of the feasible production set—we do not know whether the capital would have been in place independently of the planting decision—this suggests a complementarity between capital decisions and the reliability of water supply: either the farms did have the capital, but could not plant trees until the boards provided reliable water access, or farms, once water access improves, invest in capital and tree planting as a bundle.

To further explore the potential complementarity of capital and community enforcement, in Appendix Table A.XVI we explore the role of farm size. We define small farms as those with area below the median and large farms as those above the 80th percentile. The expansion in fruit-tree planting is concentrated among large farms: the point estimate is roughly five times larger and its Anderson-Rubin interval excludes zero, whereas for small farms the conventional standard error indicates significance at the 10 percent level but the AR interval does not exclude zero. For water-intensive species the pattern is less clean: the large-farm estimate is again the larger of the two but imprecisely estimated, while the small-farm estimate is smaller and its AR interval excludes zero. Notably, for small farms the water-intensive effect exceeds the effect on planting overall, and is the better identified of the two, implying a shift in composition toward water-intensive species rather than an expansion of planting; among large farms, water-intensive species account for roughly 70 percent of the total planting response. Micro-irrigation accompanies most of the response in both groups—89 and 64 percent of the respective planting effects—though neither estimate is individually distinguishable from zero. No upstream estimate is distinguishable from zero in any of the size subsamples.

These estimates cannot separate credit constraints from other sources of heterogeneity, but they suggest that farm scale changes the nature of adaptation. If reliability lowers the

⁶⁰We find no detectable effect on macro-irrigation. Note that these categories are not mutually exclusive, as a farm may irrigate different trees with different technologies, so the estimates need not sum to the total effect on planting.

cost of maintaining an orchard, large farms would find new planting profitable. Among small farms, those already able to invest can switch toward more water-intensive species, while the rest cannot take advantage of the change at all. This is further evidence of the complementarity between capital and community enforcement: Water Boards expand the feasible production set by making water supply reliable, but a farm's capacity to invest determines whether it can take up that opportunity or only reallocate within its existing scale.

6 Conclusions

In this paper, we examine how specialized enforcement institutions affect water markets' allocative efficiency, even in contexts with well-defined property rights that are perpetual, tradable, inheritable, and constitutionally protected against expropriation. Using a sufficient statistic approach, we find that within basins governed by Water Boards, the shadow value of water remains constant across locations. By contrast, we find evidence of misallocation in areas without such Boards, such that downstream users face higher shadow values due to upstream over-extraction. Using various identification strategies and measures, we further explore how Water Boards' enforcement creates net economic gains through extensive water reallocation that enables private investments.

Our analysis of Water Boards provides novel insights into the challenges of establishing environmental markets. The operation of these markets must overcome subtractability and high monitoring and enforcement costs—resource attributes that lead to the Tragedy of the Commons. While water is a leading example of a resource subject to this issue, similar characteristics are present in many others, including fisheries and atmospheric emissions. Surface water offers a unique analytical advantage: the directional nature of surface water flows allows for clear identification of affected agents and precise measurement of reallocation. This directionality may explain why Water Boards are effective: with a single entity maintaining both accountability and authority, it creates strong incentives for downstream users to invest in basin-wide enforcement. This highlights how market design challenges differ across resources: similar decentralized institutions may prove less effective where such

incentives are weak. In the absence of such an entity, the legal system struggles to hold agents accountable, as the individual responsibility for externalities is too hard to allocate.

Our results indicate that individual adaptation and markets cannot fully offset the lack of effective governance in ensuring efficient resource use, even in contexts with strong property rights and rule of law, causing substantial welfare losses: the estimates suggest that the impacts in water consumption after 1 year may double the revenue per hectare for the 20% of most downstream farms, without substantially affecting upstream outcomes. This reflects a fundamental tension in environmental markets: empowering users over governing authorities may reduce some market frictions but weaken enforcement, as administrative actions under changing conditions remain necessary. We show that institutional arrangements that empower local communities can resolve this tension; however, the resulting efficiency gains are unevenly distributed. These findings have important implications for the design of environmental markets, suggesting that successful implementation requires careful attention to enforcement mechanisms and their distributional consequences.

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Tables

Table I: Balance Table for Misallocation Test

Outcome: Useful Rainfall (Rainfall during the Crop Irrigation Season)						
	Full Sample		Irrigated fields sample		Placebo sample	
	(1)	(2)	(3)	(4)	(5)	(6)
Distance to coast (100km)	0.0412 (0.0539)	0.0469 (0.0610)	-0.521 (0.103)***	-0.958 (0.219)***	0.102 (0.0399)**	0.111 (0.0540)**
Water Board=1	-0.0814 (0.301)	0.0596 (0.338)	-0.415 (0.280)	-0.618 (0.399)	-0.413 (0.440)	-0.114 (0.329)
Water Board=1 $\times$ Distance to coast (100km)	0.101 (0.211)	0.111 (0.237)	0.138 (0.196)	0.381 (0.263)	0.624 (0.294)**	0.299 (0.248)
Basin FE	Yes	No	Yes	No	Yes	No
0.5 degree cell FE	No	Yes	No	Yes	No	Yes
Crop FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	79,133	79,132	15,149	15,147	63,838	63,838
R-squared	0.885	0.931	0.910	0.944	0.924	0.946
Outcome mean	0.899	0.899	0.765	0.765	0.931	0.931
Outcome SD	0.944	0.944	0.836	0.836	0.966	0.966

Notes: This balance table presents regressions of the double interaction between a Water Boards dummy and the location in the basin, measured as the distance to the sea through the river network. The outcome variable is useful rainfall: precipitation fell during the irrigation season of the crop planted in the parcel.

Table II: Shadow Value of Water: Impact of Useful Rainfall on Production for Irrigated Farms, by Treatment Status

	Outcome is log(Value Yield p/Hectare)			
	(1)	(2)	(3)	(4)
Useful pp. (ML per Ha per month)	0.464 (0.196)** [0.201]**	0.330 (0.197)* [0.200]	0.462 (0.221)** [0.221]**	0.421 (0.232)* [0.232]*
Useful pp. (ML per Ha per month) $\times$ Distance to coast (100km)	-0.225 (0.129)* [0.131]*	-0.188 (0.125) [0.125]	-0.250 (0.139)* [0.136]*	-0.272 (0.148)* [0.144]*
Water Board=1 $\times$ Useful pp. (ML per Ha per month)	-0.134 (0.240) [0.242]	-0.237 (0.217) [0.217]	-0.396 (0.251) [0.245]	-0.481 (0.258)* [0.251]*
Water Board=1 $\times$ Useful pp. (ML per Ha per month) $\times$ Distance to coast (100km)	-0.00902 (0.164) [0.166]	0.130 (0.145) [0.144]	0.235 (0.166) [0.161]	0.298 (0.170)* [0.163]*
Parcel Controls	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	No
Basin FE	No	Yes	No	No
0.5 degree cell FE	No	No	Yes	No
Crop FE	Yes	Yes	Yes	Yes
County FE	No	No	No	Yes
Observations	14,716	14,716	14,714	14,712
R-squared	0.598	0.621	0.628	0.642
Misallocation Test: Water Board=0	0.08	0.13	0.07	0.07
Misallocation Test: Water Board=1	0.01	0.42	0.87	0.76

Notes: This table presents estimates of equation 5 for irrigated parcels, with water rights, registered in canal associations. Distance to the coast measured through the river network. Controlling for logarithm of cultivated surface, logarithm of number of hired workers plus 1, dummies for educational level of manager of the farm, legal organization category of the operation, irrigation technology, and County and Crop fixed effects. Standard errors clustered at the county level (round parenthesis) and the county $\times$ irrigation season level (squared parentheses).

Table III: Placebo Exercise: Impact of Rainfall on Rainfed Farms Production

	Placebo: Outcome is log(Value Yield p/Hectare)			
	(1)	(2)	(3)	(4)
Useful pp. (ML per Ha per month)	-0.0606 (0.0502) [0.0473]	0.00566 (0.0677) [0.0602]	0.0146 (0.0493) [0.0466]	0.0830 (0.0408)** [0.0411]**
Distance to coast (100km)	-0.202 (0.104)* [0.0950]**	-0.168 (0.114) [0.105]	0.194 (0.0721)*** [0.0625]**	
Useful pp. (ML per Ha per month) $\times$ Distance to coast (100km)	0.150 (0.0439)*** [0.0409]***	0.154 (0.0447)*** [0.0417]***	0.0407 (0.0282) [0.0291]	0.0141 (0.0287) [0.0263]
Water Board=1	-0.567 (0.299)* [0.324]*	-0.811 (0.398)** [0.388]**	0.159 (0.387) [0.368]	
Water Board=1 $\times$ Useful pp. (ML per Ha per month)	0.104 (0.130) [0.140]	0.143 (0.0966) [0.112]	0.000591 (0.0593) [0.107]	0.0457 (0.0680) [0.0707]
Water Board=1 $\times$ Distance to coast (100km)	0.687 (0.208)*** [0.228]**	0.893 (0.247)*** [0.248]**	0.00530 (0.277) [0.274]	
Water Board=1 $\times$ Useful pp. (ML per Ha per month) $\times$ Distance to coast (100km)	-0.146 (0.0980) [0.102]	-0.173 (0.0739)** [0.0822]**	-0.0180 (0.0499) [0.0848]	-0.0446 (0.0542) [0.0540]
Parcel Controls	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	No
Basin FE	No	Yes	No	No
0.5 degree cell FE	No	No	Yes	No
Crop FE	Yes	Yes	Yes	Yes
County FE	No	No	No	Yes
Observations	63,838	63,838	63,838	63,836
R-squared	0.582	0.588	0.618	0.631
Misallocation Test: Water Board=0	0.00	0.00	0.15	0.63
Misallocation Test: Water Board=1	0.97	0.76	0.56	0.46

Notes: This table presents estimates of equation 5 for non-irrigated parcels, as a placebo exercise. Distance to the coast measured through the river network. Controlling for logarithm of cultivated surface, logarithm of number of hired workers plus 1, dummies for educational level of manager of the farm, legal organization category of the operation, irrigation technology, and County and Crop fixed effects. Standard errors clustered at the county level (round parenthesis) and the county $\times$ irrigation season level (squared parentheses).

Table IV: Summary Statistics: Parcel Level Dataset

Panel A: Parcels Not Under Water Board Jurisdiction						
	Mean	SD	p10	p90	Min	Max
Evapotranspiration (Summer)	3.28	1.2	1.7	4.7	0.0	7.8
Peak EVI over season	0.40	0.1	0.2	0.5	0.0	0.9
Area (m^2)	59659.00	132126.2	3920.7	150559.0	47.6	4199398.0
Latitude	-34.82	1.4	-36.8	-32.9	-37.7	-29.8
Longitude	-71.41	0.5	-72.2	-70.7	-73.0	-70.5
Mean annual prec.	1620.35	867.4	687.1	2873.4	0.0	5199.5
Mean summer prec.	184.81	145.9	40.5	406.2	0.0	1000.9
Distance to Santiago (main dom. market)	238.82	165.4	48.9	498.8	9.4	620.5
Distance to Valparaiso (main port)	305.83	187.7	89.4	592.2	15.8	707.2
Distance to San Antonio (second main port)	252.42	158.9	94.5	506.4	20.4	626.0
Distance to Coast (location in basin)	105.10	34.7	54.7	141.6	0.9	149.9
Recentered Instrument	-0.41	3.2	-4.8	4.3	-10.2	10.0
Observations	446177					

Panel B: Parcels Under Water Board Jurisdiction						
	Mean	SD	p10	p90	Min	Max
Evapotranspiration (Summer)	2.95	1.4	1.1	4.7	0.0	7.8
Peak EVI over season	0.38	0.1	0.2	0.5	-0.1	0.9
Area (m^2)	58084.77	134182.9	3748.4	138617.7	24.9	3433859.8
Latitude	-33.62	2.0	-36.7	-30.7	-37.0	-29.8
Longitude	-71.20	0.4	-72.0	-70.7	-72.3	-70.5
Mean annual prec.	1108.56	643.9	427.4	1966.6	0.0	4285.9
Mean summer prec.	121.92	95.1	34.0	237.7	0.0	688.2
Distance to Santiago (main dom. market)	269.96	173.2	46.8	504.5	20.7	589.8
Distance to Valparaiso (main port)	299.32	165.2	102.1	548.4	35.5	609.3
Distance to San Antonio (second main port)	289.47	160.5	120.4	510.0	43.2	595.3
Distance to Coast (location in basin)	103.82	34.0	48.8	140.7	0.9	150.0
Recentered Instrument	0.53	3.0	-3.2	6.0	-8.1	8.6
Observations	193035					

Notes: This table presents summary statistics for agricultural parcels, separated by Water Board jurisdiction status. Panel A shows statistics for parcels not under the jurisdiction of the Water Board, while Panel B shows statistics for parcels under jurisdiction. Evapotranspiration and Enhanced Vegetation Index (EVI) are satellite-derived variables averaged at the parcel level. Evapotranspiration measured in average millimeters per pixel during summer season (December-February) between years 1986 and 2012. Peak EVI corresponds to the maximum value of Enhanced Vegetation Index. Area measured in square meters. All distance measures in kilometers. Santiago corresponds to the main domestic market. Valparaíso and San Antonio are the main ports. Distance to coast indicates location within basin. The recentered instrument is the driving distance to the most upstream province capital in the basin, net of the average distance to the capitals selected by the counterfactual jurisdiction rules. Precipitation variables measured in millimeters.

Table V: Farm plots by Year of Board Establishment

	N plots	Percent	Cumulative perc.
1974	951	1.09	1.09
1989	1774	2.03	3.13
1993	920	1.06	4.18
1994	2338	2.68	6.86
1995	275	0.32	7.18
1996	2963	3.40	10.57
1997	1690	1.94	12.51
1998	755	0.87	13.38
1999	10055	11.53	24.91
2000	515	0.59	25.50
2001	689	0.79	26.29
2003	262	0.30	26.59
2004	1769	2.03	28.62
2005	2090	2.40	31.02
2010	28	0.03	31.05
2011	182	0.21	31.26
2012	1464	1.68	32.94
2013	2652	3.04	35.98
2018	694	0.80	36.77
2019	620	0.71	37.48
No Board	54512	62.52	100.00
Total	87198		

Notes: This table shows the total number of farm plots in our area of study, and the establishment date of a Water Board.

Table VI: Monitoring Stations by Year of Board Establishment

Year	N segments	Percent	Cumulative perc.
1974	1	0.33	0.33
1989	12	3.96	4.29
1993	10	3.30	7.59
1994	20	6.60	14.19
1995	4	1.32	15.51
1996	38	12.54	28.05
1997	6	1.98	30.03
1998	14	4.62	34.65
1999	62	20.46	55.12
2000	10	3.30	58.42
2001	11	3.63	62.05
2004	1	0.33	62.38
2005	3	0.99	63.37
2013	1	0.33	63.70
2018	5	1.65	65.35
2019	22	7.26	72.61
No Board	83	27.39	100.00
Total	303	100.00	
Observations	303		

Notes: This table shows the total number of monitoring stations in our area of study, and the establishment date of a Water Board (in case the monitoring station is in a subsubbasin within the jurisdiction of a Water Board).

Table VII: Fruit Tree Planting: Instrumental Variables Estimation at the Parcel Level

	OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream
<i>Panel A. Main result</i>				
Board	0.0282 (0.0226)	-0.0384 (0.0219)*	0.203 (0.0832)**	-0.0785 (0.114)
Observations	145,065	144,515	145,065	144,515
R-squared	0.079	0.048	0.044	0.045
Mean Dependent Var.			0.080	0.096
AR test CI			[.02974, .4307]	[-.3636, .3053]
<i>Panel B. First stage: Water Board presence</i>				
Recentered distance $\tilde{ucd}_i$			-0.0595 (0.0174)***	0.0380 (0.0145)**
Observations			145,065	144,515
R-squared			0.553	0.358
Effective F-stat			11.646	6.915

Notes: This table presents estimates of equation 11 for parcels located at elevations below 1,000 meters and within 150 km of the river mouth in the study area. The outcome variable is a dummy equal to one if the parcel has a fruit tree planted. Columns 1–2 report OLS and columns 3–4 IV estimates. Columns 1 and 3 present estimates for downstream farms (in the lowest tercile of distance to the coast), and columns 2 and 4 for upstream farms. The *first stage* block reports the coefficient on the recentered instrument $\tilde{ucd}_i$ from the first stage, a regression of Water Board presence on that instrument, together with the effective F-statistic, and is therefore reported only for the IV columns; Appendix Table A.VIII reports the first stage for the three terciles of distance to the river mouth, including the mid-section. Plot-level controls include market access measures (driving distance to Santiago and the main ports of Valparaíso and San Antonio); annual, combined winter and spring, and summer precipitation; and temperature, measured as the number of days with maximum temperatures in the ranges [25, 29], [29, 31], [31, 35], and [35, ∞^+) in Celsius degrees. All regressions include $1^\circ \times 1^\circ$ cell fixed effects and year fixed effects. Standard errors are clustered by county.

Table VIII: Water Consumption and Agricultural Productivity: Instrumental Variables Estimation at the Parcel Level

	Water consumption (evapotranspiration)				Agricultural productivity (peak EVI)			
	OLS		IV		OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream	(5) Downstream	(6) Upstream	(7) Downstream	(8) Upstream
<i>Panel A. Main result</i>								
Board	0.0157 (0.108)	0.0292 (0.0890)	1.684 (0.837)**	-1.297 (0.478)***	0.00572 (0.00853)	0.00890 (0.00754)	0.197 (0.0840)**	-0.00496 (0.0348)
Observations	143,038	144,515	143,038	144,515	145,065	144,515	145,065	144,515
R-squared	0.301	0.359	0.126	0.187	0.244	0.208	0.011	0.206
Mean Dependent Var.			2.952	3.407			0.370	0.410
AR test CI			[.3831, 4.785]	[-4.106, -.6076]			[.06819, .5084]	[-.08436, .1247]
<i>Panel B. First stage: Water Board presence</i>								
Recentered distance $\tilde{ucd}_i$			-0.0585 (0.0174)***	0.0380 (0.0145)**			-0.0595 (0.0174)***	0.0380 (0.0145)**
Observations			143,038	144,515			145,065	144,515
R-squared			0.556	0.358			0.553	0.358
Effective F-stat			11.359	6.915			11.646	6.915

Notes: This table presents estimates of equation 11 for parcels located at elevations below 1,000 meters and within 150 km of the river mouth in the study area. Columns 1–4 report estimates for average summer (January–February) evapotranspiration per plot; columns 5–8, for the maximum Enhanced Vegetation Index (EVI) reached within the year. Both are measured for the period 2005–2009. Within each outcome, the first two columns report OLS and the last two IV estimates, and within each pair the first column presents estimates for downstream farms (in the lowest tercile of distance to the coast) and the second for upstream farms. Panel B reports the corresponding first stage, a regression of Water Board presence on the recentered instrument $\tilde{ucd}_i$, and is therefore reported only for the IV columns; Appendix Table A.VIII reports the first stage for the three terciles of distance to the river mouth, including the mid-section. Plot-level controls include market access measures (driving distance to Santiago and the main ports of Valparaíso and San Antonio); annual, combined winter and spring, and summer precipitation; and temperature, measured as the number of days with maximum temperatures in the ranges [25, 29], [29, 31], [31, 35], and [35, ∞^+) in Celsius degrees. All regressions include $1^\circ \times 1^\circ$ cell fixed effects and year fixed effects. Standard errors are clustered by county.

Table IX: Fruit Tree Planting by Water Requirement: Instrumental Variables Estimation at the Parcel Level

	Water-intensive tree species				Water-saving tree species			
	OLS		IV		OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream	(5) Downstream	(6) Upstream	(7) Downstream	(8) Upstream
<i>Panel A. Main result</i>								
Board	0.0251 (0.0216)	-0.0342 (0.0160)**	0.177 (0.0787)**	-0.139 (0.0913)	0.00313 (0.00453)	-0.00421 (0.0101)	0.0268 (0.0201)	0.0601 (0.0503)
Observations	145,065	144,515	145,065	144,515	145,065	144,515	145,065	144,515
R-squared	0.079	0.026	0.044	-0.005	0.064	0.041	0.062	0.026
Mean Dependent Var.			0.062	0.055			0.019	0.041
AR test CI			[.01235, .3914]	[-.4361, .09919]			[-.01965, .07636]	[-.02715, .3058]
<i>Panel B. First stage: Water Board presence</i>								
Recentered distance $\tilde{ucd}_i$			-0.0595 (0.0174)***	0.0380 (0.0145)**			-0.0595 (0.0174)***	0.0380 (0.0145)**
Observations			145,065	144,515			145,065	144,515
R-squared			0.553	0.358			0.553	0.358
Effective F-stat			11.646	6.915			11.646	6.915

Notes: This table presents estimates of equation 11 for parcels located at elevations below 1,000 meters and within 150 km of the river mouth in the study area. The outcome variables are dummies equal to one if the parcel has a water-intensive (columns 1–4) or water-saving (columns 5–8) fruit tree species planted; species are classified by their mid-season crop coefficient (K_c) following Allen et al. (1998, Table 12), with water-intensive species those above the sample median. Within each outcome, the first two columns report OLS and the last two IV estimates, and within each pair the first column presents estimates for downstream farms (in the lowest tercile of distance to the coast) and the second for upstream farms. Panel B reports the corresponding first stage, a regression of Water Board presence on the recentered instrument $\tilde{ucd}_i$, and is therefore reported only for the IV columns; Appendix Table A.VIII reports the first stage for the three terciles of distance to the river mouth, including the mid-section. Plot-level controls include market access measures (driving distance to Santiago and the main ports of Valparaíso and San Antonio); annual, combined winter and spring, and summer precipitation; and temperature, measured as the number of days with maximum temperatures in the ranges [25, 29], [29, 31], [31, 35], and [35, ∞^+) in Celsius degrees. All regressions include $1^\circ \times 1^\circ$ cell fixed effects and year fixed effects. Standard errors are clustered by county.

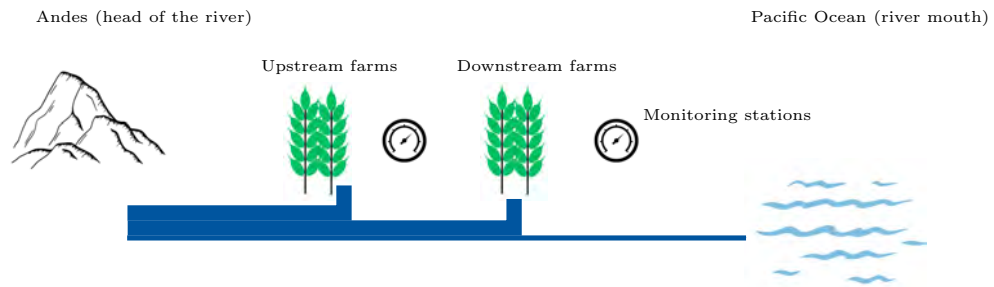
Table X: Fruit Tree Planting and Irrigation Technology: Instrumental Variables Estimation at the Parcel Level

	Micro-irrigation				Macro-irrigation				Traditional irrigation			
	OLS		IV		OLS		IV		OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream	(5) Downstream	(6) Upstream	(7) Downstream	(8) Upstream	(9) Downstream	(10) Upstream	(11) Downstream	(12) Upstream
<i>Panel A. Main result</i>												
Board	0.0216 (0.0189)	-0.0168 (0.0100)*	0.146 (0.0726)**	0.0153 (0.0459)	0.0000818 (0.000355)	-0.0000519 (0.000506)	0.000785 (0.000816)	0.000551 (0.00217)	0.00712 (0.00768)	-0.0216 (0.0155)	0.0614 (0.0262)**	-0.0985 (0.0756)
Observations	145,065	144,515	145,065	144,515	145,065	144,515	145,065	144,515	145,065	144,515	145,065	144,515
R-squared	0.073	0.021	0.050	0.017	0.001	0.001	0.001	0.001	0.037	0.031	0.024	0.013
Mean Dependent Var.			0.059	0.042			0.000	0.001			0.020	0.052
AR test CI			[-0.1126, .3368]	[-0.07971, .2034]			[-0.009217, .002918]	[-0.004864, .00759]			[.009884, .1374]	[-.3269, .1141]
<i>Panel B. First stage: Water Board presence</i>												
Recentered distance $\tilde{ucd}_i$			-0.0595 (0.0174)**	0.0380 (0.0145)**			-0.0595 (0.0174)**	0.0380 (0.0145)**			-0.0595 (0.0174)**	0.0380 (0.0145)**
Observations			145,065	144,515			145,065	144,515			145,065	144,515
R-squared			0.553	0.358			0.553	0.358			0.553	0.358
Effective F-stat			11.646	6.915			11.646	6.915			11.646	6.915

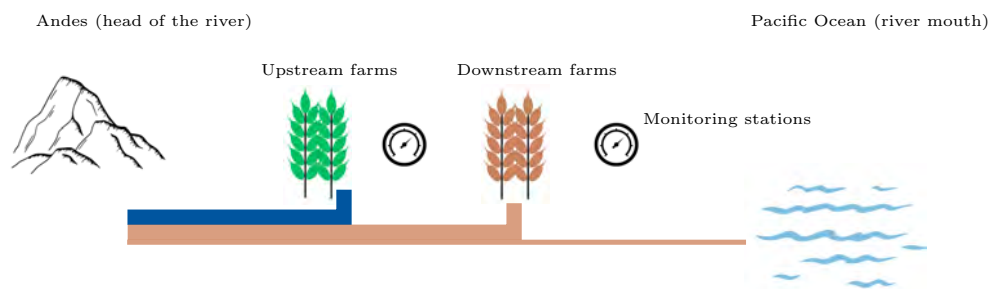
Notes: This table presents estimates of equation 11 for parcels located at elevations below 1,000 meters and within 150 km of the river mouth in the study area. The outcome variables are dummies equal to one if the parcel has a tree irrigated with micro-irrigation such as dripping (columns 1–4), macro-irrigation such as large-scale sprinklers (columns 5–8), or traditional furrow irrigation (columns 9–12). Within each outcome, the first two columns report OLS and the last two IV estimates, and within each pair the first column presents estimates for downstream farms (in the lowest tercile of distance to the coast) and the second for upstream farms. Panel B reports the corresponding first stage, a regression of Water Board presence on the recentered instrument $\tilde{ucd}_i$, and is therefore reported only for the IV columns; Appendix Table A.VIII reports the first stage for the three terciles of distance to the river mouth, including the mid-section. Plot-level controls include market access measures (driving distance to Santiago and the main ports of Valparaíso and San Antonio); annual, combined winter and spring, and summer precipitation; and temperature, measured as the number of days with maximum temperatures in the ranges [25, 29], [29, 31], [31, 35], and [35, ∞^+) in Celsius degrees. All regressions include $1^\circ \times 1^\circ$ cell fixed effects and year fixed effects. Standard errors are clustered by county.

Figures

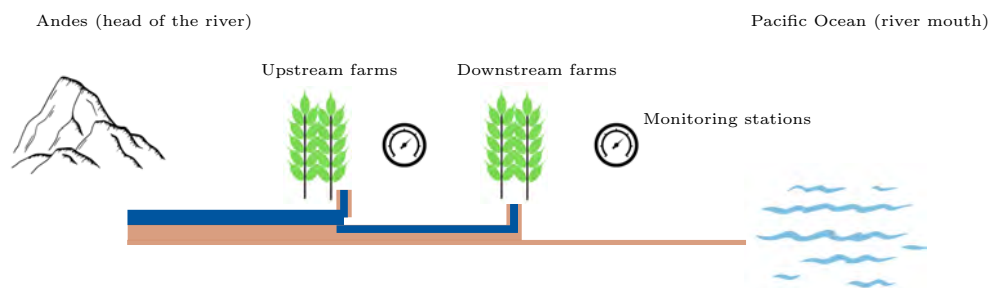
Figure I: How Property Rights Enforcement Affects Water Allocation and Agricultural Outcomes



(a) No drought

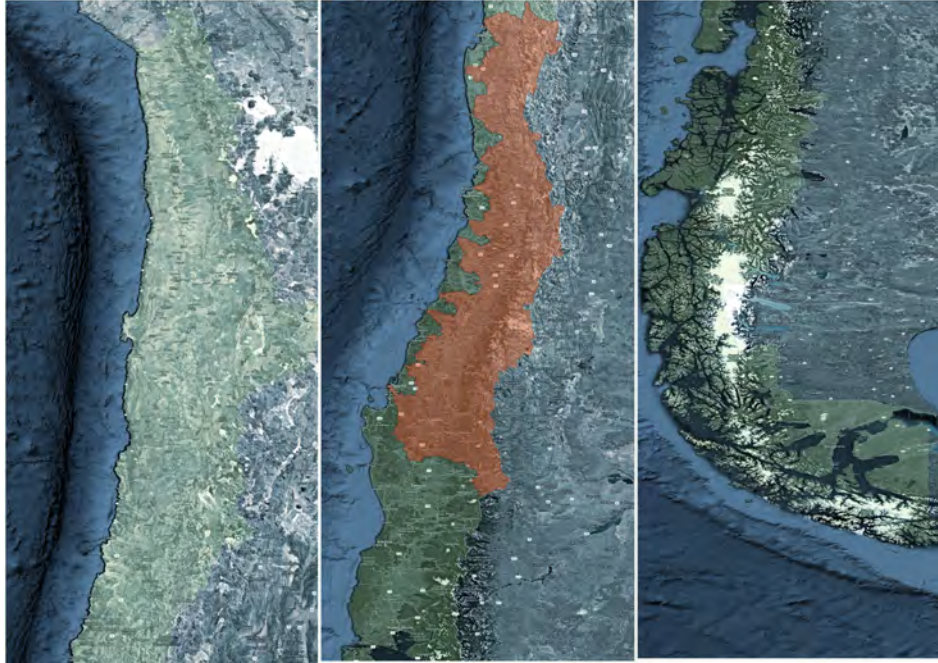


(b) Drought, and no enforcement of water rights



(c) Drought, and Water Boards enforce water rights

Figure II: Area of Study



Notes: Left, center, and right panels correspond to the northern, central, and southern areas of Chile. The colored region represents the total area of study.

Figure III: Basins and Rivers in Area of Study



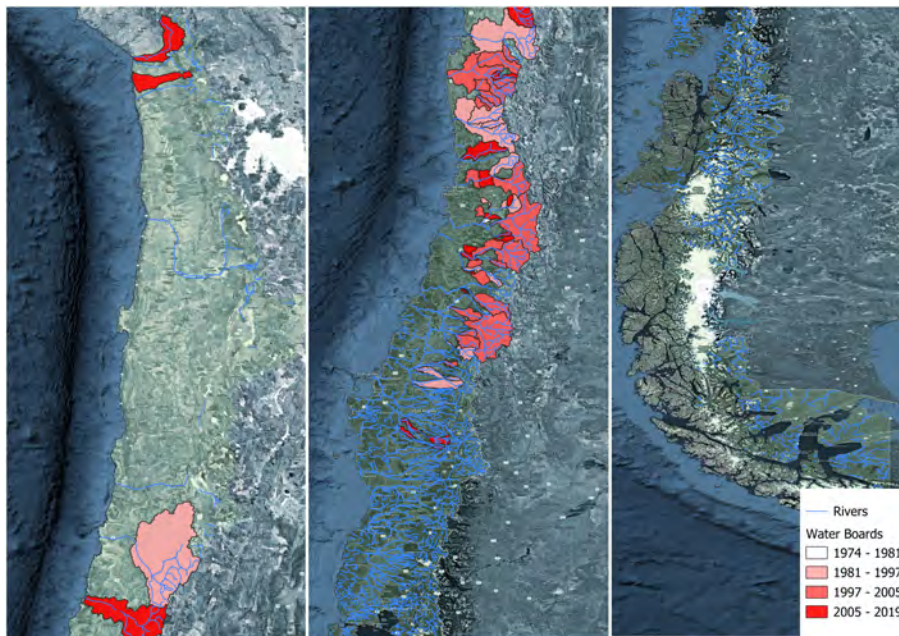
Notes: This map zooms into the colored area of map II. Blue lines represent rivers and orange lines represent their corresponding basin boundaries.

Figure IV: Example of Canal Gate



Notes: This picture shows a canal gate. Water Boards have the legal right to open and close them with locks during droughts, to implement a system of irrigation shifts as a water allocation enforcement mechanism.

Figure V: Water Boards Jurisdictions



Notes: Left, center, and right panels correspond to the northern, central, and southern areas of Chile. The colored areas represent the boundaries of existing Water Boards jurisdictions and their year of establishment.

Figure VI: Administrative and Legal Hierarchy of Institutions over Water Rights Issues

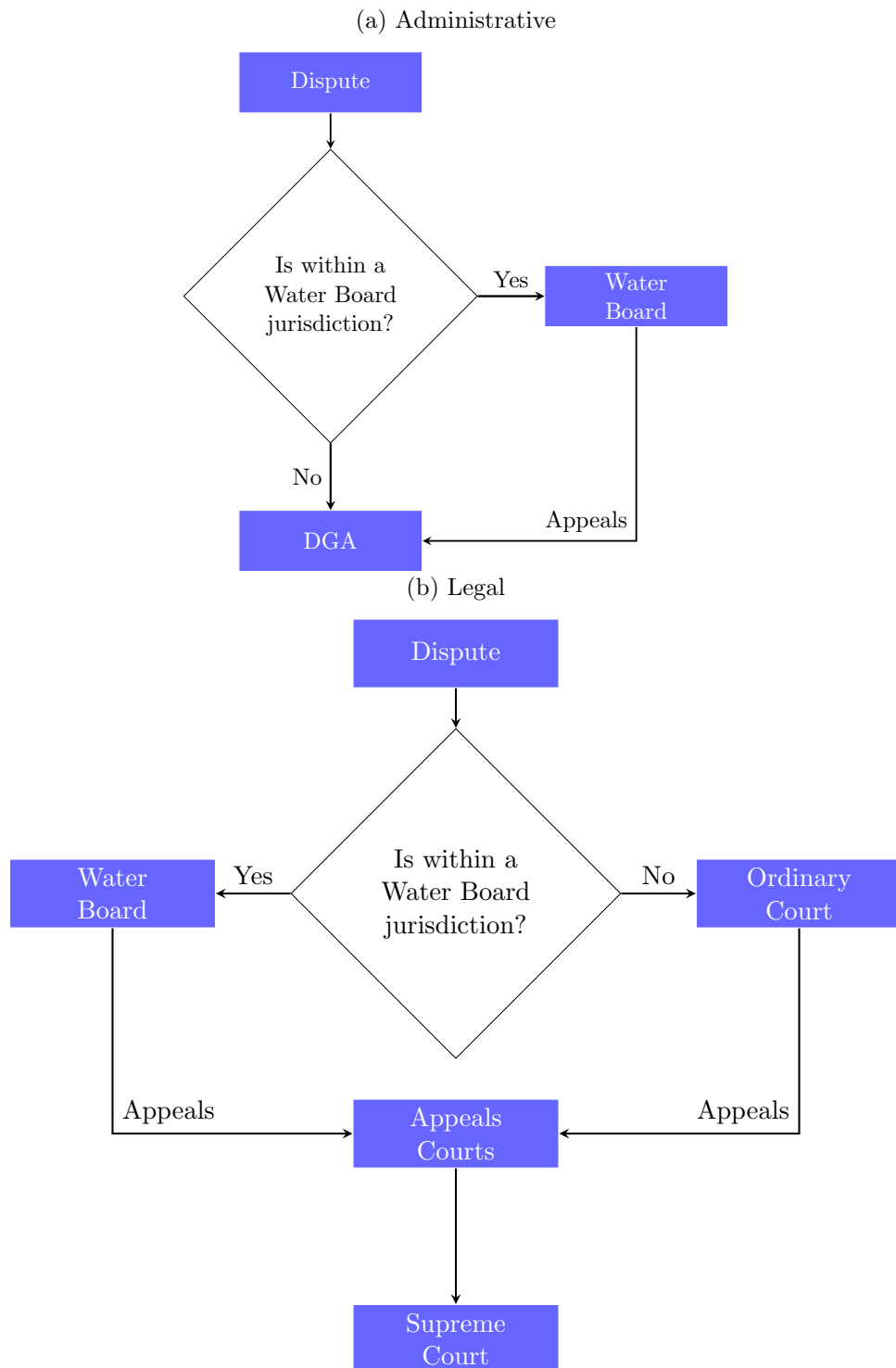
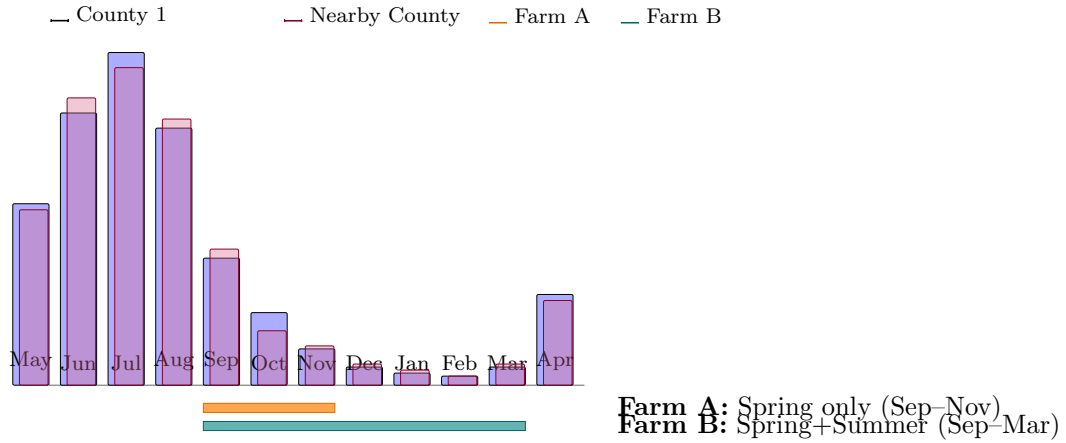


Figure VII: Data Sources and Corresponding Analyses

Data Sources	Analysis (Section)	Unit
2007 Agricultural Census	Misallocation Test (Section 4)	Farm Level
Land Plots (SII/CIREN) Satellite Data (LANDSAT 5) Fruit Cadaster (CIREN/ODEPA)	DID & IV Analysis (Section 5)	Plot Level
Streamflow Data (DGA)	DID Analysis (Section 5)	Basin Level

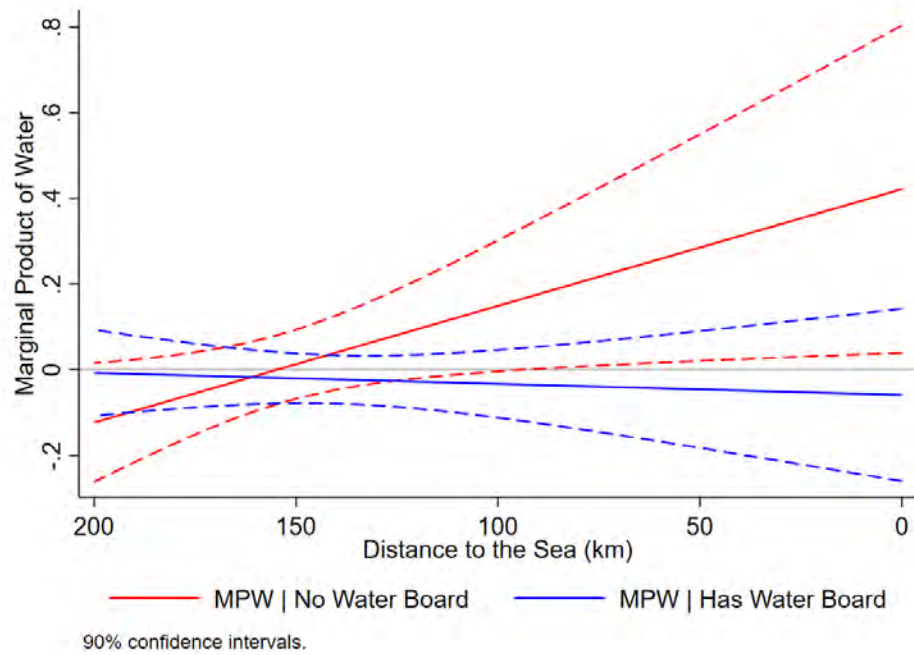
Notes: All analyses use water organization data (DGA), as well as climate data and basin boundaries from CR², as common inputs. The table presents additional data sources, along with their corresponding analyses and units of observation.

Figure VIII: Construction of Useful Rainfall



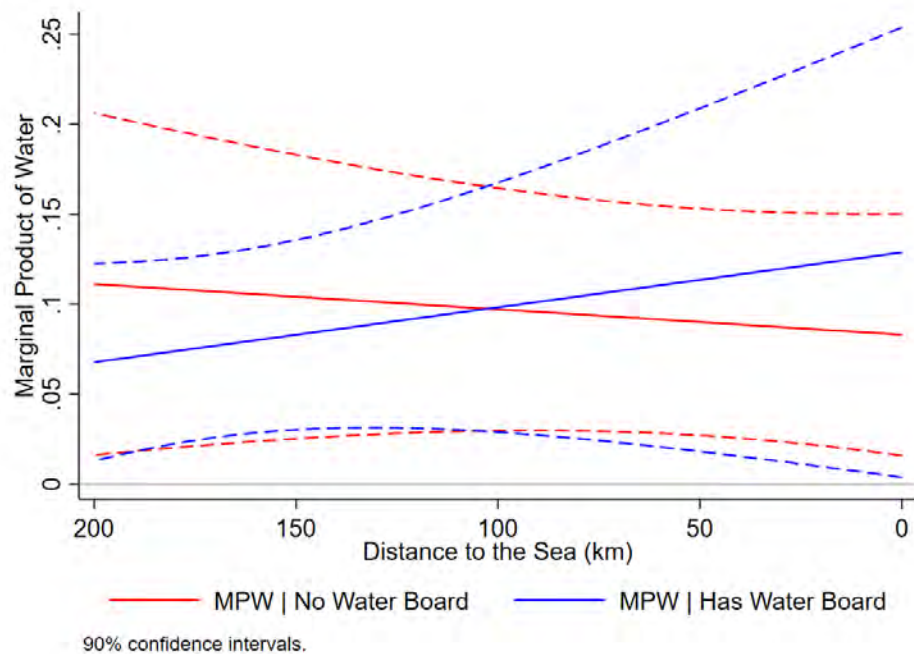
Notes: This figure illustrates the construction of our measure of useful rainfall, and the sources of variation. We define useful rainfall as the rainfall falling in a farmer’s county when their crop needs irrigation. In this example, Farm A and Farm B are located in County 1, with rainfall over the agricultural year depicted with the blue bars. While Farm A chose a crop that follows the yellow irrigation schedule, Farm B chose one that follows the green schedule. Therefore, all the rainfall received by both farms in the months of December, January and February is useful for farm A but not farm B. Another source of variation is provided by the space: consider another farm following the same irrigation schedules as farm A, but located in a nearby county, facing rainfall illustrated in red bars. That farm will have a marginally different water input than A, given the different rainfall received on each month in their county.

Figure IX: Main Results: Effect on $\log(\text{Production})$ of Rainfall during the Irrigation Season by Longitude and Treatment Status for Irrigated Parcels Registered in Canal



Notes: Graphical representation of results in Table II.

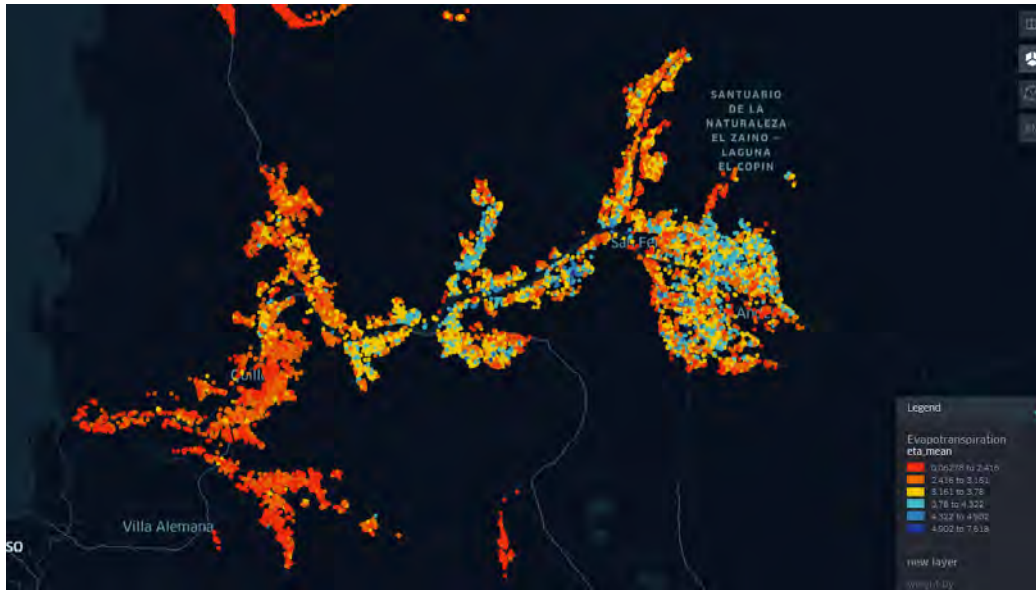
Figure X: Placebo Exercise: Effect on $\log(\text{Production})$ of Rainfall during the Irrigation Season by Longitude and Treatment Status for Rainfed Parcels



Notes: Graphical representation of results in Table III.

Figure XI: Example: Water Consumption and Agricultural Yield Estimates for Farms in Aconcagua Basin

(a) Water Consumption

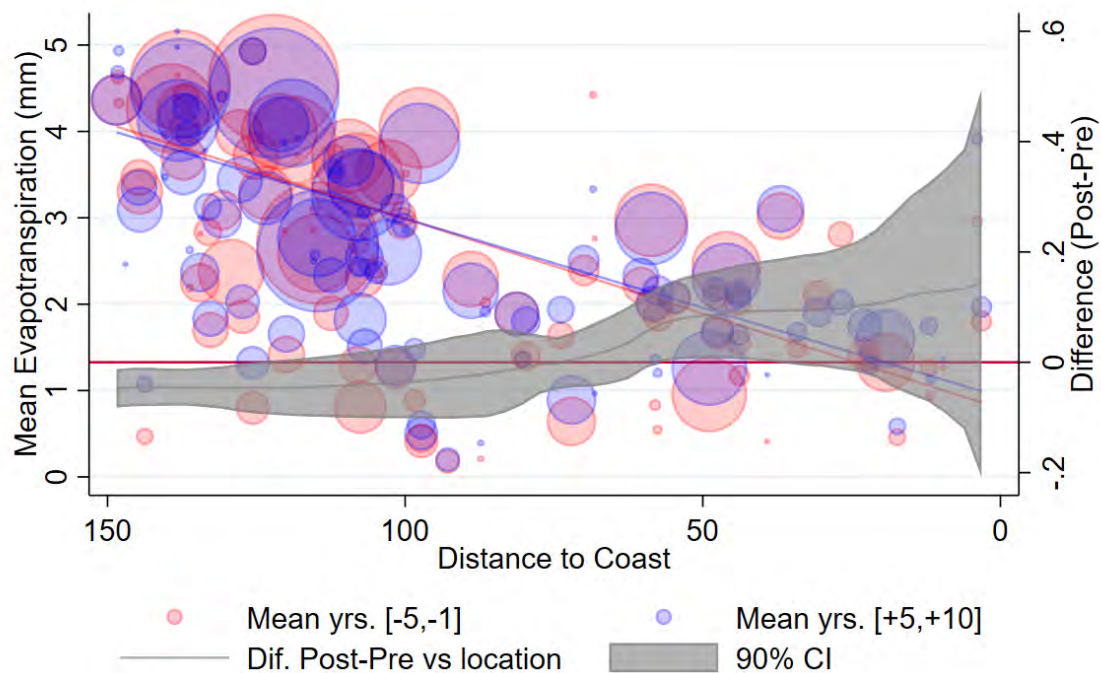


(b) Agricultural Yield



Notes: Panel (a) shows estimated water consumption (evapotranspiration) and panel (b) shows agricultural yield (using Enhanced Vegetation Index) for farms in the Aconcagua Basin.

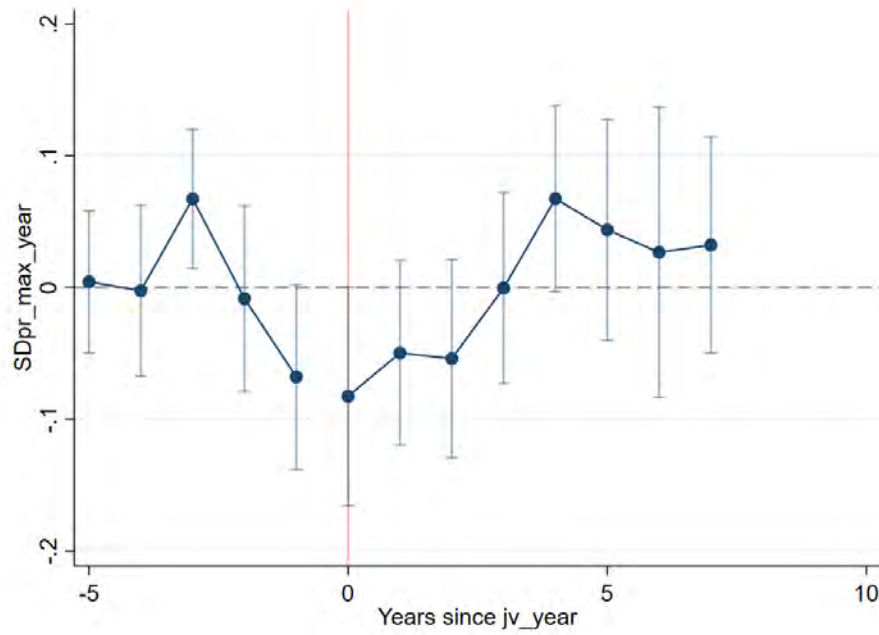
Figure XII: Average evapotranspiration for treated farms, conditional on position in the basin, before and after board establishment.



Mean by 3rd-level basin. Size of circle reflects number of plots.
Local linear regression for subsubbasins adopting boards before 2010,
weighted by number of plots.

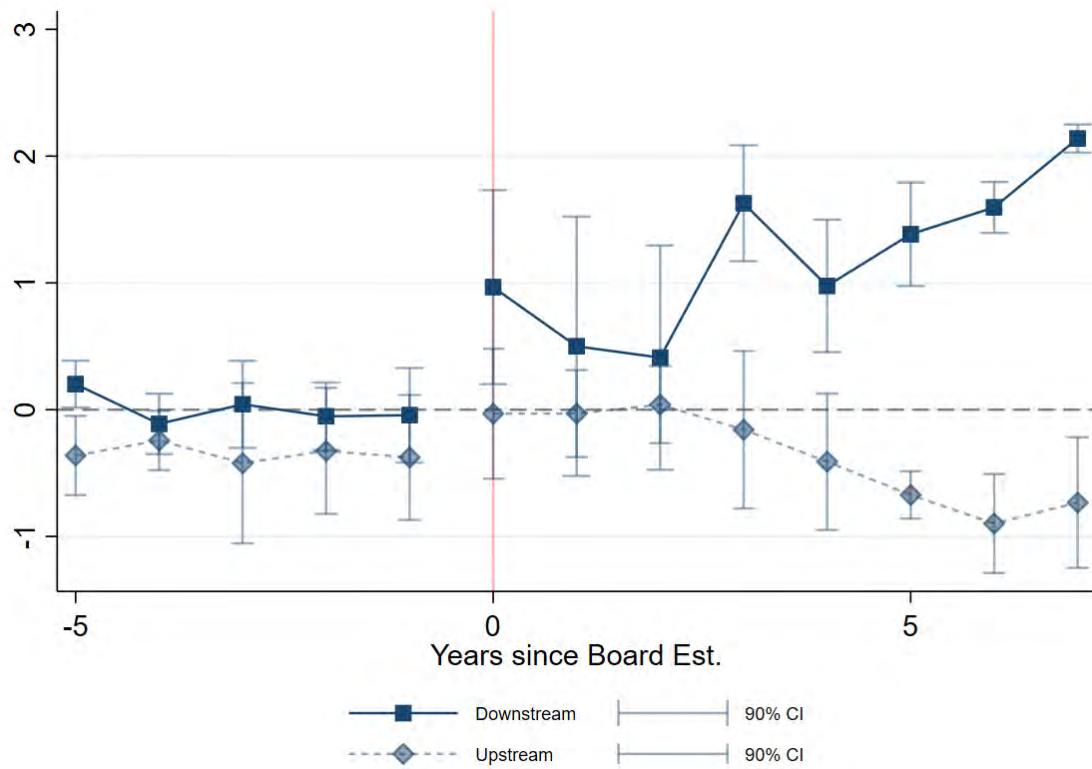
Notes: This figure depicts the average evapotranspiration as a function of the position in the basin, measured as the distance to the coast. Each circle represents one third-level basin, with size proportional to the number of plots it contains. In red, we present the average evapotranspiration over the 5 years prior to board establishment, while in blue, we present the average between 5 and 10 years after establishment. The red (blue) solid line represents the fitted regression line of average evapotranspiration in the pre (post) period on distance to the coast. The gray line is a local polynomial regression of the difference between the before and after averages for each third-level basin, with the gray translucent area depicting 90% confidence intervals.

Figure XIII: Precipitation dynamics around Water Board events



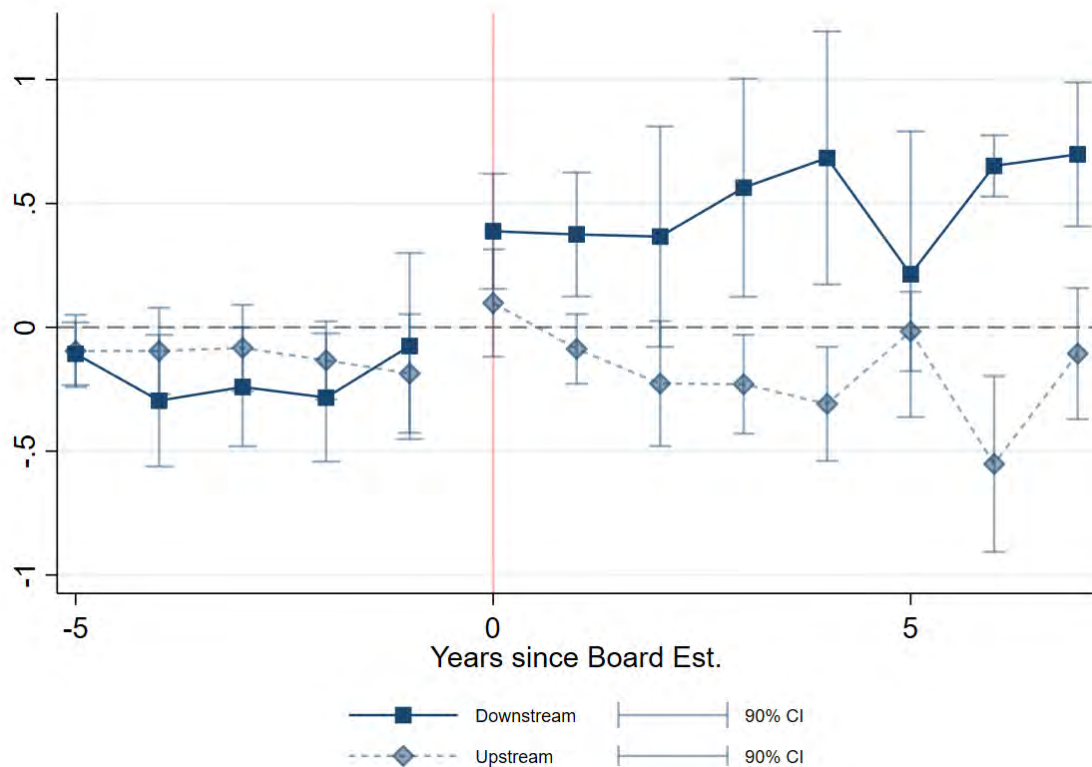
Notes: This figure presents the dynamics of standardized precipitation around Water Board establishment events, using the staggered difference-in-differences imputation estimator by Borusyak et al. (2024). The x-axis shows years relative to the establishment year, with year 0 indicating the establishment year. The sample includes farms located in basins with Water Boards established between 1981-2010, covering 5 years pre- and 7 years post-establishment. Controlling for plot-level fixed effects, and flexible time trends by latitude-longitude cells ($2^\circ \times 2^\circ$) and quantile of location in the basin. Standard errors estimated using Borusyak et al. (2024) leave-one-out procedure, clustering at the 3rd level basin.

Figure XIV: Differences-in-Differences estimates of Impacts of Water Boards on Standardized Water Consumption.



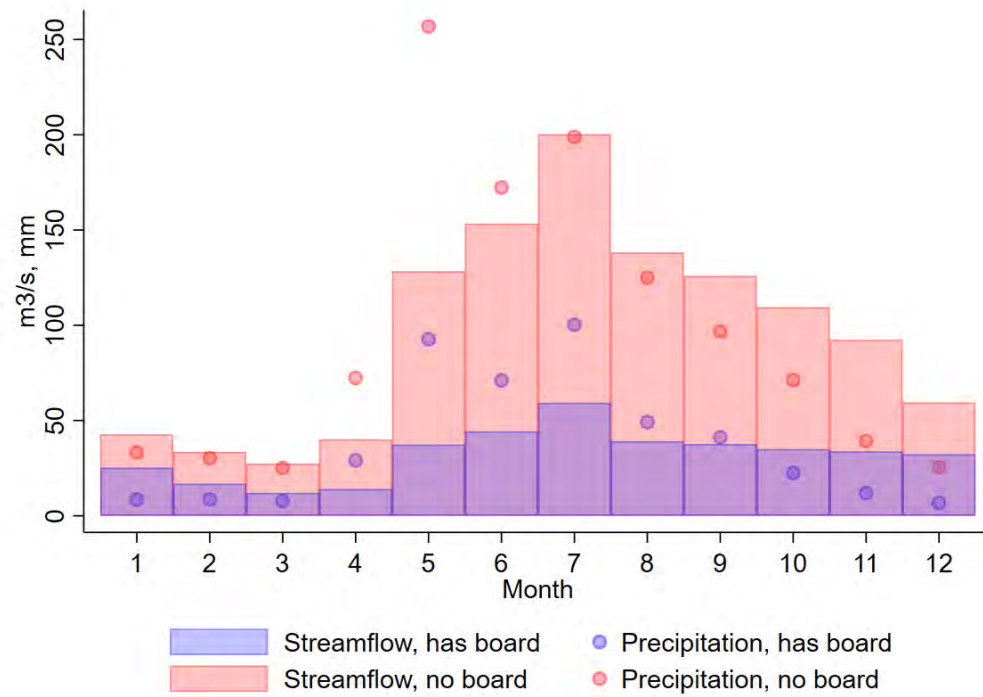
Notes: This figure presents difference-in-differences estimates of the impact of Water Board establishment on standardized summer evapotranspiration, using the staggered difference-in-differences imputation estimator by Borusyak et al. (2024). We estimate heterogeneous effects by location within the river basin. The solid blue line with squares represents downstream farms and the dotted line with diamonds represents upstream farms; mid-basin farms are reported separately in the Appendix. The x-axis shows years relative to Water Board establishment, with year 0 marking the establishment year. The sample includes farms located in basins with Water Boards established between 1981-2010, covering 5 years pre- and 7 years post-establishment. Controlling for plot-level fixed effects and flexible time trends by latitude-longitude cells ($2^\circ \times 2^\circ$). Standard errors estimated using Borusyak et al. (2024) leave-one-out procedure, clustering at the 3rd level basin.

Figure XV: Differences-in-Differences estimates of Impacts of Water Boards on Standardized Agricultural Yield.



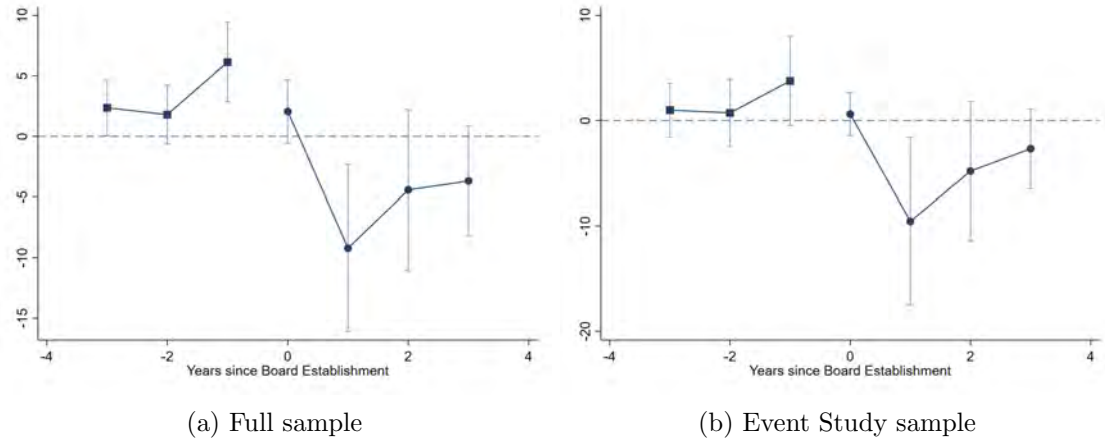
Notes: This figure presents difference-in-differences estimates of the impact of Water Board establishment on standardized agricultural yield (peak EVI over the season), using the staggered difference-in-differences imputation estimator by Borusyak et al. (2024). We estimate heterogeneous effects by location within the river basin. The solid blue line with squares represents downstream farms and the dotted line with diamonds represents upstream farms; mid-basin farms are reported separately in the Appendix. The x-axis shows years relative to Water Board establishment, with year 0 marking the establishment year. The sample includes farms located in basins with Water Boards established between 1981-2010, covering 5 years pre- and 7 years post-establishment. Controlling for plot-level fixed effects and flexible time trends by latitude-longitude cells ($2^\circ \times 2^\circ$). Standard errors estimated using Borusyak et al. (2024) leave-one-out procedure, clustering at the 3rd level basin.

Figure XVI: Streamflow and Precipitation Within and Outside Water Board Jurisdictions, Pre-1985



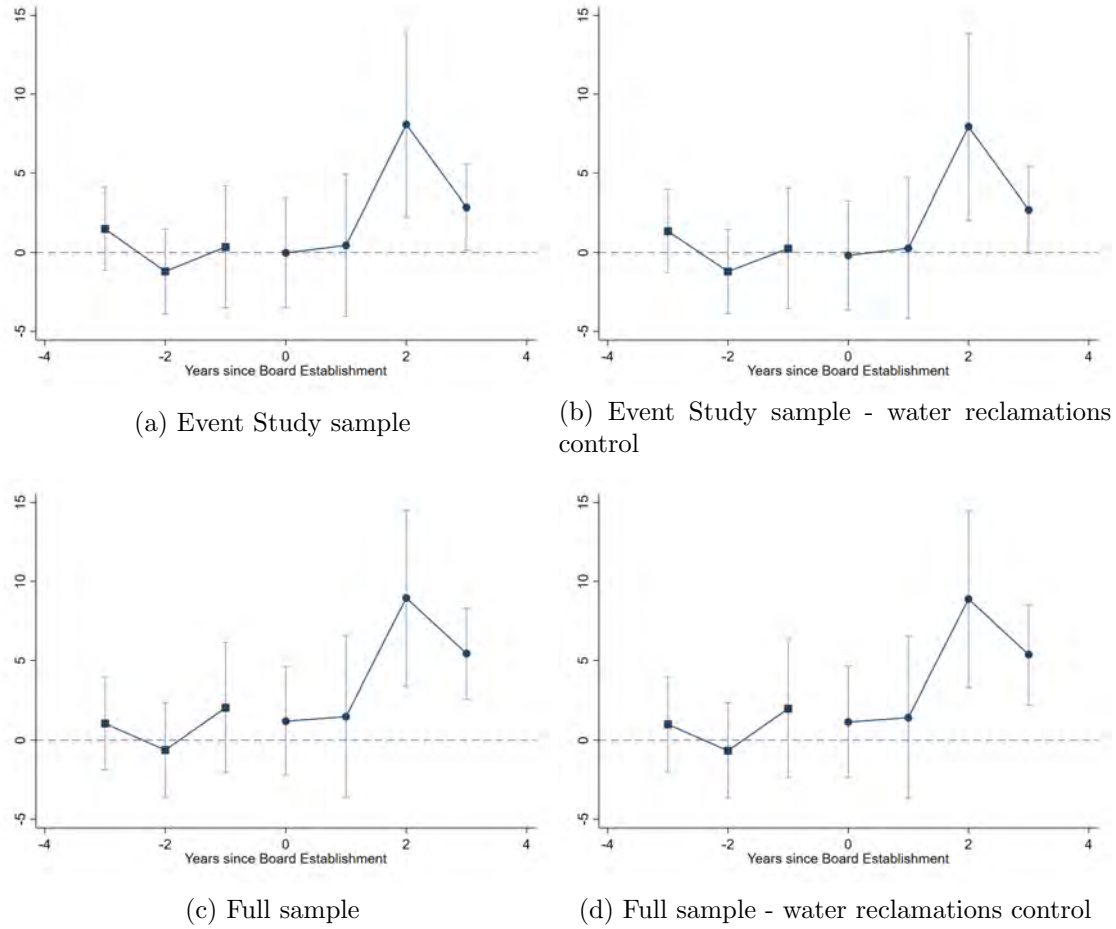
Notes: Average streamflow and precipitation in monitoring stations within and outside Water Board jurisdictions before 1985. Source: Authors' calculation.

Figure XVII: Precipitations and Board Establishment Events (Placebo) Event



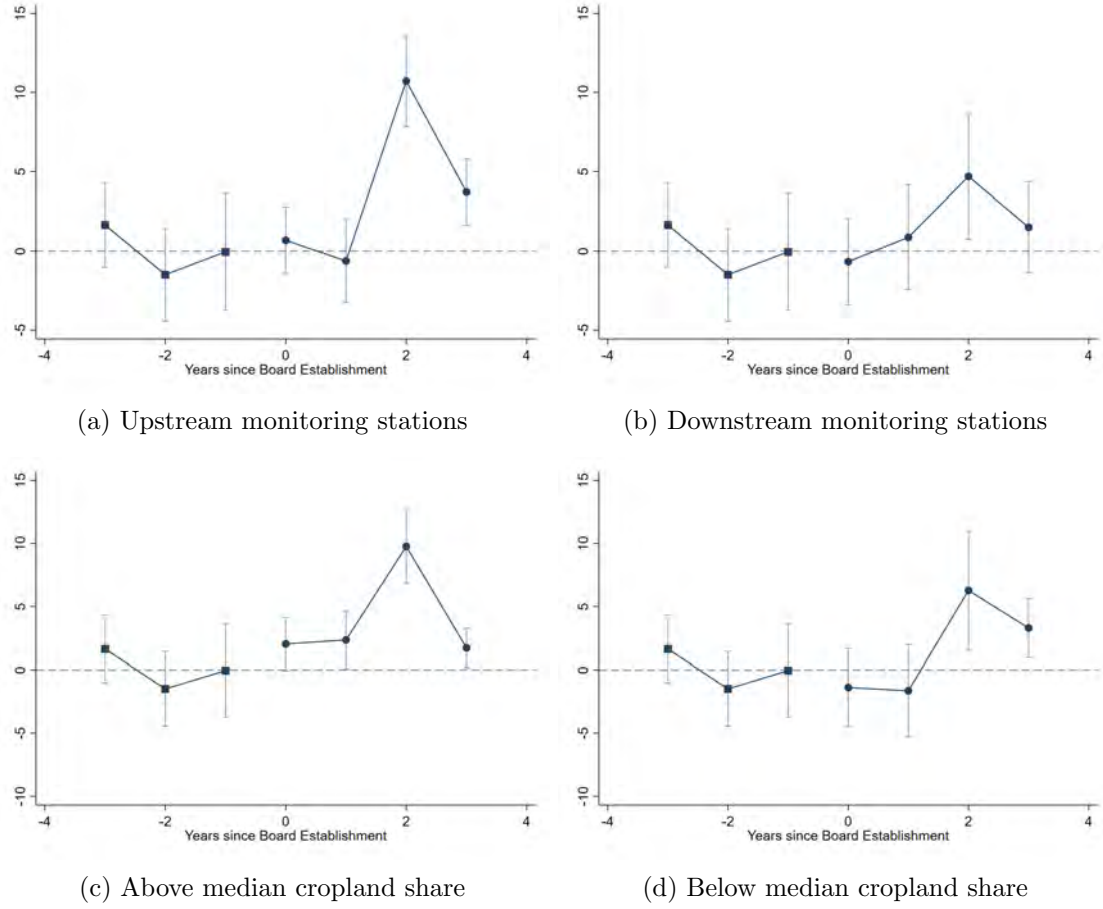
Notes: This figure presents the dynamics of precipitation around Water Board establishment events, using the Event Study estimator by Borusyak, Jaravel and Spiess (2024). The Full Sample correspond to monitoring stations located in the Study Region, while the Event Study sample only includes eventually treated monitoring stations (i.e. monitoring stations located in subsubbasins within Water Boards jurisdictions). Controlling for monitoring station-month and year fixed effects. Standard errors estimated using Borusyak, Jaravel and Spiess (2024) leave-one-out procedure, clustering at the hydrological basin level and allowing for effects heterogeneity based on 4 groups by longitudinal distance to the coast and agricultural land share in the drainage area of each monitoring station.

Figure XVIII: Dynamic Effects of Board Establishment Events on Dry Season Streamflow



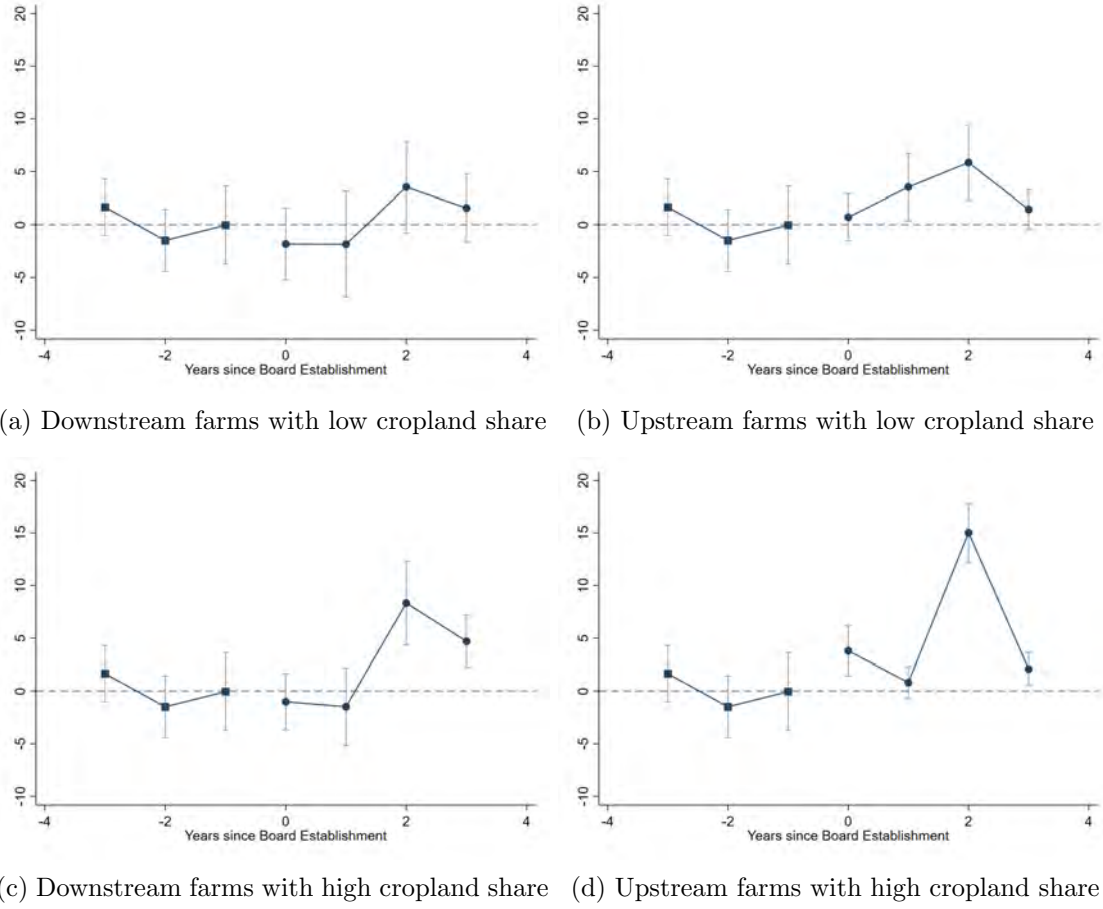
Notes: This figure presents our main estimates of the impact of Water Board establishment events on streamflows, using the Event Study estimator by Borusyak, Jaravel and Spiess (2024). The Full Sample correspond to monitoring stations located in the Study Region, while the Event Study sample only includes eventually treated monitoring stations (i.e. monitoring stations located in subsubbasins within Water Boards jurisdictions). Controlling for monthly climatic measures in the drainage area of each monitoring station (linear and squared terms for precipitation and average maximum temperature, average minimum temperature, and potential evapotranspiration) and monitoring station-month and year fixed effects. Standard errors estimated using Borusyak, Jaravel and Spiess (2024) leave-one-out procedure, clustering at the hydrological basin level and allowing for effects heterogeneity based on 4 groups by longitudinal distance to the coast and agricultural land share in the drainage area of each monitoring station.

Figure XIX: Heterogeneous Effects on Dry Season Streamflow by Location and Cropland Share



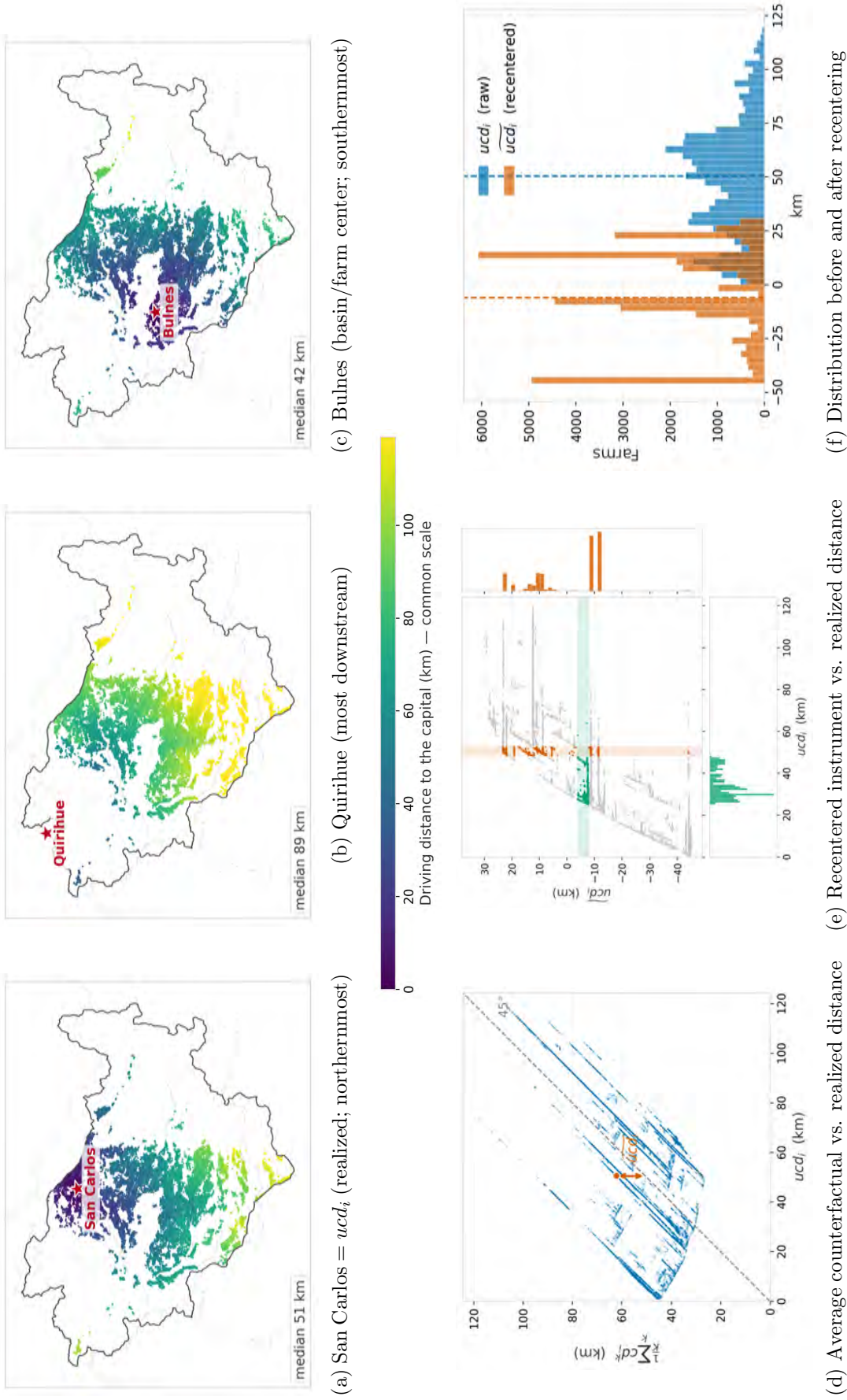
Notes: This figure presents our heterogeneous effects estimates of the impact of Water Board establishment events on streamflows, using the Event Study estimator by Borusyak, Jaravel and Spiess (2024). All estimates correspond to the Event Study sample (eventually treated monitoring stations, i.e. monitoring stations located in subsubbasins within Water Boards jurisdictions). Upper panels present heterogeneous impacts according to longitudinal distance to the coast: upstream (Panel a) versus downstream (Panel b) monitoring stations. Lower panels present heterogeneous impacts according to share of cropland in the drainage area: monitoring stations with above median cropland share (Panel c) versus those with below median cropland share (Panel d). Controlling for monthly climatic measures in the drainage area of each monitoring station (linear and squared terms for precipitation and average maximum temperature, average minimum temperature, and potential evapotranspiration) and monitoring station-month and year fixed effects. Standard errors estimated using Borusyak, Jaravel and Spiess (2024) leave-one-out procedure, clustering at the hydrological basin level and allowing for effects heterogeneity based on 4 groups by longitudinal distance to the coast and agricultural land share in the drainage area of each monitoring station.

Figure XX: Heterogeneous Effects on Dry Season Streamflow by Location-Cropland Combinations



Notes: This figure presents our heterogeneous effects estimates of the impact of Water Board establishment events on streamflows, using the Event Study estimator by Borusyak, Jaravel and Spiess (2024). All estimates correspond to the Event Study sample (eventually treated monitoring stations, i.e. monitoring stations located in subsubbasins within Water Boards jurisdictions). The panels show the interaction between geographical location and agricultural intensity: downstream farms with low cropland share (Panel a), upstream farms with low cropland share (Panel B), downstream farms with high cropland share (Panel c), and upstream farms with high cropland share (Panel d). Controlling for monthly climatic measures in the drainage area of each monitoring station (linear and squared terms for precipitation and average maximum temperature, average minimum temperature, and potential evapotranspiration) and monitoring station-month and year fixed effects. Standard errors estimated using Borusyak, Jaravel and Spiess (2024) leave-one-out procedure, clustering at the hydrological basin level and allowing for effects heterogeneity based on 4 groups by longitudinal distance to the coast and agricultural land share in the drainage area of each monitoring station.

Figure XXI: Example of Construction of the Recentered Instrument: the Itata Basin



Notes: Construction of the recentered instrument in the Itata basin (34,537 irrigated plots; three candidate provincial capitals; ucd_i is the driving distance to San Carlos). Panels (a)–(c) show the driving distance to each candidate capital under the realized and counterfactual jurisdictional rules, on the common scale shown below the maps. Panel (d) plots ucd_i on the horizontal axis against the average counterfactual distance; the vertical gap to the 45° line is the recentered instrument $\tilde{ucd}_i = ucd_i - \frac{1}{\#K} \sum_{k \in K} cd_i^k$ (the two are correlated at 0.529). Panel (e) plots the recentered instrument against the realized distance. Panel (f) shows the instrument before and after recentering (standard deviation 26.3 $\rightarrow$ 22.4 km).

A Appendix

Table A.I: Irrigation Calendar, by Crop.

Irrigated crops	Crop Needs Irrigation in Month											
	J	F	M	A	M	J	J	A	S	O	N	D
Wheat						x	x	x	x	x	x	x
Rice	x	x	x								x	x
Maize	x	x	x								x	x
Barley						x	x	x	x	x	x	
Other cereals	x	x	x								x	x
Vegetables	x	x	x								x	x
Fruits	x	x	x	x	x	x	x	x	x	x	x	x
Grapes	x	x	x	x	x	x	x	x	x	x	x	x
Citrus	x	x	x	x	x	x	x	x	x	x	x	x
Oil crops	x	x	x								x	x
Potatoes	x	x	x								x	x
Pulses	x	x	x								x	x
Sugar beet	x	x	x							x	x	x
Fodder temporary						x	x	x	x	x	x	
Tobacco	x	x	x								x	x
Pasture permanent	x	x	x	x	x	x	x	x	x	x	x	x

Source: FAO, INE.

Table A.II: Misallocation Test: Poisson Estimates

	Outcome is Value Yield p/Hectare			
	(1)	(2)	(3)	(4)
Useful pp. (ML per Ha per month)	0.605 (0.242)** [0.242]**	0.624 (0.252)** [0.249]**	0.693 (0.300)** [0.293]**	0.659 (0.326)** [0.321]**
Useful pp. (ML per Ha per month) $\times$ Distance to coast (100km)	-0.332 (0.174)* [0.174]*	-0.364 (0.175)** [0.172]**	-0.377 (0.213)* [0.208]*	-0.403 (0.228)* [0.223]*
Water Board=1 $\times$ Useful pp. (ML per Ha per month)	-0.405 (0.293) [0.295]	-0.581 (0.266)** [0.267]**	-0.562 (0.324)* [0.318]*	-0.576 (0.352) [0.346]*
Water Board=1 $\times$ Useful pp. (ML per Ha per month) $\times$ Distance to coast (100km)	0.201 (0.210) [0.212]	0.355 (0.189)* [0.190]*	0.326 (0.226) [0.222]	0.319 (0.246) [0.242]
Parcel Controls	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	No
Basin FE	No	Yes	No	No
0.5 degree cell FE	No	No	Yes	No
Crop FE	Yes	Yes	Yes	Yes
County FE	No	No	No	Yes
Observations	14,838	14,838	14,836	14,834
Misallocation Test: Water Board=0	0.06	0.04	0.08	0.08
Misallocation Test: Water Board=1	0.19	0.89	0.53	0.35

Notes: This table replicates Table II, estimating equation 5 by Poisson pseudo-maximum likelihood, with the value of output per hectare in levels rather than in logs. The sample, controls and fixed effects are identical to those in Table II: irrigated parcels, with water rights, registered in canal associations. Distance to the coast measured through the river network. Controlling for logarithm of cultivated surface, logarithm of number of hired workers plus 1, dummies for educational level of manager of the farm, legal organization category of the operation, irrigation technology, and County and Crop fixed effects. Standard errors clustered at the county level (round parenthesis) and the county $\times$ irrigation season level (squared parentheses). The last two rows report the p-values of the test of equality of the marginal product of water across locations.

Table A.III: Misallocation Test: Additional robustness checks

	Baseline	Crop intercepts	Crop elasticities	Potential Confounders		
	(1)	(2)	(3)	(4)	(5)	(6)
Useful pp. (ML per Ha per month)	0.421 (0.232)* [0.232]*	0.611 (0.262)** [0.252]**		0.424 (0.233)* [0.232]*	0.424 (0.231)* [0.231]*	0.551 (0.251)** [0.254]**
Useful pp. (ML per Ha per month) $\times$ Distance to coast (100km)	-0.272 (0.148)* [0.144]*	-0.552 (0.217)** [0.196]***	-0.203 (0.110)* [0.108]*	-0.272 (0.148)* [0.144]*	-0.269 (0.147)* [0.143]*	-0.398 (0.173)** [0.172]**
Water Board=1 $\times$ Useful pp. (ML per Ha per month)	-0.481 (0.258)* [0.251]*	-0.559 (0.176)*** [0.195]***	-0.410 (0.223)* [0.218]*	-0.489 (0.260)* [0.252]*	-0.485 (0.256)* [0.249]*	-0.505 (0.263)* [0.254]**
Water Board=1 $\times$ Useful pp. (ML per Ha per month) $\times$ Distance to coast (100km)	0.298 (0.170)* [0.163]*	0.345 (0.127)*** [0.152]**	0.262 (0.147)* [0.141]*	0.304 (0.172)* [0.164]*	0.302 (0.168)* [0.162]*	0.318 (0.173)* [0.166]*
Crop FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Crop FE $\times$ 0.5 degree cell FE	No	Yes	No	No	No	No
Crop FE $\times$ Useful Rainfall	No	No	Yes	No	No	No
Fertirrigation $\times$ Distance $\times$ Useful Rain.	No	No	No	No	Yes	No
Groundwater Well $\times$ Distance $\times$ Useful Rain.	No	No	No	Yes	No	No
Potential Productivity $\times$ Distance $\times$ Useful Rain.	No	No	No	No	No	Yes
Observations	14,712	14,036	14,712	14,712	14,712	14,688
R-squared	0.642	0.722	0.674	0.642	0.642	0.642
Misallocation Test: Water Board=0	0.07	0.01	0.07	0.07	0.07	0.02
Misallocation Test: Water Board=1	0.76	0.34	0.52	0.71	0.69	0.45

Notes: This table presents different robustness checks, including new control variables and interactions of useful rainfall with some relevant controls. We regress $\log(Y)$ of rainfall during the irrigation season by longitude and treatment status for irrigated parcels registered in canal associations.

Table A.IV: Anticipation of Useful Rainfall: Temporary Labor Demand

	Non-Irrigation Season		Irrigation Season	
	Fall (1)	Winter (2)	Spring (3)	Summer (4)
Useful Rainfall	0.0204 (0.134)	-0.0336 (0.144)	-0.622 (0.298)**	-0.395 (0.217)*
Observations	15,446	15,573	15,829	15,828
Pseudo- R^2	0.979	0.979	0.976	0.958

Notes: This table presents the results of estimating Poisson models of temporary labor hired on each season on our measure of useful rainfall, for the main sample (described in Section 4). Useful Rainfall is measured in ML per hectare per month. All columns control for log-farm size, county and crop fixed-effects. The outcomes are temporary labor in: Fall 2006 in column 1, Winter 2006 in column 2, Spring 2006 in column 3, and Summer 2007 in column 4.

Table A.V: Robustness — Alternative Instrument: Expanded Set of Counterfactual Rules

	Fruit-tree planting				Water consumption (evapotranspiration)				Agricultural yield (peak EVI)			
	OLS		IV		OLS		IV		OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream	(5) Downstream	(6) Upstream	(7) Downstream	(8) Upstream	(9) Downstream	(10) Upstream	(11) Downstream	(12) Upstream
<i>Panel A. Main result</i>												
Board	0.0282 (0.0226)	-0.0384 (0.0219)*	0.212 (0.0832)**	-0.129 (0.112)	0.0157 (0.108)	0.0292 (0.0890)	1.969 (1.005)*	-1.302 (0.497)***	0.00572 (0.00853)	0.00890 (0.00754)	0.235 (0.101)**	-0.00978 (0.0337)
Observations	145,065	144,515	145,065	144,515	143,038	144,515	143,038	144,515	145,065	144,515	145,065	144,515
R-squared	0.079	0.048	0.040	0.034	0.301	0.359	0.061	0.186	0.244	0.208	-0.090	0.204
Mean Dependent Var.			0.080	0.096			2.952	3.407			0.370	0.410
AR test CI			[.04804, .4686]	[-.4869, .1688]			[.4711, 6.144]	[-4.316, -6.007]			[.09056, .6648]	[-.09314, .1068]
<i>Panel B. First stage: Water Board presence</i>												
Recentered distance $\tilde{ucd}_i$			-0.0478 (0.0149)***	0.0348 (0.0133)**			-0.0468 (0.0148)***	0.0348 (0.0133)**			-0.0478 (0.0149)***	0.0348 (0.0133)**
Observations			145,065	144,515			143,038	144,515			145,065	144,515
R-squared			0.552	0.362			0.555	0.362			0.552	0.362
Effective F-stat			10.303	6.865			9.979	6.865			10.303	6.865

Notes: Estimates replicate the main specification using an instrument recentered over an expanded set of counterfactual jurisdictional rules, which adds the capital of the province where each farm is located to the five rules listed in Section 5.3. Columns 1–4 report fruit-tree planting, columns 5–8 water consumption, and columns 9–12 agricultural yield; within each outcome, the first two columns report OLS and the last two IV estimates, for downstream and upstream farms. The corresponding benchmark estimates are reported in the main text. Panel B reports the first stage for this alternative instrument, which is the only place in the paper where it is shown. Estimates are less precise under the expanded rule set, as recentering over an additional rule absorbs part of the instrument's variation; point estimates are unchanged in sign and magnitude, and Anderson–Rubin confidence intervals continue to exclude zero for downstream farms. Standard errors are clustered by county; Anderson–Rubin confidence intervals are reported for the IV estimates.

Table A.VI: Robustness — Excluding the Maule Region

	Any fruit tree		Water-intensive species		Water-saving species		Micro-irrigation		Traditional irrigation	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Downstream	Upstream	Downstream	Upstream	Downstream	Upstream	Downstream	Upstream	Downstream	Upstream
<i>Panel A. IV estimates</i>										
Board	0.218 (0.0760)***	-0.145 (0.191)	0.192 (0.0712)***	-0.118 (0.132)	0.0262 (0.0118)**	-0.0273 (0.0763)	0.174 (0.0677)***	0.0463 (0.0658)	0.0456 (0.0211)**	-0.191 (0.182)
Observations	116,595	106,615	116,595	106,615	116,595	106,615	116,595	106,615	116,595	106,615
AR test CI	[.05963, .4637]	$(-\infty, \infty)$	[.05013, .433]	$(-\infty, \infty)$	[-.01108, .05125]	$(-\infty, \infty)$	[.03729, .3993]	$(-\infty, \infty)$	[-.007585, .1021]	$(-\infty, \infty)$
<i>Panel B. First stage: Water Board presence</i>										
Recentered distance $\tilde{ucd}_i$	-0.0853 (0.0287)***	0.0247 (0.0159)	-0.0853 (0.0287)***	0.0247 (0.0159)	-0.0853 (0.0287)***	0.0247 (0.0159)	-0.0853 (0.0287)***	0.0247 (0.0159)	-0.0853 (0.0287)***	0.0247 (0.0159)
Observations	116,595	106,615	116,595	106,615	116,595	106,615	116,595	106,615	116,595	106,615
R-squared	0.595	0.369	0.595	0.369	0.595	0.369	0.595	0.369	0.595	0.369
Effective F-stat	8.801	2.428	8.801	2.428	8.801	2.428	8.801	2.428	8.801	2.428

Notes: Estimates replicate the main specification excluding the Maule region, the region for which the fruit census wave used extends to 2013. Panel A reports IV estimates for the five outcomes, each for downstream and upstream farms; Panel B reports the corresponding first stage, which is common to all five columns because the estimation sample is identical across outcomes. Downstream, the estimates are close to the benchmark and the Anderson–Rubin intervals continue to exclude zero. Upstream, the instrument loses power once the Maule is dropped—the effective F-statistic falls to 2.4—and the Anderson–Rubin intervals are unbounded, so the upstream estimates are uninformative in this subsample. Standard errors are clustered by county.

Table A.VII: Balance: Instrument and Plot-Level Covariates

	(1) Downstream	(2) Upstream
Climatic covariates		
Mean summer prec.	7.6 (5.9)	-.84 (3.2)
Mean winter and spring prec.	-3 (31)	-4.6 (20)
Mean annual prec.	7.3 (38)	-4.8 (25)
Days between 25 and 29 °C	-.53 (.52)	-.37 (.54)
Days between 29 and 31 °C	-.18 (.24)	-.3 (.29)
Days between 31 and 35 °C	-.12 (.26)	-.41 (.27)
Days above °C	-.048 (.036)	-.035 (.029)
Economic covariates		
Max. potential yield value per hectare (USD/ha)	-553 (355)	-16 (34)
Aver. potential yield value per hectare (USD/ha)	-142 (97)	-4.2 (8)
Distance to Santiago (main dom. market)	2.3 (2)	.57 (3.1)
Distance to Valparaiso (main port)	-2.9 (2.3)	-1.1 (3.1)
Distance to San Antonio (second main port)	.72 (2.2)	-1.9 (3.2)

Notes: This table reports coefficients from regressions of the recentered instrument ucd_i , defined in equation 11, on plot-level covariates, for parcels located at elevations below 1,000 meters and within 150 km of the river mouth in the study area. Columns present estimates for downstream farms (in the lowest tercile of distance to the coast) and upstream farms, respectively. The covariates are plot-level climatic and economic characteristics: mean annual, winter–spring, and summer precipitation; the number of days in each maximum-temperature bin; the maximum and average potential yield value per hectare; and market access (driving distance to Santiago and to the main ports of Valparaíso and San Antonio). Standard errors are clustered by county.

Table A.VIII: First Stage: Water Board Presence and the Recentered Instrument

	First stage: Water Board		
	(1) Downstream	(2) Mid section	(3) Upstream
Recentered distance $\tilde{ucd}_i$	-0.0595 (0.0174)***	0.00427 (0.0189)	0.0380 (0.0145)**
Observations	145,065	144,910	144,515
R-squared	0.553	0.315	0.358

Notes: First-stage regression of Water Board presence on the recentered instrument $\tilde{ucd}_i$, by tercile of distance to the river mouth. The coefficient enters with opposite signs downstream and upstream and is small and statistically insignificant in the mid-section. This is the only table that reports the mid-section tercile: the outcome tables report the downstream and upstream terciles only, precisely because the first stage is not monotonic across the three. Standard errors are clustered by county.

Table A.IX: Crop Choice: Instrumental Variables Estimation at the Parcel Level

	Season length (months)				Summer peak (peak after December)			
	OLS		IV		OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream	(5) Downstream	(6) Upstream	(7) Downstream	(8) Upstream
<i>Panel A. Main result</i>								
Board	0.137 (0.0794)*	-0.0283 (0.0503)	0.724 (0.337)**	-0.0787 (0.223)	0.0467 (0.0286)	-0.00679 (0.0253)	0.234 (0.127)*	0.0573 (0.120)
Observations	145,065	144,515	145,065	144,515	145,065	144,515	145,065	144,515
R-squared	0.057	0.121	0.031	0.120	0.056	0.121	0.039	0.118
Mean Dependent Var.			6.924	7.060			0.280	0.353
AR test CI			[.05146, 1.684]	[-.606, .7224]			[-.03081, .5833]	[-.1817, .566]
<i>Panel B. First stage: Water Board presence</i>								
Recentered distance $\tilde{ucd}_i$			-0.0595 (0.0174)***	0.0380 (0.0145)**			-0.0595 (0.0174)***	0.0380 (0.0145)**
Observations			145,065	144,515			145,065	144,515
R-squared			0.553	0.358			0.553	0.358
Effective F-stat			11.646	6.915			11.646	6.915

Notes: This table presents estimates of equation 11 for parcels located at elevations below 1,000 meters and within 150 km of the river mouth in the study area. The outcome variables are the number of months between the start of the season and the month of peak Enhanced Vegetation Index (EVI) (columns 1–4, season length), and a dummy equal to one if the EVI peak occurs in the Summer months (columns 5–8, summer peak). Within each outcome, the first two columns report OLS and the last two IV estimates, and within each pair the first column presents estimates for downstream farms (in the lowest tercile of distance to the coast) and the second for upstream farms. Panel B reports the corresponding first stage, a regression of Water Board presence on the recentered instrument $\tilde{ucd}_i$, and is therefore reported only for the IV columns. Plot-level controls include market access measures (driving distance to Santiago and the main ports of Valparaíso and San Antonio); annual, combined winter and spring, and summer precipitation; and temperature, measured as the number of days with maximum temperatures in the ranges [25, 29], [29, 31], [31, 35], and [35, ∞^+) in Celsius degrees. All regressions include $1^\circ \times 1^\circ$ cell fixed effects and year fixed effects. Standard errors are clustered by county.

Table A.X: Randomization Inference for the Main Estimates

	(1) Downstream	(2) Upstream
Outcomes		
Evapotranspiration (mm)	1.684 (2.00) [-1.87, 1.92]** [-2.50, 2.52] [-1.89, 1.94]*	-1.297 (-2.70) [-2.46, 2.77]** [-3.64, 3.56] [-2.81, 2.55]
Peak EVI	0.197 (2.33) [-1.98, 2.06]*** [-2.76, 2.82] [-2.03, 2.26]*	-0.005 (-0.14) [-0.14, 0.14] [-0.14, 0.14] [-0.15, 0.15]
Has fruit trees	0.203 (2.43) [-1.60, 1.67]*** [-1.81, 1.79]*** [-1.85, 1.82]***	-0.079 (-0.69) [-0.81, 0.75] [-0.72, 0.72] [-0.69, 0.72]

Notes: This table reports randomization inference for the main IV estimates on the three key outcomes: water consumption (average summer Evapotranspiration, in mm), agricultural productivity (peak EVI), and fruit tree planting (a dummy for whether the parcel has fruit trees). Column 1 reports downstream parcels and column 2 upstream parcels. For each outcome, the first row reports the IV point estimate with its t -statistic in round parentheses; the three rows below report the 5th–95th percentile range of the placebo t -statistic distribution when treatment is randomized at the basin, sub-basin, and sub-sub-basin level, respectively. Stars denote that the observed t -statistic falls outside the placebo distribution at the 10% (*), 5% (**), and 1% (***) two-sided levels.

Table A.XI: Robustness — Fruit-Tree Planting: Alternative Control Sets

	Benchmark				+ Soil quality and groundwater rights				+ Land potential productivity			
	OLS		IV		OLS		IV		OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream	(5) Downstream	(6) Upstream	(7) Downstream	(8) Upstream	(9) Downstream	(10) Upstream	(11) Downstream	(12) Upstream
<i>Panel A. Main result</i>												
Board	0.0282 (0.0226)	-0.0384 (0.0219)*	0.203 (0.0832)**	-0.0785 (0.114)	0.0190 (0.0211)	-0.0344 (0.0215)	0.190 (0.0773)**	-0.0550 (0.117)	0.0283 (0.0226)	-0.0383 (0.0219)*	0.207 (0.0847)**	-0.0792 (0.113)
Observations	145,065	144,515	145,065	144,515	141,675	143,795	141,675	143,795	145,055	144,515	145,055	144,515
R-squared	0.079	0.048	0.044	0.045	0.091	0.056	0.058	0.055	0.079	0.048	0.043	0.045
Mean Dependent Var.			0.080	0.096			0.080	0.096			0.080	0.096
AR test CI			[.02974, .4307]	[-.3636, .3053]			[.03632, .4002]	[-.3537, .392]			[.02966, .439]	[-.3629, .299]
<i>Panel B. First stage: Water Board presence</i>												
Recentered distance $\tilde{ucd}_i$			-0.0595 (0.0174)***	0.0380 (0.0145)**			-0.0631 (0.0174)***	0.0368 (0.0148)**			-0.0590 (0.0174)***	0.0381 (0.0144)**
Observations			145,065	144,515			141,675	143,795			145,055	144,515
R-squared			0.553	0.358			0.541	0.358			0.553	0.358
Effective F-stat			11.646	6.915			13.122	6.213			11.476	6.965

Notes: Effect of Water Board presence on fruit-tree planting under three control sets. Columns 1–4 report the benchmark specification used in the main text; columns 5–8 add soil-quality quartiles and groundwater rights; columns 9–12 add the potential land-productivity value as a control. Within each control set, the first two columns report OLS and the last two IV estimates, and within each pair the first column presents estimates for downstream farms and the second for upstream farms. Panel B reports the corresponding first stage, a regression of Water Board presence on the recentered instrument $\tilde{ucd}_i$, and is therefore reported only for the IV columns. Standard errors are clustered by county; Anderson–Rubin confidence intervals are reported for the IV estimates.

Table A.XII: Robustness — Growing-Season Length: Alternative Control Sets

	Benchmark				+ Soil quality and groundwater rights				+ Land potential productivity			
	OLS		IV		OLS		IV		OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream	(5) Downstream	(6) Upstream	(7) Downstream	(8) Upstream	(9) Downstream	(10) Upstream	(11) Downstream	(12) Upstream
<i>Panel A. Main result</i>												
Board	0.137 (0.0794)*	-0.0283 (0.0503)	0.724 (0.337)**	-0.0787 (0.223)	0.105 (0.0620)*	-0.00977 (0.0463)	0.560 (0.291)*	0.0532 (0.236)	0.132 (0.0791)*	-0.0278 (0.0502)	0.691 (0.333)**	-0.0815 (0.222)
Observations	145,065	144,515	145,065	144,515	141,675	143,795	141,675	143,795	145,055	144,515	145,055	144,515
R-squared	0.057	0.121	0.031	0.120	0.078	0.131	0.063	0.130	0.058	0.121	0.035	0.120
Mean Dependent Var.			6.924	7.060			6.924	7.060			6.924	7.060
AR test CI			[.05146, 1.684]	[-.606, .7224]			[-.03811, 1.326]	[-.4342, 1.176]			[.02399, 1.645]	[-.6109, .703]
<i>Panel B. First stage: Water Board presence</i>												
Recentered distance ucd_i			-0.0595 (0.0174)***	0.0380 (0.0145)**			-0.0631 (0.0174)***	0.0368 (0.0148)**			-0.0590 (0.0174)***	0.0381 (0.0144)**
Observations			145,065	144,515			141,675	143,795			145,055	144,515
R-squared			0.553	0.358			0.541	0.358			0.553	0.358
Effective F-stat			11.646	6.915			13.122	6.213			11.476	6.965

Notes: Effect of Water Board presence on growing-season length under three control sets. Columns 1–4 report the benchmark used in the main text; columns 5–8 add soil-quality quartiles and groundwater rights; columns 9–12 add the potential land-productivity value as a control. Within each control set, the first two columns report OLS and the last two IV estimates, and within each pair the first column presents estimates for downstream farms and the second for upstream farms. Panel B reports the corresponding first stage, a regression of Water Board presence on the recentered instrument ucd_i , and is therefore reported only for the IV columns. Standard errors clustered by county; Anderson–Rubin confidence intervals are reported for the IV estimates.

Table A.XIII: Robustness — Summer Peak (Greenness): Alternative Control Sets

	Benchmark				+ Soil quality and groundwater rights				+ Land potential productivity			
	OLS		IV		OLS		IV		OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream	(5) Downstream	(6) Upstream	(7) Downstream	(8) Upstream	(9) Downstream	(10) Upstream	(11) Downstream	(12) Upstream
<i>Panel A. Main result</i>												
Board	0.0467 (0.0286)	-0.00679 (0.0253)	0.234 (0.127)*	0.0573 (0.120)	0.0353 (0.0213)	0.00136 (0.0234)	0.169 (0.109)	0.118 (0.131)	0.0449 (0.0285)	-0.00646 (0.0253)	0.221 (0.126)*	0.0556 (0.119)
Observations	145,065	144,515	145,065	144,515	141,675	143,795	141,675	143,795	145,055	144,515	145,055	144,515
R-squared	0.056	0.121	0.039	0.118	0.076	0.129	0.067	0.121	0.057	0.121	0.042	0.119
Mean Dependent Var.			0.280	0.353			0.280	0.353			0.280	0.353
AR test CI			[-0.03081, .5833]	[-.1817, .566]			[-.06475, .4424]	[-.1125, .8474]			[-.04373, .567]	[-.1845, .5523]
<i>Panel B. First stage: Water Board presence</i>												
Recentered distance ucd_i			-0.0595 (0.0174)***	0.0380 (0.0145)**			-0.0631 (0.0174)***	0.0368 (0.0148)**			-0.0590 (0.0174)***	0.0381 (0.0144)**
Observations			145,065	144,515			141,675	143,795			145,055	144,515
R-squared			0.553	0.358			0.541	0.358			0.553	0.358
Effective F-stat			11.646	6.915			13.122	6.213			11.476	6.965

Notes: Effect of Water Board presence on the summer-peak (greenness) indicator under three control sets. Columns 1–4 report the benchmark used in the main text; columns 5–8 add soil-quality quartiles and groundwater rights; columns 9–12 add the potential land-productivity value as a control. Within each control set, the first two columns report OLS and the last two IV estimates, and within each pair the first column presents estimates for downstream farms and the second for upstream farms. Panel B reports the corresponding first stage, a regression of Water Board presence on the recentered instrument $\tilde{ucd}_i$, and is therefore reported only for the IV columns. Standard errors clustered by county; Anderson–Rubin confidence intervals are reported for the IV estimates.

Table A.XIV: Robustness — Water Consumption (Evapotranspiration): Alternative Control Sets

	Benchmark				+ Soil quality and groundwater rights				+ Land potential productivity			
	OLS		IV		OLS		IV		OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream	(5) Downstream	(6) Upstream	(7) Downstream	(8) Upstream	(9) Downstream	(10) Upstream	(11) Downstream	(12) Upstream
<i>Panel A. Main result</i>												
Board	0.0157 (0.108)	0.0292 (0.0890)	1.684 (0.837)**	-1.297 (0.478)***	-0.0210 (0.101)	0.0487 (0.0851)	1.574 (0.755)**	-1.147 (0.450)**	0.00574 (0.109)	0.0293 (0.0892)	1.622 (0.837)*	-1.296 (0.476)***
Observations	143,038	144,515	143,038	144,515	139,693	143,795	139,693	143,795	143,030	144,515	143,030	144,515
R-squared	0.301	0.359	0.126	0.187	0.298	0.371	0.131	0.232	0.305	0.359	0.140	0.187
Mean Dependent Var.			2.952	3.407			2.953	3.407			2.952	3.407
AR test CI			[.3831, 4.785]	[-4.106, -.6076]			[.3857, 4.19]	[-4.249, -.5036]			[.3176, 4.741]	[-4.07, -.6072]
<i>Panel B. First stage: Water Board presence</i>												
Recentered distance $\tilde{ucd}_i$			-0.0585 (0.0174)***	0.0380 (0.0145)**			-0.0620 (0.0173)***	0.0368 (0.0148)**			-0.0580 (0.0174)***	0.0381 (0.0144)**
Observations			143,038	144,515			139,693	143,795			143,030	144,515
R-squared			0.556	0.358			0.545	0.358			0.556	0.358
Effective F-stat			11.359	6.915			12.776	6.213			11.186	6.965

Notes: Effect of Water Board presence on summer evapotranspiration under three control sets. Columns 1–4 report the benchmark used in the main text; columns 5–8 add soil-quality quartiles and groundwater rights; columns 9–12 add the potential land-productivity value as a control. Within each control set, the first two columns report OLS and the last two IV estimates, and within each pair the first column presents estimates for downstream farms and the second for upstream farms. Panel B reports the corresponding first stage, a regression of Water Board presence on the recentered instrument $\tilde{ucd}_i$, and is therefore reported only for the IV columns. Standard errors clustered by county; Anderson–Rubin confidence intervals are reported for the IV estimates.

Table A.XV: Robustness — Agricultural Yield (Peak EVI): Alternative Control Sets

	Benchmark				+ Soil quality and groundwater rights				+ Land potential productivity			
	OLS		IV		OLS		IV		OLS		IV	
	(1) Downstream	(2) Upstream	(3) Downstream	(4) Upstream	(5) Downstream	(6) Upstream	(7) Downstream	(8) Upstream	(9) Downstream	(10) Upstream	(11) Downstream	(12) Upstream
<i>Panel A. Main result</i>												
Board	0.00572 (0.00853)	0.00890 (0.00754)	0.197 (0.0840)**	-0.00496 (0.0348)	0.000500 (0.00740)	0.0120 (0.00647)*	0.174 (0.0715)**	0.0247 (0.0330)	0.00483 (0.00861)	0.00898 (0.00752)	0.193 (0.0844)**	-0.00537 (0.0345)
Observations	145,065	144,515	145,065	144,515	141,675	143,795	141,675	143,795	145,055	144,515	145,055	144,515
R-squared	0.244	0.208	0.011	0.206	0.278	0.247	0.078	0.246	0.247	0.208	0.024	0.206
Mean Dependent Var.			0.370	0.410			0.370	0.410			0.370	0.410
AR test CI			[.06819, .5084]	[-.08436, .1247]			[.06512, .4268]	[-.03407, .2068]			[.06284, .5076]	[-.08449, .121]
<i>Panel B. First stage: Water Board presence</i>												
Recentered distance $\tilde{ucd}_i$			-0.0595 (0.0174)***	0.0380 (0.0145)**			-0.0631 (0.0174)***	0.0368 (0.0148)**			-0.0590 (0.0174)***	0.0381 (0.0144)**
Observations			145,065	144,515			141,675	143,795			145,055	144,515
R-squared			0.553	0.358			0.541	0.358			0.553	0.358
Effective F-stat			11.646	6.915			13.122	6.213			11.476	6.965

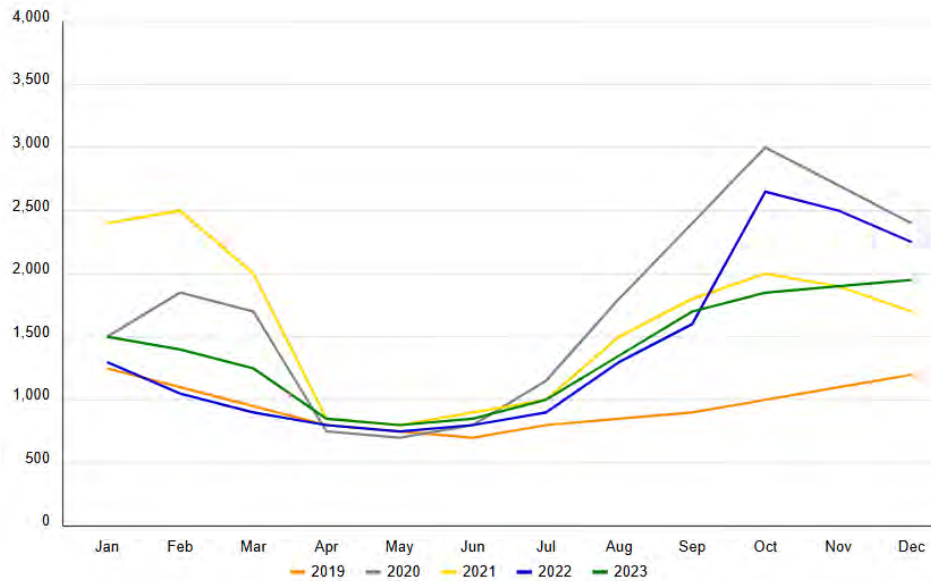
Notes: Effect of Water Board presence on peak EVI under three control sets. Columns 1–4 report the benchmark used in the main text; columns 5–8 add soil-quality quartiles and groundwater rights; columns 9–12 add the potential land-productivity value as a control. Within each control set, the first two columns report OLS and the last two IV estimates, and within each pair the first column presents estimates for downstream farms and the second for upstream farms. Panel B reports the corresponding first stage, a regression of Water Board presence on the recentered instrument $\tilde{ucd}_i$, and is therefore reported only for the IV columns. Standard errors clustered by county; Anderson–Rubin confidence intervals are reported for the IV estimates.

Table A.XVI: Heterogeneity by Farm Size: Fruit-Tree Investment (IV Estimates)

	Any fruit tree				Water-intensive species				Micro-irrigation			
	Small farms		Large farms		Small farms		Large farms		Small farms		Large farms	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
<i>Panel A. Main result (IV estimates)</i>												
Board	0.0638 (0.0374)*	-0.0894 (0.0667)	0.326 (0.185)*	-0.0512 (0.193)	0.0809 (0.0374)**	-0.0887 (0.0618)	0.230 (0.169)	-0.201 (0.130)	0.0567 (0.0353)	-0.0377 (0.0273)	0.208 (0.161)	0.103 (0.0999)
Observations	77,225	69,995	25,560	29,985	77,225	69,995	25,560	29,985	77,225	69,995	25,560	29,985
AR test CI	[-0.1161, .1709]	[-.4157, .1067]	[.02566, 1.175]	[-.4271, .7239]	[.01102, .1967]	[-.415, .07975]	[-.05578, .9698]	[-.5742, .1504]	[-.0173, .154]	[-.1908, .03208]	[-.07303, .8924]	[-.06439, .5697]
<i>Panel B. First stage: Water Board presence</i>												
Recentered distance $\tilde{ucd}_i$	-0.0612 (0.0182)**	0.0372 (0.0164)**	-0.0490 (0.0175)**	0.0404 (0.0148)**	-0.0612 (0.0182)**	0.0372 (0.0164)**	-0.0490 (0.0175)**	0.0404 (0.0148)**	-0.0612 (0.0182)**	0.0372 (0.0164)**	-0.0490 (0.0175)**	0.0404 (0.0148)**
Observations	77,225	69,995	25,560	29,985	77,225	69,995	25,560	29,985	77,225	69,995	25,560	29,985
R-squared	0.545	0.352	0.579	0.337	0.545	0.352	0.579	0.337	0.545	0.352	0.579	0.337
Effective F-stat	11.308	5.161	7.822	7.436	11.308	5.161	7.822	7.436	11.308	5.161	7.822	7.436

Notes: This table reports IV estimates of equation 11 for small and large farm subsamples (small farms are those below the median of farm area; large farms those above the 80th percentile). Columns 1–4 report fruit-tree planting, columns 5–8 water-intensive species, and columns 9–12 micro-irrigation. Within each outcome, the first two columns report small farms and the next two large farms, for the downstream and upstream terciles. Panel B reports the corresponding first stage, estimated separately for each size subsample. The coefficient on Board is shown with its standard error in parentheses (clustered by county); Anderson–Rubin confidence intervals are reported below each estimate. Because the size subsamples have weak first stages, inference should rely on the Anderson–Rubin intervals rather than the point estimates.

Figure A.I: Water Distribution by El Arrayan Water Board to Governed Canals, 2019-2023

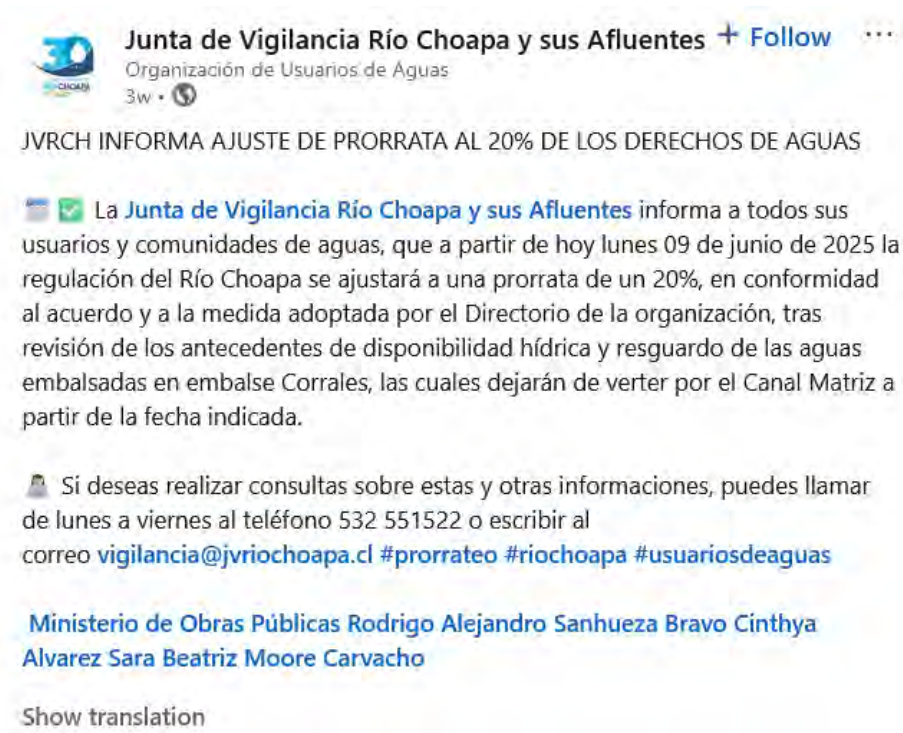


Notes: Panel (a) shows average streamflow and precipitation in monitoring stations within and outside Water Board jurisdictions before 1985. Source: Authors' calculation. Panel (b) displays water distribution by El Arrayan Water Board to governed canals from 2019-2023, measured in liters per second. Source: <https://jmapocho.cl/reparto-total/>. Captured on November 7, 2023.

Figure A.II: Example: Water Right Title

1	AMELIA GALVEZ CARVALLO
2	CONSERVADOR ARCHIVERO
3	SAN BERNARDO
4	Fs. 100 N° 197
5	Año 1999/gcg.
6	COPIA DE INSCRIPCION
7	(Registro de Propiedad de Aguas)
8	
9	N° 197 de 1999 MADECO S. A., Rut N° 91.021.000-9, con domicilio
10	DERECHO a DE en calle Ureta Cox número novecientos treinta,
11	APROVECHAMIENTO de AGUAS SUB- comuna de San Miguel, es ruella de un derecho de
12	TERRANEAS
13	aprovechamiento consuntivo de aguas subterráneas,
14	MADECO S. A.
15	de ejercicio permanente y continuo, de sesenta y
16	cuatro litros por segundo, que se captarán por
17	elevación mecánica desde un pozo de ciento
18	veinticinco metros de profundidad, ubicado el predio
19	de la interesada, Lote de terreno que formaba
20	parte del predio denominado hoy, fundo La Divi-
21	sa, Rol de Avalúo número cuatro mil quinientos
22	quién dieciséis, a doscientos cincuenta y cinco
23	metros al norte del deslinde sur y dieciocho
24	como ocho metros al oriente del eje de la calle
25	La Divisa. El Área de protección del pozo queda
26	definida por un círculo de doscientos metros de
27	radio con centro en el eje del pozo, la que no
28	podrá abarcar más del cincuenta por ciento de la
29	superficie de las propiedades vecinas. Adquirió
30	este derecho de aprovechamiento, por consti-
	tución que le hizo la Dirección General de
	Aguas, de conformidad a los Artículos 60, 61,
	141, 149 y 150 del Código de Aguas y en
	virtud de la Resolución N° 275 de la Dirección
	General de Aguas del Ministerio de Obras Públi-

Figure A.III: Water Rights Proration Adjustment



Notes: Screenshot taken on July 29, 2025, from a LinkedIn post by Junta de Vigilancia Río Choapa y sus Afluentes.

Figure A.IV: General Assembly of Water Users



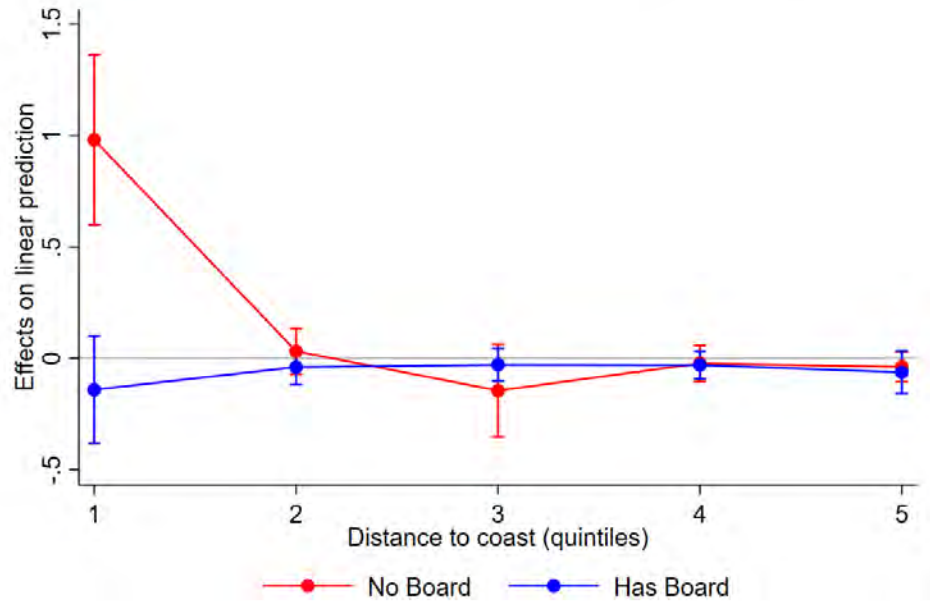
Notes: Screenshot taken on July 29, 2025, from a LinkedIn post by Junta de Vigilancia Río Choapa y sus Afluentes.

Figure A.V: Irrigation Infrastructure Maintenance



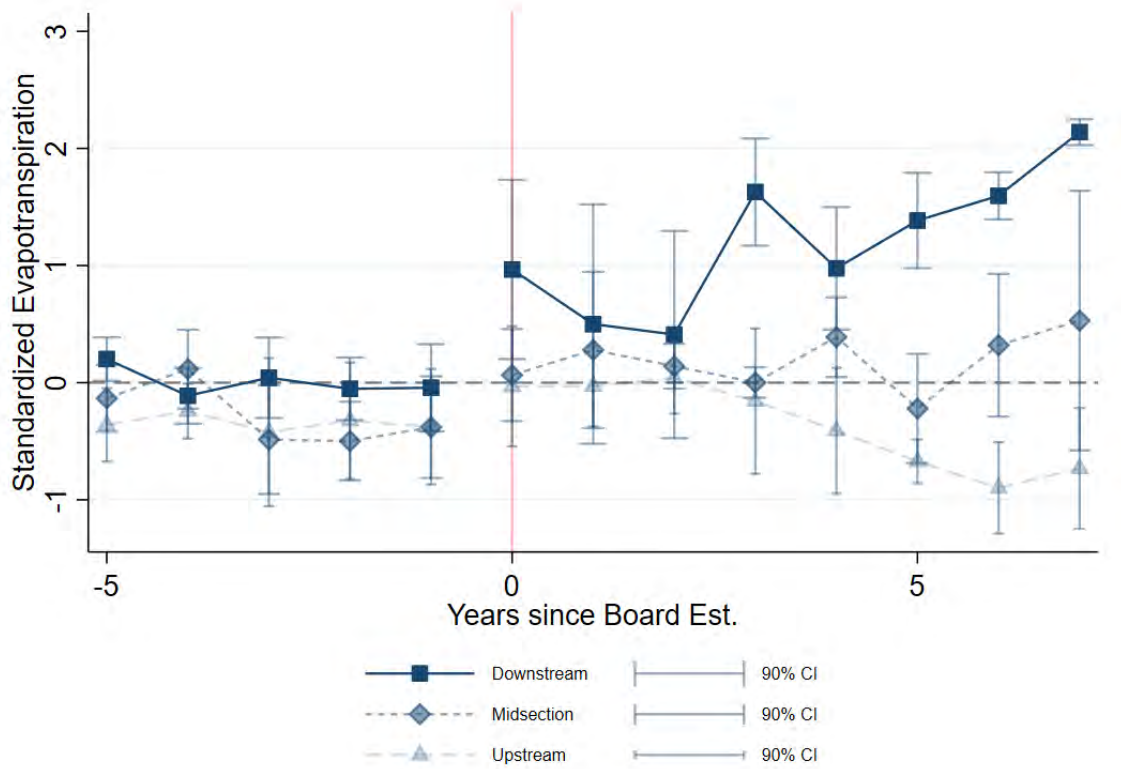
Notes: Photograph of a cleaning campaign, as featured on the official website of the Junta de Vigilancia del Río Maipo, depicting canal maintenance activities. Image retrieved on July 29, 2025, from their website: <https://jvriomaipo.cl/limpieza-basura-canales-regadio/>.

Figure A.VI: Robustness: Effect on $\log(\text{Production})$ of Rainfall during the Irrigation Season by Longitude Quintile and Treatment Status for Irrigated Parcels Registered in Canal



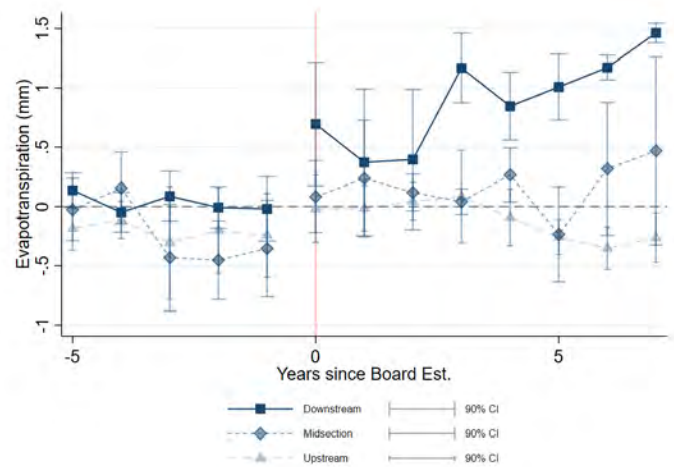
Notes: This figure presents a robustness check on the functional form of equation 5, replacing the continuous distance to the coast with quintile indicators.

Figure A.VII: Differences-in-Differences estimates of Impacts of Water Boards on Standardized Water Consumption, by tercile of distance to the coast.



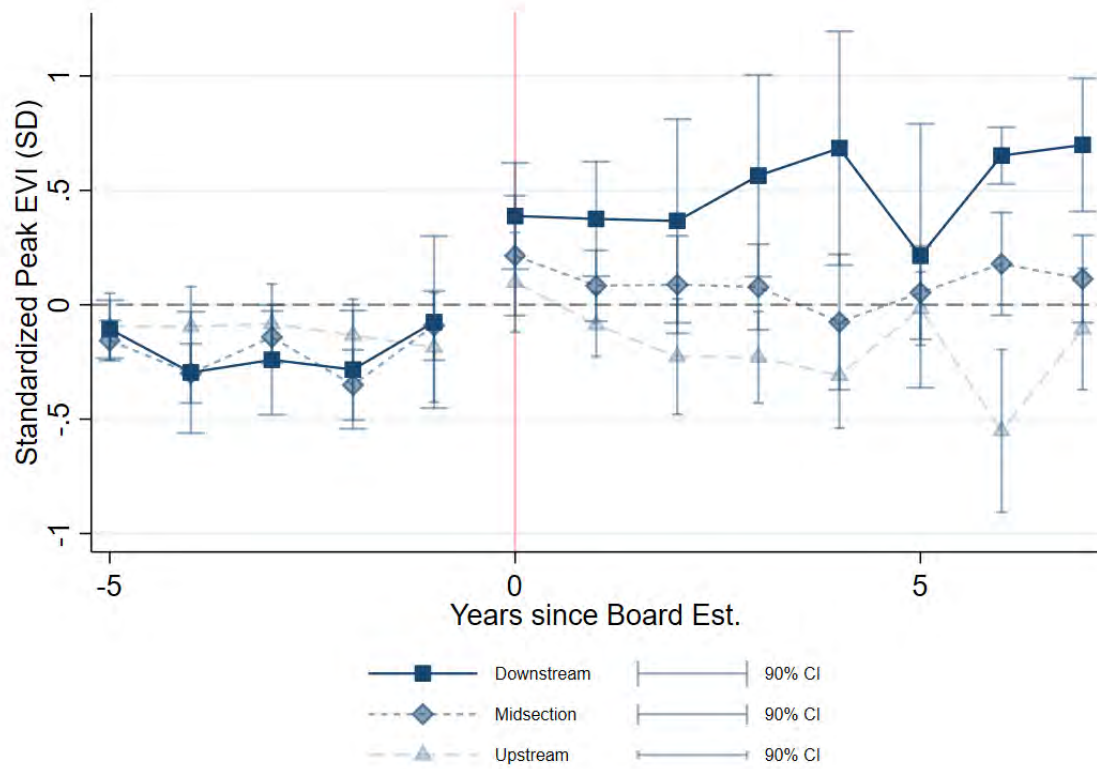
Notes: This figure presents difference-in-differences estimates of the impact of Water Board establishment on standardized summer evapotranspiration, using the staggered difference-in-differences imputation estimator by Borusyak et al. (2024). It presents heterogeneous effects by location within the river basin, where the solid dark-blue line with squares represents downstream farms, the dashed line with diamonds shows mid-basin farms, and the dashed light-blue line with triangles represents upstream farms. The x-axis shows years relative to Water Board establishment, with year 0 marking the establishment year. The sample includes farms located in basins with Water Boards established between 1981-2010, covering 5 years pre- and 7 years post-establishment. Controlling for plot-level fixed effects and flexible time trends by latitude-longitude cells ($2^\circ \times 2^\circ$). Standard errors estimated using Borusyak et al. (2024) leave-one-out procedure, clustering at the 3rd level basin.

Figure A.VIII: Differences-in-Differences estimates of Impacts of Water Boards on Evapotranspiration (non-standardized)



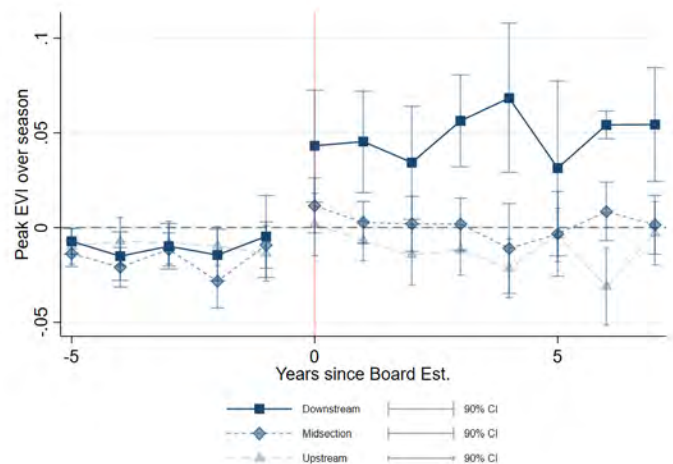
Notes: This figure presents difference-in-differences estimates of the impact of Water Board establishment on non-standardized Evapotranspiration, using the staggered difference-in-differences imputation estimator by Borusyak et al. (2024). It presents heterogeneous effects by location within the river basin, where the solid dark-blue line with squares represents downstream farms, the dashed line with diamonds shows mid-basin farms, and the dashed light-blue line with triangles represents upstream farms. The x-axis shows years relative to Water Board establishment, with year 0 marking the establishment year. The sample includes farms located in basins with Water Boards established between 1981-2010, covering 5 years pre- and 7 years post-establishment. Controlling for plot-level fixed effects and flexible time trends by latitude-longitude cells ($2^\circ \times 2^\circ$). Standard errors estimated using Borusyak et al. (2024) leave-one-out procedure, clustering at the 3rd level basin.

Figure A.IX: Differences-in-Differences estimates of Impacts of Water Boards on Standardized Agricultural Yield, by tercile of distance to the coast.



Notes: This figure presents difference-in-differences estimates of the impact of Water Board establishment on standardized agricultural productivity (EVI), using the staggered difference-in-differences imputation estimator by Borusyak et al. (2024). It presents heterogeneous effects by location within the river basin, where the solid dark-blue line with squares represents downstream farms, the dashed line with diamonds shows mid-basin farms, and the dashed light-blue line with triangles represents upstream farms. The x-axis shows years relative to Water Board establishment, with year 0 marking the establishment year. The sample includes farms located in basins with Water Boards established between 1981-2010, covering 5 years pre- and 7 years post-establishment. Controlling for plot-level fixed effects and flexible time trends by latitude-longitude cells ($2^\circ \times 2^\circ$). Standard errors estimated using Borusyak et al. (2024) leave-one-out procedure, clustering at the 3rd level basin.

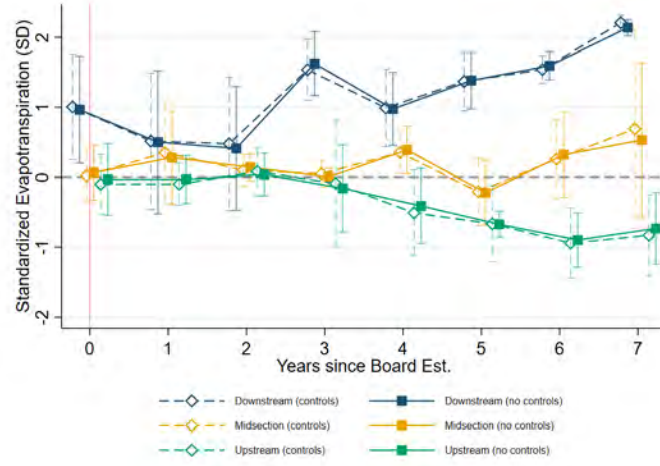
Figure A.X: Differences-in-Differences estimates of Impacts of Water Boards on Agricultural Yield (non-standardized)



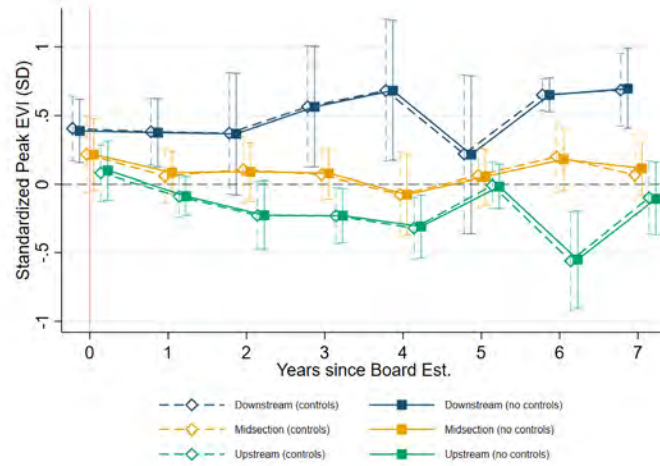
Notes: This figure presents difference-in-differences estimates of the impact of Water Board establishment on non-standardized agricultural productivity (EVI), using the staggered difference-in-differences imputation estimator by Borusyak et al. (2024). It presents heterogeneous effects by location within the river basin, where the solid dark-blue line with squares represents downstream farms, the dashed line with diamonds shows mid-basin farms, and the dashed light-blue line with triangles represents upstream farms. The x-axis shows years relative to Water Board establishment, with year 0 marking the establishment year. The sample includes farms located in basins with Water Boards established between 1981-2010, covering 5 years pre- and 7 years post-establishment. Controlling for plot-level fixed effects and flexible time trends by latitude-longitude cells ($2^\circ \times 2^\circ$). Standard errors estimated using Borusyak et al. (2024) leave-one-out procedure, clustering at the 3rd level basin.

Figure A.XI: Comparison of Differences-in-Differences estimates of Impacts of Water Boards on Standardized Water Consumption and Agricultural Yield, with vs. without controls

(a) Water Consumption: Heterogeneous Impacts, by tercile of distance to the coast.

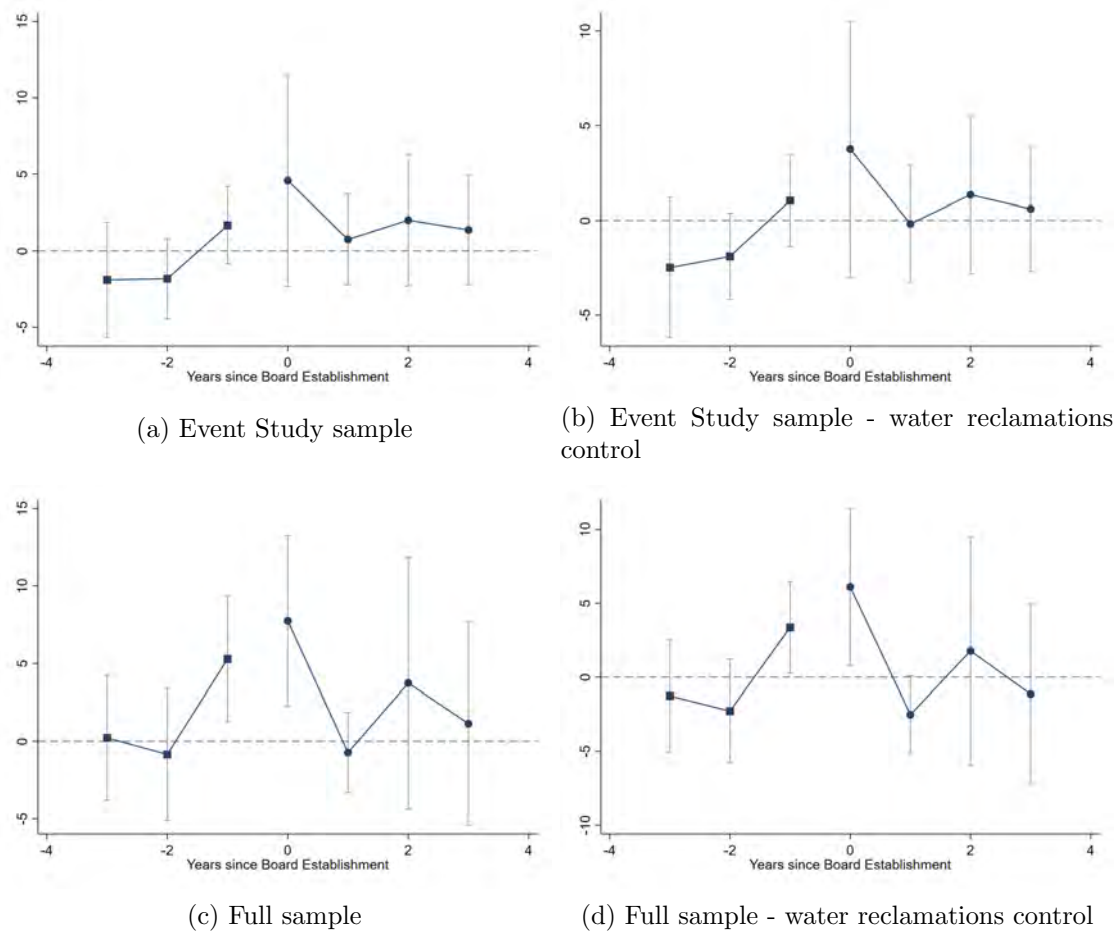


(b) Agricultural Yield: Heterogeneous Impacts, by tercile of distance to the coast.



Notes: This figure presents heterogeneous effects by location within the river basin for standardized Water Consumption (Panel A) and standardized agricultural yield (Panel B), using the staggered difference-in-differences imputation estimator by Borusyak et al. (2024). In both panels, blue denotes downstream farms, orange mid-basin farms, and green upstream farms; solid lines with filled squares report estimates without the full set of controls, and dashed lines with hollow diamonds report estimates with them. The x-axis shows years relative to Water Board establishment, with year 0 marking the establishment year. The sample includes farms located in basins with Water Boards established between 1981–2010, covering 5 years pre- and 7 years post-establishment. All estimates include plot-level fixed effects and flexible time trends by latitude-longitude cells ($2^\circ \times 2^\circ$); the specifications labelled *controls* additionally include annual climatic characteristics (total precipitation, summer precipitation, and temperature ranges). Standard errors are estimated using the Borusyak et al. (2024) leave-one-out procedure, clustering at the 3rd level basin.

Figure A.XII: Dynamic Effects of Board Establishment Events on Streamflow - Full Year



Notes: This figure presents our main estimates of the impact of Water Board establishment events on streamflows, using the Event Study estimator by Borusyak, Jaravel and Spiess (2024). The Full Sample correspond to monitoring stations located in the Study Region, while the Event Study sample only includes eventually treated monitoring stations (i.e. monitoring stations located in subsubbasins within Water Boards jurisdictions). Controlling for monthly climatic measures in the drainage area of each monitoring station (linear and squared terms for precipitation and average maximum temperature, average minimum temperature, and potential evapotranspiration) and monitoring station-month and year fixed effects. Standard errors estimated using Borusyak, Jaravel and Spiess (2024) leave-one-out procedure, clustering at the hydrological basin level and allowing for effects heterogeneity based on 4 groups by longitudinal distance to the coast and agricultural land share in the drainage area of each monitoring station.

B Appendix: A Model of Water Rights Enforcement

This model illustrates the informational complexities on enforcing water rights (private property rights over water usage), in the presence of unverifiable water extraction and unobservable productivity differences. The spirit of the model is similar to a classic Production in Teams model (e.g., Holmstrom, 1982), where unobserved productivity differences and unverifiable behavior introduce inefficiencies in the provision of a costly action. A distinctive feature of this model is the asymmetry across agents in terms of the order in which they can take action—in this case, to extract a share of a common resource. Taking action first gives some agents an advantage over others, by implicitly giving them priority in access. While in this paper, this model is used to analyze the equilibrium allocation of water from rivers, the lessons apply to other common pool resources where priority by distance or time creates asymmetries.⁶¹

The appendix proceeds as a taxonomy of three regimes, ordered by the strength of the institutions in place. Section B.1 sets out the environment, the legal system, and the evidentiary rule the court applies. Section B.2 characterizes *effective open access*: when the award from a successful suit does not cover its cost, no suit is ever filed, the evidentiary rule is never actually applied, and extraction is based purely on the relative position of the farmers. Section B.3 characterizes *decentralized enforcement*: when litigation is worth pursuing, a credible court disciplines only those agents who cannot hide, so upstream agents with low productivity over-extract strategically at exactly the rate that keeps them indistinguishable in court from lawful high-productivity neighbors. In this equilibrium, a subset of agents over-extract, another receives no water at all, and no legal action is taken, as responsibility for the shortage downstream is dispersed among many upstream users, akin to a directional “Tragedy of the Commons”. Comparing the two regimes shows that, first, open access and decentralized enforcement in the presence of informational constraints are not equivalent; and second, that a court system facing verifiability and attribution challenges can leave farmers worse off, on average, than no court at all. Section B.4 presents an equilibrium under *centralized enforcement*, in which the basin hires a river manager who is legally

⁶¹Examples include migratory fisheries across international borders, pollution across generations, and pollution within a season.

empowered and legally responsible to allocate water consistent with the private property rights in place. In this equilibrium there is an increase in the share of downstream users with water access, as they can sue a single river manager whenever they do not receive their allotment; the “Tragedy of the Commons” is replaced by a classic principal-agent problem.⁶² Section B.5 microfound the manager’s reduced-form effort as a choice of whom to inspect, and shows that under some assumptions an optimal inspection set is contiguous and starts at the head of the river.

B.1 The environment

B.1.1 Farmers, water and technology

Consider a river basin, containing only one linear stream that only has one head (the upstream extreme of the basin) and one mouth (the downstream extreme of the basin). The basin is populated by a continuum of atomistic farmers of total mass M , indexed by $i \in [0, M]$, with the index reflecting their relative position across the river (from the river head to the river mouth).⁶³

There are two states of nature: Normal ($s = N$) and Drought ($s = D$), that happen with probability $\Pr(s = N) = q \in (0, 1)$. The total amount of water available to be distributed in the river is determined by the state of nature, with values $A^N = A(s = N)$ and $A^D = A(s = D) = A^N \kappa$, with $\kappa \in (0, 1)$. Farmer i holds water rights $\alpha(i) \geq 0$, a measurable function with $\int_0^M \alpha(j) dj = 1$, allowing them to extract $a(i|s = N) = \alpha(i)A^N$ units of water in a Normal state and $a(i|s = D) = \alpha(i)A^D$ in a state of Drought.

The infrastructure for water extraction in place does not physically allow farmers to extract more than their corresponding water rights in normal times. Denote by $w(i) \geq 0$ the effective water extraction. The physical constraints faced by the farmer are

$$w(i) \leq \min \left\{ \alpha(i)A^N, \max \left\{ A^s - \int_0^i w(j) dj, 0 \right\} \right\}.$$

⁶²In principle, this corresponds to a multiple-principals-single-agent problem but given the legal environment, an opportunistic upstream principal (a farmer) may not exert any power over the agent (a river manager), as there is no lawsuit available to punish the agent due to not being able to over-extract water.

⁶³In a more general case, this index reflects the priority with which agents can extract the resource under analysis.

In other words, the farmer cannot extract more water than 1) what is their legal extraction right in normal times, and 2) what is left from upstream users.

Assumption 1 (Technology and types). *The production function of a farmer is $y(i) = \varepsilon(i)f(w(i))$, where*

- *$f(\cdot)$ is a continuous, increasing function, such that $f(0) = 0$ and that satisfies homogeneity of degree $\beta \in (0, 1]$.*
- *$\varepsilon(i)$ takes two values: a fraction p of farmers are high-productivity, with $\varepsilon(i) = 1$, and the remaining fraction $(1 - p)$ are low-productivity, with $\varepsilon(i) = \mu < 1$. We assume a law of large numbers applies, so that these shares hold for the basin as a whole and also among the farmers located upstream of any given point.*
- *Who belongs to each group is private information, known to the farmer at the beginning of the season; the shares p and $1 - p$ are common knowledge.*⁶⁴

So a low-productivity farmer i will have a production function $y(i) = \mu f(w(i))$ while a high-productivity farmer j will produce $y(j) = f(w(j))$.

B.1.2 The legal system

Agents who believe their interests have been damaged can sue; these lawsuits are ruled by a judge who rules based on the evidence presented in court. The court observes, for each litigant i , the pair $(y(i), \alpha(i))$ and the realized state s ; it observes neither the extraction $w(i)$ nor the type $\varepsilon(i)$. Farmers can observe all other farmers' extraction, but such observations are inadmissible in court.

A farmer who does not get their lawful allotment can sue any agent they believe responsible, at cost $\gamma > 0$; a victorious plaintiff collects an award $B > 0$, and a convicted

⁶⁴Productivity is modelled as a single per-season draw for tractability. The conclusions extend to richer specifications in which productivity has several components, some of them persistent, provided the persistent components are common knowledge: all the mimicking equilibrium requires is that *some* non-persistent component remain private at the moment of extraction. If instead a persistent component were private and inferable from past output, courts and neighbours would update on it across seasons, and it would introduce a ratchet effect: some farmers will have incentives to restrain extraction today to avoid revealing a high type, precisely in order to keep tomorrow's evidentiary threshold loose. We abstract from this dynamic force, which could introduce additional distortions. The former implies that μ should be interpreted as a measure of dispersion over this particular productivity component rather than any properly observable difference in productivity.

defendant pays a fine F , independently of the number of suits. We assume $F > f(\alpha(i)A^N) - f(\alpha(i)A^D)$ for all i , i.e. the fine exceeds the maximal feasible gain from any detectable over-extraction. The comparison between B and γ partitions the environment without a river manager into the two regimes studied in Sections B.2 and B.3.⁶⁵

Assumption 2 (Evidentiary standard). *The court rules for the plaintiff if and only if (i) the plaintiff suffered a shortfall relative to their lawful allotment, and (ii) the defendant's guilt is established with certainty ("beyond reasonable doubt"), i.e., the evidence is inconsistent with every lawful history of the defendant. The court does not require proof of a causal link between the defendant's extraction and the plaintiff's individual shortfall.*

With multiple atomistic agents potentially violating the allocation, no individual deviation is pivotal for any individual shortfall, so a causation requirement would make enforcement infeasible. Assumption 2 rules that requirement out.⁶⁶

Lemma 1 (The court's rule). *Under Assumptions 1–2, the court convicts a defendant i in state s if and only if*

$$y(i) > y^T(\alpha(i) | s) \equiv f(\alpha(i)A^s). \quad (12)$$

Proof. A lawful history has $w(i) \leq \alpha(i)A^s$, so the largest output consistent with lawfulness is attained at $\varepsilon(i) = 1$, $w(i) = \alpha(i)A^s$, i.e. $f(\alpha(i)A^s)$. Any $y(i) > f(\alpha(i)A^s)$ has probability zero under every lawful history, establishing guilt with certainty. Any $y(i) \leq f(\alpha(i)A^s)$ is consistent with the lawful history $(\varepsilon(i) = 1, w(i) = f^{-1}(y(i)))$, so guilt is not established with certainty. $\square$

The rule admits a simple reading. Because the court cannot observe productivity, the standard of proof forces it to evaluate every defendant under the technology most favorable to them: a farmer is convicted only if their output exceeds what a high-productivity farmer could have produced while extracting exactly their lawful allotment.

⁶⁵The knife-edge case $B = \gamma$ is not treated here. At that point a potential plaintiff is exactly indifferent, so every filing probability $\pi \in [0, 1]$ is sequentially rational, introducing multiple equilibria. Although potentially interesting, given its practical irrelevance, and as none of the results below depend on it, we omit its discussion.

⁶⁶For example, in a suit over pollution damages, the standard adopted here requires an exposed plaintiff to show that they suffered harm and that the defendant emitted a pollutant in excess of its permit. A causation requirement would additionally require showing that the pollutant reaching the plaintiff was the one emitted by that particular defendant.

In the Normal state the infrastructure constraint already coincides with the lawful allotment, so the rule has bite only in Droughts. Throughout we consider droughts severe enough that the infrastructure cap does not bind in the Drought state.

B.2 Effective open access

The evidentiary rule of Lemma 1 disciplines behavior only if somebody is willing to invoke it. Suppose first that they are not.

Proposition 1 (Effective open access). *Suppose $B < \gamma$. Then filing any suit is strictly dominated (it nets $B - \gamma < 0$ if won, $-\gamma$ if lost), so no suit is ever filed and the threshold rule of Lemma 1 is never actually enforced. Every farmer then extracts up to the physical cap, $w(i) = \min\{\alpha(i)A^N, \text{remaining water}\}$, independently of productivity. In a Drought, the served region $[0, \tilde{t}^{OA})$ satisfies $\int_0^{\tilde{t}^{OA}} \alpha(j) dj = \kappa$; under the uniform allocation, the served share is exactly κ .*

Although formally the rights remain in place, they do not affect behavior. Water is purely allocated based on the position along the basin. This limiting case allows for an interesting comparison with the equilibrium of the next section, where legal action is worthwhile but limited verifiability and attribution allow it to discipline only those who cannot hide.

B.3 Decentralized enforcement

Suppose now that suing is worthwhile once guilt is certain: $B > \gamma$. Decentralized enforcement is now credible, and the threshold rule binds, but, as we show, it binds asymmetrically across types.

B.3.1 Beliefs and the extraction decision

To define a Bayesian Nash equilibrium, we need to specify beliefs off the equilibrium path. If $B > \gamma$, then by Lemma 1 suing a farmer whose output exceeds threshold (12) is a dominant strategy for any farmer damaged by over-extraction: conviction is certain, and the award is collected regardless of how many others file. Consider beliefs such that any farmer who

produces above the threshold (12) is certain to be sued.⁶⁷ Under these beliefs, farmer i 's problem in state s is

$$\begin{aligned} \max_{w(i)} \quad & \varepsilon(i)f(w(i)) - F \times \mathbf{1}[\varepsilon(i)f(w(i)) > y^T(\alpha(i)|s)] \text{ subject to} \\ & w(i) \geq 0, \quad w(i) \leq \alpha(i)A^N, \quad w(i) \leq A^D - \int_0^i w(j) dj. \end{aligned}$$

The fine makes any detectable over-extraction unprofitable, so the optimal extraction satisfies

$$w^*(i) = \min \left\{ \underbrace{f^{-1}(y^T(\alpha(i)|s)/\varepsilon(i))}_{\text{(A) Legal deterrence}}, \underbrace{\alpha(i)A^N}_{\text{(B) Infrastructure}}, \underbrace{A^D - \int_0^i w(j) dj}_{\text{(C) Availability}} \right\}. \quad (13)$$

For a high-productivity farmer, any over-extraction during a drought is immediately detectable: term (A) equals $\alpha(i)A^D$. For a low-productivity farmer, it is possible to over-extract and still pass for a lawful high-productivity one, at a rate

$$\rho \equiv \mu^{-1/\beta} - 1 > 0,$$

which is increasing in the productivity gap (decreasing in μ) and in the curvature of the technology (decreasing in β).⁶⁸ To make progress and illustrate the mechanism, we will assume a uniform allocation of water rights, $\alpha(i) = 1/M$, for the remainder of the analysis.⁶⁹

Proposition 2 (Equilibrium under decentralized enforcement). *Sustained by the beliefs above, there is an equilibrium in which: (i) high-productivity farmers with water access extract $w(i) = \alpha(i)A^D$; (ii) low-productivity farmers with water access extract $w(i) = \alpha(i)A^D(1 + \rho)$; (iii) no lawsuits are filed on the equilibrium path, since over-extraction is calibrated to be indistinguishable in court from lawful high-productivity output; and (iv) a*

⁶⁷Since all water belongs to someone in the model, any over-extraction means some damage to at least one user.

⁶⁸Therefore, our assumption that infrastructure is not binding implies $\kappa \leq \mu^{1/\beta}$.

⁶⁹We discuss a hypothetical socially optimal allocation in Section B.3.4. In general, Proposition 2 is valid under any allocation of water rights such that the allocation is statistically independent of the productivity shock.

share $\tilde{i}/M$ of farmers, with $\tilde{i}$ defined by

$$\int_0^{\tilde{i}} \alpha(j) dj = \frac{1}{1 + (1 - p)\rho}, \quad (14)$$

have access to water, with farmers $i > \tilde{i}$ receiving none. Under the uniform allocation $\alpha(i) = 1/M$, the served share is $\tilde{i}/M = 1/(1 + (1 - p)\rho) < 1$.

Proof. Optimality of (13) under the stated beliefs gives (i)–(ii) directly. No farmer wants to deviate: crossing the threshold triggers a fine exceeding any feasible gain, and staying at the cap is a strict best response given water remains. For (iii), a low-productivity farmer’s output at the cap is $\mu f(\alpha(i)A^D(1 + \rho)) = f(\alpha(i)A^D)$ by construction of ρ , exactly matching a high-productivity farmer’s lawful output; a suit against either type is therefore indistinguishable from a suit against a lawful high type and cannot be won. Water is exhausted moving downstream where cumulative extraction $A^D(p + (1 - p)(1 + \rho)) \int_0^i \alpha(j) dj = A^D(1 + (1 - p)\rho) \int_0^i \alpha(j) dj$ reaches A^D , giving (14). $\square$

The share of served farmers is decreasing in $1 - p$ (the share of low-productivity farmers, who over-extract) and in ρ (the over-extraction rate itself). The two regimes are observationally distinct: under credible litigation, drought extraction is sorted by productivity and served farmers’ output is equalized at $f(\alpha(i)A^D)$; under effective open access, extraction is purely positional and uncorrelated with productivity.

B.3.2 Comparison between decentralized enforcement and open access

Because legal deterrence binds only on the high types—who cannot hide—partial enforceability diverts water toward low types, whereas open access allocates it by position alone, uncorrelated with productivity. Equilibrium output under decentralized enforcement can therefore be *lower* than under no enforcement at all. Writing $\bar{\varepsilon} \equiv p + (1 - p)\mu$ for mean productivity, the comparison admits a closed form.

Proposition 3 (Decentralized enforcement can perform worse than open access). *Fix the uniform allocation $\alpha(i) = 1/M$ and let $\beta < 1$. Average output under effective open access*

exceeds average output under decentralized enforcement if and only if

$$\kappa > \kappa^* \equiv \left[\bar{\varepsilon}(1 + (1 - p)\rho) \right]^{-\frac{1}{1-\beta}}. \quad (15)$$

Values of κ satisfying both (15) and the maintained assumption $\kappa \leq \mu^{1/\beta}$ exist if and only if $\bar{\varepsilon}(1 + (1 - p)\rho) > \mu^{1-1/\beta}$, which fails both at $p = 0$ and at $p = 1$.

Proof. Under decentralized enforcement the served set is $[0, \tilde{i})$ with $\tilde{i}/M = 1/(1 + (1 - p)\rho)$ by Proposition 2, and every served farmer produces $f(A^D/M)$: a high type directly, and a low type because $\mu(1 + \rho)^\beta = 1$ by the definition of ρ . Farmers $i > \tilde{i}$ produce $f(0) = 0$. Hence $W^{DE} = f(A^D/M)/(1 + (1 - p)\rho)$, the superscript standing for decentralized enforcement.

Under open access every farmer extracts the physical cap A^N/M until the stream is exhausted, so the served set is an initial segment of measure κM , since $\kappa M \cdot A^N/M = \kappa A^N = A^D$. Homogeneity gives $f(A^N/M) = f(\kappa^{-1}A^D/M) = \kappa^{-\beta}f(A^D/M)$, extraction is independent of type, and by Assumption 1 the mean type on an initial segment is $\bar{\varepsilon}$. Hence $W^{OA} = \bar{\varepsilon}\kappa^{1-\beta}f(A^D/M)$.

Dividing both by $f(A^D/M) > 0$, $W^{OA} > W^{DE}$ iff $\kappa^{1-\beta} > [\bar{\varepsilon}(1 + (1 - p)\rho)]^{-1}$. Since $\beta < 1$ the map $x \mapsto x^{1-\beta}$ is strictly increasing on $(0, \infty)$, which gives (15).

For the second claim, raise both sides of $\kappa^* = [\bar{\varepsilon}(1 + (1 - p)\rho)]^{-1/(1-\beta)} < \mu^{1/\beta}$ to the power $1 - \beta > 0$, which preserves the inequality, to obtain $[\bar{\varepsilon}(1 + (1 - p)\rho)]^{-1} < \mu^{(1-\beta)/\beta}$; taking reciprocals reverses it, giving $\bar{\varepsilon}(1 + (1 - p)\rho) > \mu^{-(1-\beta)/\beta} = \mu^{1-1/\beta} = \mu(1 + \rho)$. At $p = 1$ both $\bar{\varepsilon}$ and $1 + (1 - p)\rho$ equal one while $\mu^{1-1/\beta} > 1$, since $1 - 1/\beta < 0$ and $\mu < 1$; at $p = 0$ we have $\bar{\varepsilon} = \mu$ and $1 + (1 - p)\rho = 1 + \rho$, so both sides coincide. $\square$

That the condition fails at both extremes is the substance of the result rather than a technicality: courts are harmful only because deterrence binds asymmetrically, and asymmetry requires both types to be present. With $p = 1$ nobody can mimic, so the threshold rule implements the lawful allocation exactly and cannot be beaten; with $p = 0$ everyone mimics identically, so there is no misallocation across types and the rule merely caps extraction and serves more farmers.

The two regimes ration on different margins. Under effective open access every farmer

who reaches the stream takes their full normal-times cap, so rationing is entirely extensive: the scarcer the water, the fewer farmers it reaches, but who is served is uncorrelated with productivity in this model. Under decentralized enforcement nobody extracts the physical maximum, so rationing is partly intensive: the served share rises from κ to $[1 + (1 - p)\rho]^{-1}$, but the water in use is tilted toward the low-productivity farmers, who take $(1 + \rho)$ times their allotment while high-productivity ones are held to exactly theirs. Which distortion is worse depends on drought severity—which determines the share of farmers irrigated under open access—and on the concavity of the production function, which governs both the over-extraction rate ρ and the loss from concentrating water in few hands.

Decentralized enforcement does not implement the lawful allocation because limited verifiability and difficulties in attribution rule out individual accountability for those who over-extract—which is what centralized enforcement supplies below.

B.3.3 The economic cost of decentralized enforcement relative to the lawful allocation

Fix the uniform allocation $\alpha(i) = 1/M$. Under the lawful allocation, average Drought output is $\bar{\varepsilon}f(A^D/M)$. In the equilibrium of Proposition 2, served high types produce $f(A^D/M)$, served low types produce $f(A^D/M)$ as well (by construction of ρ), and the served share is $1/(1 + (1 - p)\rho)$, so average output is

$$W^{DE}(\rho) = \frac{1}{1 + (1 - p)\rho} f\left(\frac{A^D}{M}\right).$$

Proposition 4 (Strict output loss). *For all $p \in (0, 1)$, $\mu \in (0, 1)$, $\beta \in (0, 1]$, $W^{DE}(\rho) < \bar{\varepsilon}f(A^D/M)$.*

Proof. $W^{DE}(\rho) < \bar{\varepsilon}f(A^D/M) \iff 1 < \bar{\varepsilon}(1 + (1 - p)\rho) \iff (1 - p)(1 - \mu) < \bar{\varepsilon}(1 - p)\rho$, which holds since $\bar{\varepsilon}\rho \geq \mu\rho = \mu(\mu^{-1/\beta} - 1) \geq \mu^{1-1/\beta} - \mu \geq 1 - \mu$ using $\beta \leq 1$, and the first inequality ($\bar{\varepsilon} \geq \mu$) is strict for $p \in (0, 1)$. $\square$

Corollary 1 (Productivity gaps amplify losses). *As $\mu \downarrow 0$ at fixed p , lawful output falls linearly (through $\bar{\varepsilon}$) while equilibrium output under decentralized enforcement vanishes. A*

wider productivity gap does not merely lower average productivity; it widens the range over which low types can mimic, so upstream low types absorb an ever larger share of the water and both margins of misallocation (exclusion and diversion toward low types) worsen together.

B.3.4 Optimal water rights allocation

The previous sections presented results for an egalitarian allocation ($\alpha(i) = 1/M$). A natural question is what the efficient allocation of rights would be, and whether a planner could implement it. We characterize the efficient rights across the two types in terms of the parameters of the model, and explain why it is informationally infeasible; decentralizing it requires tradable rights, which in turn requires that those rights be enforceable.

The socially optimal allocation equalizes the marginal value of water across users: for a high- and a low-productivity user, $f'(\alpha^H A^D) = \mu f'(\alpha^L A^D)$, and homogeneity of degree $\beta - 1$ of f' gives

$$\frac{\alpha^H}{\alpha^L} = \mu^{\frac{1}{\beta-1}} > 1 \quad \text{if} \quad (\beta < 1),$$

so under strictly decreasing returns to scale the optimal allocation gives more water rights to high-productivity farmers.⁷⁰

This allocation is informationally infeasible for a planner: since the marginal cost of extraction is zero, a low-productivity type always prefers, and can always claim, the high type's assignment; there is no direct mechanism that separates the types. Moreover, any type-contingent assignment would need to rely on information that we are assuming private. Decentralizing the implementation of the first-best allocation, therefore, requires a market for water rights in which prices, not a planner, elicit productivities. This, in turn, requires that the rights being traded be enforceable. This motivates the analysis under centralized enforcement or governance.

⁷⁰At $\beta = 1$ marginal products are constant and the planner's problem has no interior solution: all water goes to the high types.

B.4 Centralized enforcement: the River Manager

Consider an alternative legal arrangement: the farmers in the basin can hire a River Manager who (i) has the power to open and close the outlets of each farm, and (ii) is legally responsible for any missing water.

These two provisions are interdependent: the second makes the first enforcement rather than predation; the first makes the second accountability instead of scapegoating. In principle, nothing in the capacity to close a canal intake points towards implementing the lawful allocation. What pins enforcement at $w(i) = \alpha(i)A^D$ is two-sided liability: the Manager does not farm, so gains nothing from over-cutting, while the Manager's total liability is increasing in the mass of farmers left short of their allotment, since every farmer denied water can sue and collect B . Decentralized forcible enforcement has no such interior stopping point.⁷¹

B.4.1 Reduced form

We maintain $B > \gamma$ throughout this section.⁷² If the Manager does not exert the effort of opening and closing the outlets, with some probability the farmers can over-extract as in Proposition 2. We model enforcement effort in reduced form: the manager makes an effort $e \in [0, 1]$ at cost $C(e) = \frac{1}{2}ce^2$, such that with probability e the lawful allocation is realized, and with probability $(1 - e)$ farmers extract as in the decentralized equilibrium. This is consistent with the standard enforcement and monitoring literature (Becker, 1968; Holmström, 1979), where costly effort raises the probability of the desired outcome without guaranteeing it.

⁷¹This allows us to discuss briefly what providing the previous technology to the farmers themselves would imply: allowing them to go to another farmer's intake and block it. A farmer whose intake was closed by another may sue, but that lawsuit would face the same inherent challenges as the lawsuits studied in Propositions 1 and 2. In particular, a low-productivity farmer may block another's gate up to the point where their production mimics that of high-productivity farmers. The visible signal of a gate being blocked does not identify who blocked it. More importantly, who would actually benefit from this depends on parameters, but in most cases it would be only a local subset of the farmers close enough to $\tilde{i}$: farmers farther away would see the released water flow to farmers located above them, having to repeat the enforcement action iteratively (and so, increasing its cost).

⁷²As before, if $B < \gamma$, then we will have the effective open access equilibrium: there is no lawsuit that can be used by the farmers to discipline the principal, and so there is no enforcement effort. In that case, the farmers are better off without any river manager.

B.4.2 Lawsuits under centralized enforcement

If a farmer receives less water than their lawful stream, they individually can sue the manager, who is legally responsible for implementing the lawful allocation. If the Manager exerts effort and successfully implements the lawful allocation, a record of the amount of water allocated to each farm is created and can be used in court; an insufficient effort level generates incomplete records, from which the court infers the unlawful allocation was implemented. Records are not a separate technology layered on top of enforcement: in practice the Manager enforces by locking each farm's intake, so that enforcing and generating admissible evidence are the same act. The lock physically caps extraction, its absence at a given intake is observable, and a lock that has been cut is evidence of over-extraction attributable to the single farm that intake serves—which is exactly the attribution that mimicry destroys in Section B.3. Therefore, in a lawsuit involving a farmer and the Manager, the court observes $\{\alpha_i, y_i, w_i\}$ under the lawful equilibrium and $\{\alpha_i, y_i, \emptyset\}$ under the unlawful one. The court rules in favor of farmer i if

$$w_i < \alpha_i A^D \text{ (with proper records), or } y_i < \mu f(\alpha_i A^D) \text{ (with incomplete records).}$$

A victorious farmer receives the same award B as before. Under the lawful allocation no farmer suffers a shortfall, so no suit can succeed.⁷³

The critical difference relative to Section B.3 is the identity of the defendant. Under decentralized enforcement, a dry farmer's claim is against a continuum of upstream neighbors, each of whom over-extracted strategically. Consolidating responsibility in a single Manager aggregates the multiple claims of the infinitesimal downstream farmers into one claim of substantial size, which is what makes the litigation threat operative.

⁷³We can also consider liquidity constraints on the litigation decision: if farmers cannot access credit markets they must pay legal costs upfront, which translates into the condition $f(w_i(\mathbf{w}_{-i})) > \gamma$ i.e. the farmers are only able to pay with what they were able to produce. Under heterogeneous farmers, where only some of them are able to afford the lawsuit, the integral will only include those farmers able to afford it.

B.4.3 The problem of the River Manager

The manager is paid a salary S .⁷⁴ Since it is never optimal for farmers to sue under the lawful allocation, and every dry farmer sues under the unlawful one, the manager's problem in a Drought is

$$\max_{e \in [0,1]} S - (1 - e)(M - \tilde{i})B - \frac{1}{2}ce^2,$$

with interior solution $e^* = (M - \tilde{i})B/c$ whenever this lies in $(0, 1)$, and $e^* = 1$ if the total indemnity $(M - \tilde{i})B$ exceeds c .⁷⁵ Since $\tilde{i}$ does not depend on e ,⁷⁶ tighter exclusion under decentralized enforcement (a smaller $\tilde{i}$) strictly increases the manager's optimal effort by raising the indemnity at stake.

Overall, given the right set of incentives, the establishment of a River Manager strictly increases both the served share and expected output relative to Proposition 2: expected output is a convex combination of the lawful allocation and the equilibrium of Proposition 2, and the former is strictly larger by Proposition 4.

B.5 Microfoundation: upstream contiguous enforcement

The reduced-form effort e above stands in for a concrete choice: the share of farmers to inspect. We will show here that there is a microfoundation for this. First, we will show heuristically how a river manager would allocate effort along the river. Second, we will provide a general proof of this. Finally, we will characterize the optimal enforcement strategy.

Consider first a manager who inspects finitely many non-overlapping segments of the river, ensuring that the farmers in them do not over-extract. Let $I = [i^1, i^2] \cup [i^3, i^4] \cup \dots \subseteq [0, M]$ denote the union of the inspected segments, with masses $m_1, m_2, \dots$, and fix the egalitarian allocation $\alpha(i) = 1/M$. By Assumption 1 and Proposition 2, an inspected farmer with access to water extracts A^D/M , while an uninspected one extracts $(1 + (1 - p)\rho)A^D/M$ on average. Since uninspected low types over-extract, water runs out at some $\tilde{i} \leq M$,

⁷⁴We assume throughout it is profitable for all farmers to pay this salary, and that S satisfies the Manager's participation constraint.

⁷⁵If $e^* \leq 0$ —which can only happen with equality when legal costs are too high to discipline the Manager—farmers are better off not hiring the Manager.

⁷⁶This is true only in this reduced-form formulation, as it will be shown in Section B.5.

implicitly defined by total extraction exhausting the stream:

$$\int_0^{\tilde{i}} \left[1 + (1-p)\rho \mathbf{1}[j \notin I] \right] dj = M.$$

Writing m' for the inspected mass upstream of $\tilde{i}$ —the inspected farmers who actually receive water—this condition reads

$$\tilde{i} + (\tilde{i} - m')(1-p)\rho = M \iff \tilde{i} = m' + \frac{M - m'}{1 + (1-p)\rho},$$

which is increasing in m' (as $(1-p)\rho > 0$). The fraction of farmers receiving water is therefore an increasing function of the *inspected mass of farmers that receive water*, and of nothing else about the inspection plan: any inspection falling downstream of $\tilde{i}$ is wasted effort, while the location of the inspected segments within the served region is irrelevant. It is thus weakly optimal to inspect farmers as upstream as possible,⁷⁷ and auditing a contiguous mass m starting from the head of the river, $I = [0, m]$, is always among the optimal plans: it is the one placement that can never leave inspected mass stranded beyond the served frontier.

The next lemma states this for an arbitrary inspection set, not only finite unions of segments. Suppose now that the inspected set $I \subseteq [0, M]$ is an arbitrary measurable set, at a cost $C(|I|)$ that depends only on the inspected mass, where $|X|$ denotes the mass of a set of farmers $X \subseteq [0, M]$. Because the river is directional, an inspection at j relieves the water constraint only for farmers downstream of j : the water remaining at position i (while positive) is

$$A^R(i) = A^N \kappa \left[1 - \frac{i + (i - |I \cap [0, i]|)(1-p)\rho}{M} \right],$$

strictly increasing in $|I \cap [0, i]|$, the inspected mass strictly upstream of i , since $(1-p)\rho > 0$.

Lemma 2 (Contiguity of the optimal inspection set). *Fix an inspected mass m . The contiguous upstream set $I^* = [0, m]$ maximizes $|I \cap [0, i]|$ simultaneously at every position i , among all I with $|I| = m$.*

⁷⁷Weakly, because there is no difference between inspecting two segments upstream, as long as both of them receive water. What matters is the total mass of inspected farmers that receive water.

Proof. For any measurable I with $|I| = m$, $|I \cap [0, i]| \leq \min\{m, i\}$, since the intersection can exceed neither the total inspected mass nor the length of the interval. The set $I^* = [0, m]$ attains $|I^* \cap [0, i]| = \min\{m, i\}$ for every i . Cost is unchanged, as it depends only on $|I| = m$. $\square$

Since the water available at every position is increasing in the inspected mass upstream of it, and the cost depends only on the total mass, $[0, m]$ is an optimal placement of any inspected mass m —not as an assumption but as a property of the objective, although not uniquely so, by the indifference noted above. What is pinned down is that no optimal plan inspects downstream of the served frontier. This licenses the reduced form of Section B.4: enforcement, in principle an arbitrary measurable set, collapses without loss to a single scalar, the inspected mass.

By Lemma 2, then, let the inspected set be $[0, m]$ with $m \in [0, M]$, at cost $C(m) = \frac{1}{2}cm^2$. Farmers in $[0, m]$ extract A^D/M ; beyond m , a fraction p extract A^D/M and a fraction $(1 - p)$ extract $(A^D/M)(1 + \rho)$. Water is exhausted at

$$\tilde{i}(m) = m + \frac{M - m}{1 + (1 - p)\rho}, \quad \text{so that} \quad M - \tilde{i}(m) = (M - m) \frac{(1 - p)\rho}{1 + (1 - p)\rho}. \quad (16)$$

The served boundary is *linear* in m , with $\tilde{i}(0) = M/(1 + (1 - p)\rho)$ recovering Proposition 2 and $\tilde{i}(M) = M$. Note that $\tilde{i}(m) = M$ only at $m = M$: partial enforcement never restores the lawful allocation, because every uninspected low type over-extracts. This is the microfounded counterpart of $e < 1$ in the reduced form—the Manager improves access without restoring it.

By Section B.4.2, a farmer left dry recovers B ; writing $\pi \in (0, 1]$ for the probability that a dry farmer files—equal to one absent the liquidity constraints of footnote 73—the expected liability per dry farmer is πB . The manager solves

$$\max_{m \in [0, M]} S - \pi B(M - m) \frac{(1 - p)\rho}{1 + (1 - p)\rho} - \frac{1}{2}cm^2.$$

Because $\tilde{i}$ is linear in m , the marginal benefit of inspection is constant, and the solution is

closed-form:

$$m^* = \min \left\{ \pi B \frac{(1-p)\rho}{c(1+(1-p)\rho)}, M \right\}.$$

Since that marginal benefit is strictly positive while the marginal cost vanishes at $m = 0$, $m^* > 0$ always: with centralized enforcement, water always reaches more users than without. Monitoring is increasing in the filing probability π and the award B , in the over-extraction rate ρ , and in the share of low types $(1-p)$ —precisely the parameters that make the ungoverned equilibrium worse—and decreasing in the unit cost of inspection c .

Inspection raises access; whether it raises output follows from Proposition 4. Inspected farmers are held to their allotment and produce $\varepsilon(i)f(A^D/M)$; uninspected served farmers all produce $f(A^D/M)$, whether they mimic or not. Since $\tilde{i}(m)$ is linear in m , so is average output, which runs from W^{DE} at $m = 0$ to the lawful allocation $\bar{\varepsilon}f(A^D/M)$ at $m = M$; it is increasing precisely under Proposition 4. This is not immediate, since inspecting a low type *lowers* their output from $f(A^D/M)$ to $\mu f(A^D/M)$, reallocating that water downstream. Since $m^* > 0$, centralized enforcement strictly raises average output, though this comparison does not net out the cost of inspection $C(m)$. When c is small enough that $m^* = M$, the Manager implements the lawful allocation and no suit is filed in equilibrium: enforcement rests entirely on the off-path threat of liability.

B.6 Discussion

Two modeling choices deserve comment. First, extraction is fully consumptive: there are no return flows. In real rivers part of upstream over-extraction re-enters the stream, attenuating the downstream incidence of ρ ; the model therefore bounds the extensive-margin losses from above. Second, the two-point type distribution delivers exact mimicry; with a continuum of types, each type would be capped at a type-specific level, the lowest types would over-extract the most, and observed output would compress toward the threshold—the qualitative conclusions survive, at the cost of closed forms.

C Appendix: Measuring Distances

This appendix details the calculations developed to determine spatial relationships between various entities (e.g., gauges, municipality irrigated areas, farms) along a river system, which we use to determine relative positions along rivers, i.e. downstream and upstream locations. It also describe measures of distance along road networks used to control for market access and create instrumental variable.

C.1 Distance to the River Mouth

This algorithm estimates how far each entity is from the river mouth, measured along the river's path. This measure is used later as in input to determine the relative position of entities in the river network. Along the way, it provides a first order approximation of the relative position of each entity in the river system: places close to the river mouth can be roughly said to be downstream while places far from the mouth can roughly said to be upstream.

C.1.1 Inputs

- A set of georeferenced points of interest. This is straightforward when the object of interest is approximately a point (e.g. a gauge). If the object of interest has a non-negligible area (e.g. farm, municipality's irrigated area), we use its point of inaccessibility as reference point.
- River network layer.
- River mouth coordinate.

C.1.2 Steps

1. Convert the river network into a set of points.
2. For each point of interest, find the closest point in the river. Call each of these points a river connecting point (rcp).

3. Rasterize the river. In the resulting raster, every pixel crossed by the river has a value of 1. The rest of the raster has null values.
4. Estimate the least cost path connecting each rcp to the mouth of the river, over the rasterized version of the river.
5. For each rcp, recover the length of its corresponding least cost path.
6. For each object of interest, assign the distance to the outfall from its corresponding rcp.

C.1.3 Output

For each object of interest we have its distance to the river mouth over the river network. This corresponds to the distance to the outfall from the point of the river that is closest to the object of interest. This process is illustrated in figure [C.I](#).

C.2 Distance Calculation over Road Network

This algorithm estimates distances between a set of points (e.g., farms) and key locations such as the upstream provincial capital within the basin, ports, and the national capital. The calculation is performed using the built-in least-cost-path tool in QGIS. The road network is represented by a rasterized layer of paved roads from 2014. To reflect the varying quality of road types, paved roads are assigned half the travel cost of other road types (e.g., dirt roads, stabilized roads, gravel roads). The algorithm then calculates the shortest path over the road network, ensuring that distances reflect actual travel paths rather than straight-line distances.

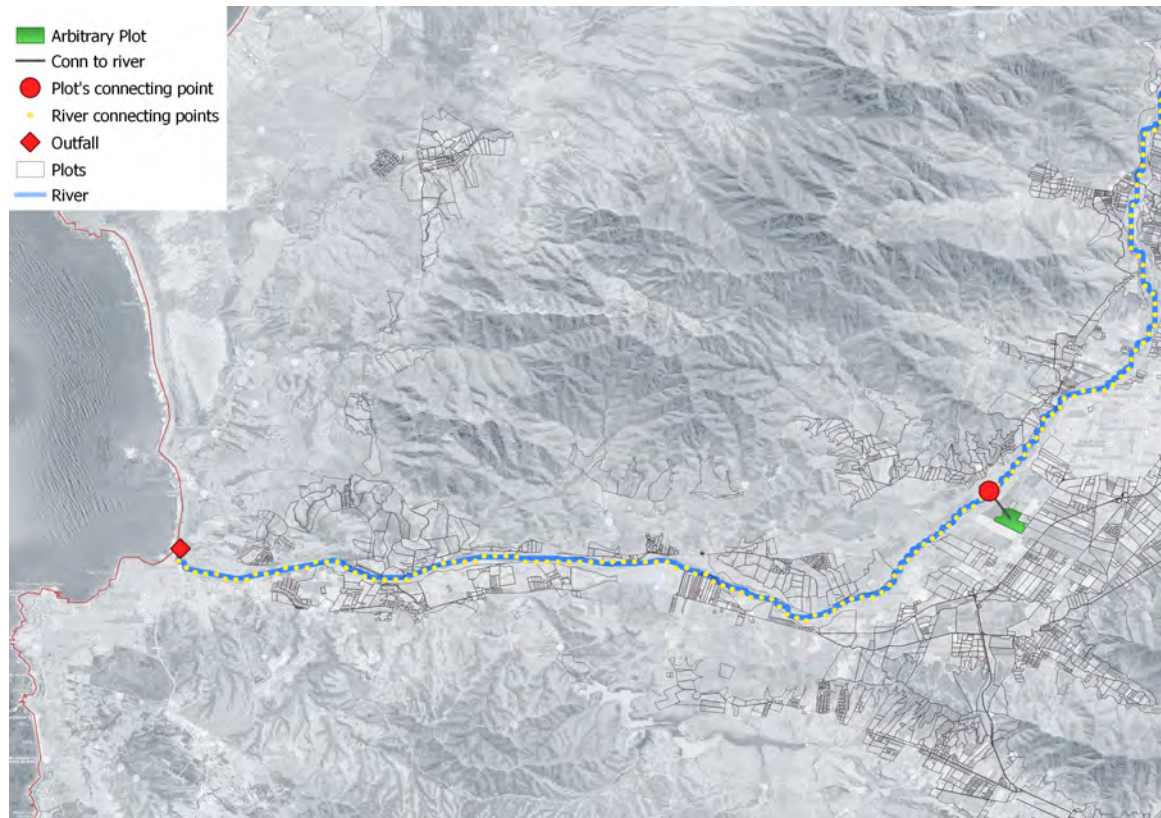


Figure C.I: Algorithm Illustration. An arbitrary area—shown in green—is connected to the closest point in the river, marked by a red circle. The distance assigned to this arbitrary object will be the distance along the river from the connecting point (red circle) to the river mouth or outfall (red square).